



## **ENERGY, WATER, AND CARBON AUDIT REPORT**

### **FOUR SEASONS RESORT AT MALDIVES, KUDA HURAA**

**MARCH, 2023**

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| <b>EEG</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              |                |                                         |
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## Abbreviations

|                 |                                                               |
|-----------------|---------------------------------------------------------------|
| AC              | Air Conditioning                                              |
| BMS             | Building Management System                                    |
| CHW             | Chilled Water System                                          |
| CO <sub>2</sub> | Carbon Dioxide                                                |
| COP             | Coefficient of Performance ( $KW_{cooling}/KW_{electrical}$ ) |
| EEM             | Energy Efficiency Measures                                    |
| ESCO            | Energy Services Company                                       |
| HVAC            | Heating Ventilation Air Conditioning                          |
| IRR             | Internal Rate of Return                                       |
| KVA             | kilo Volt Ampere                                              |
| KW              | Kilowatt                                                      |
| KWh             | Kilowatt-hour                                                 |
| LED             | Light Emitting Diode                                          |
| m               | Meter                                                         |
| m <sup>2</sup>  | Square meter                                                  |
| M3              | Cubic meter                                                   |
| MDB             | Main Distribution Board                                       |
| °C              | Degree Celsius                                                |
| PF              | Power Factor                                                  |
| RT              | Refrigerated Tons                                             |
| TR              | Transformer                                                   |
| UPS             | Uninterruptible Power Supply                                  |
| VAV             | Variable Air Volume                                           |
| VFD             | Variable Frequency Drive                                      |
| W               | Watt                                                          |
| MJ              | Mega Joules                                                   |
| USD             | United State Dollar                                           |
| FSRMKH          | Four Seasons Resort Maldives, at Kuda Huraa                   |

## Conversion Factors

| from               | to                       | multiply by |
|--------------------|--------------------------|-------------|
| meter              | Feet                     | 3.2808399   |
| meter              | Inches                   | 39.370079   |
| kg                 | Pounds                   | 2.2046226   |
| tons               | Kg                       | 1000        |
| liter              | Gallons (US beer)        | 0.264179    |
| liter              | Cubic feet               | 0.035315    |
| kWh                | MJ                       | 3.6         |
| kWh                | BTU                      | 3412        |
| W/m <sup>2</sup>   | BTU/ft <sup>2</sup>      | 0.317       |
| kJ/kg K            | BTU/lb °F                | 0.2388      |
| W (heat flow rate) | BTU/h                    | 3.412       |
| W/m <sup>2</sup> k | BTU/ft <sup>2</sup> h °F | 0.1761      |
| kJ/kg              | BTU/lb                   | 0.4299      |
| MJ/m <sup>3</sup>  | BTU/ft <sup>3</sup>      | 26.84       |
| Mm Hg (mercury)    | Pa                       | 133.332     |
| Ft of water        | kPa                      | 2.98898     |
| m <sup>3</sup> /kg | ft <sup>3</sup> /lb      | 16.02       |
| m/s                | ft/s                     | 3.281       |

# 1. Introduction

Following an agreement with the Hotel Management, Energy Efficiency Group **EEG** has undertaken an ASHRAE Level II energy audit of the Four Seasons Resort Maldives, Kuda Huraa **(in this report the facility shall be abbreviated by FSRMKH).**

The audits were carried out by combining thorough field assessments and further extensive analysis in the office. A close cooperation was made with the facility department throughout the audit who showed great support and provided quality assistance.

The present report is divided into four main sections. The first section includes a general description of the facility as well as the utilities accounting section (energy baseline), the second section covers the energy systems description as well as the development of the energy balance. The third section covers the water end uses while the fourth section includes the Energy and Water Efficiency Measures with their rationale, technical description and economical analysis.

## 1.1. Energy Efficiency Rationale

Simply put, Energy Efficiency is using less energy to accomplish the same amount of work needed. Ever since energy emerged as a top priority for policymakers, it has become a critical economic issue. In other words, unsustainable energy supply has real implications. Efficient buildings offer considerable opportunities to save money while reducing greenhouse gas emissions and taking leadership in sustainable development.

Within all sectors, energy costs may be a small percentage of turnovers, but reducing those costs can directly impact the profitability as they go straight to the bottom line, making businesses more competitive, especially with the rising of energy prices. The implementation of energy efficiency and sustainability measures increases levels of occupants' comfort as well as improving staff's general morale.

Energy audits include a review of all energy systems such as cooling, lighting, heating, ventilation, and building envelope. An analysis then serves the purpose of identifying where and how a facility uses energy and developing the set of Energy Efficiency Measures [EEMs] that could optimize the overall energy costs and which can range from small energy projects to major capital investments on operations and maintenance.

The audit focuses on a first phase on Actions with little or no cost which are improvements in current operations and practices that involve little or no cost of execution [LCNC]. Then it develops a road map for additional Energy efficiency measures that would cover all loads and infrastructure and will have different levels of CAPEX (Capital Expenditures) and Prioritization.

## 1.2. Energy Audit Goals and Tasks

EEG's energy audit is considered an Investment Grade Energy audit which exceeds the ASHRAE Level II Procedure for Commercial Building Energy Auditing,

### 1.2.1. End Goals

The goals of an energy audit are to:

- ✓ Identify the types and costs of energy/utilities use.
- ✓ Understand how the energy is being used – and possibly wasted.
- ✓ Determine and analyze various EEM alternatives such as improved operational techniques and/or new equipment that could substantially reduce energy costs.
- ✓ Perform an economic analysis on those alternatives and determine which ones are cost effective for the business or industry involved.

### 1.2.2. Methodology

The energy audit's methodology includes the following set of tasks:

- ✓ Gathering all historical utilities billing database.
- ✓ Gathering all facility layout, description, load data, operational hours.
- ✓ Undertake an in-depth field survey and auditing covering all sections and utilities end uses of the facility, completing the load inventory, analyzing equipment status & operational performances...
- ✓ Perform all needed real time measurements and data logging.
- ✓ Build an Energy Simulation of the facility using an Energy Analysis software. This modeling will allow us to have a detailed load and cost breakdown (Energy balance) along an in-depth study on the consumption of the facility.
- ✓ Investigate potential Energy Efficiency Measures (EEMs) from the low cost/no cost ones to those with low CAPEX and quick returns and then those with higher CAPEX and longer returns.
- ✓ Develop the chosen EEMs with their impact on the energy consumption and related financials.
- ✓ Prepare a comprehensive report that would be divided into 2 main parts and the appendixes:
- ✓ In depth analysis of the present situation including utilities analysis, energy baseline and energy balance, detailed systems description...
- ✓ EEMs: Technical analysis, savings development, financial analysis.
- ✓ Appendixes: Full load inventory list and any supporting technical documentation.

## 1.3. Water Audit Goals

### 1.3.1. End Goals

- ✓ A water audit is a comprehensive evaluation of an organization's water usage and management practices. The primary goals of a water audit are:
- ✓ Measurement and Analysis: To accurately measure the quantity of water used and identify the sources of water usage.
- ✓ Cost savings: To identify opportunities for reducing water usage and costs, including finding and fixing leaks, implementing water-saving technologies, and reducing waste.
- ✓ Water Conservation: To promote the efficient and responsible use of water, reducing the impact on the environment and preserving water resources for future generations.
- ✓ Compliance: To ensure compliance with local and national regulations regarding water usage and discharge, avoiding penalties and fines.
- ✓ Sustainability: To support an organization's sustainability goals and demonstrate a commitment to environmental responsibility.
- ✓ Continuous Improvement: To identify areas for improvement in water management practices and establish a continuous improvement program.

Overall, the goals of a water audit are to increase efficiency, reduce costs, and minimize the environmental impact of an organization's water usage.

## 1.4. Carbon Audit Goals

### 1.4.1. End Goals

A carbon audit is a comprehensive evaluation of an organization's carbon footprint, including its greenhouse gas emissions. The primary goals of a carbon audit are:

- ✓ **Measurement and Analysis:** To accurately measure the organization's carbon footprint and identify the sources of greenhouse gas emissions.
- ✓ **Emissions Reduction:** To identify opportunities for reducing greenhouse gas emissions and establish a plan for reducing emissions over time.
- ✓ **Cost Savings:** To identify opportunities for reducing energy costs and improving energy efficiency, which can also result in reduced greenhouse gas emissions.
- ✓ **Compliance:** To ensure compliance with local and national regulations regarding greenhouse gas emissions and energy usage, avoiding penalties and fines.
- ✓ **Sustainability:** To support an organization's sustainability goals and demonstrate a commitment to environmental responsibility.
- ✓ **Continuous Improvement:** To identify areas for improvement in carbon management practices and establish a continuous improvement program.
- ✓ **Transparency and Communication:** To increase transparency and communicate the organization's efforts to reduce greenhouse gas emissions to stakeholders.

Overall, the goals of a carbon audit are to increase efficiency, reduce costs, minimize the environmental impact of an organization's activities, and demonstrate a commitment to sustainability and climate action.

## 2. Facility General Description

### 2.1. Introduction

Four Seasons Resort Maldives Kuda Huraa is a luxury resort located on a private island in the Maldives covering an area of approximately 32,746 m<sup>2</sup> out of which 12,976 m<sup>2</sup> is conditioned and includes a range of accommodations, restaurants, and recreational facilities.

Figure 1:FSRMKH aerial view



- 1: SPA Island
- 2: Main Jetty
- 3: Staff housing, technical area
- 4: main Pool
- 5: Gym
- 6: Dive Center and Dive Shop
- 7: Kids Club
- 8: Beach Pavilion and Beach Bungalow guestrooms
- 9: Water Bungalow guestrooms

The resort features a total of 96 private villas, each with its own pool. The villas are designed in traditional Maldivian style and feature high ceilings, thatched roofs, and wooden furnishings and are grouped into 3 main types: Beach Pavilions, Beach Bungalows, and water Bungalow, in addition to 1 Royal Villa.

Figure 2: Water Bungalow (Left), Beach Bungalow (Right)



FSRMKH offers a wide range of activities and amenities for guests, including two swimming pool, a fitness center, SPA, Diving Center, and a Kids Club. It encloses also a variety of F&B outlets : Baraabaru, Café Huraa, Kandu Grill, Reef Club, and Sunset Lounge.

Figure 3: SPA Island (left), Gym (Right)



- Top: SPA Island (Left), Fitness Center (Right)
- Bottom: Kids Club

Finally, the expansive BOH area includes a on site staff accomodations composed of multiple buildings with a variety of rooms' types and all the related services such as an outdoor sports court, fitness room and Canteen.

## 2.1. Building Envelope

A building envelope is a major component of buildings energy performance and the physical separator between conditioned and unconditioned environments mainly including the resistance to heat and noise transfer. The building envelope includes all elements of the outer shell that maintain a dry, heated, or cooled indoor environment and facilitate its climate control. It is considered inefficient when it allows high rates of heat transfer between interior spaces and the outdoor environment. Eliminating heat gain and heat loss completely is not possible, but they can be minimized to improve efficiency. Below are some thermal images taken during the field visit in November 2022 revealing the performance of the building envelope.

- **Architecture:**

The resort's architecture is designed to blend with the natural environment of the island and features traditional Maldivian design elements, such as thatched roofs, natural sustainable materials, and open-air spaces.

Figure 4: Coconut leaves roof bungalows



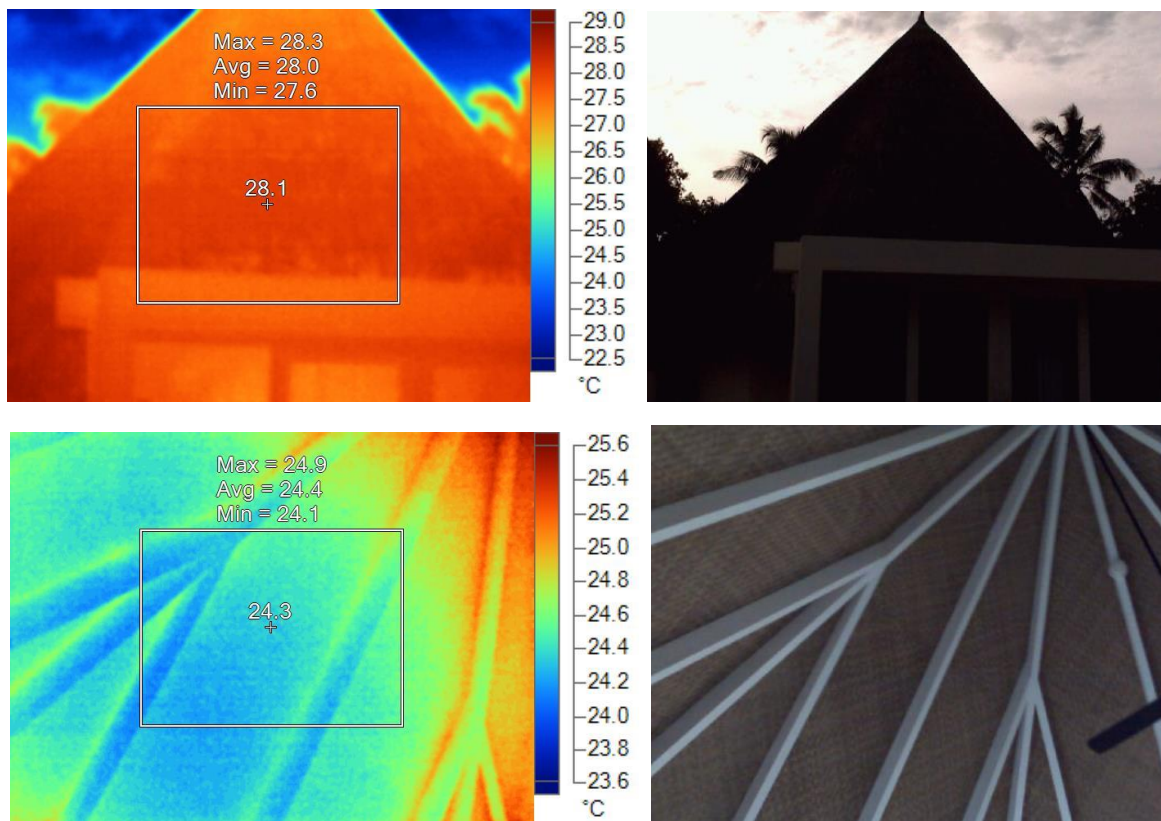
To keep the interiors cool and comfortable, guestrooms are insulated with materials such as spray foam or rigid foam, which helps to prevent heat transfer from the exterior to the interior and reducing by that the amount of energy needed to cool the space.

The roofs of the buildings are also designed to be reflective, which helps to reduce heat gain and keep the interiors cool. The roofs of Villa Guestrooms in FSMKH are made from thatch, which reflect the sun's rays and reduce the amount of heat absorbed .

Figure 5: Thatched roofs



Figure 6: Thermal Image (left) and original image (right) of the building envelope of the bungalow- interior and exterior

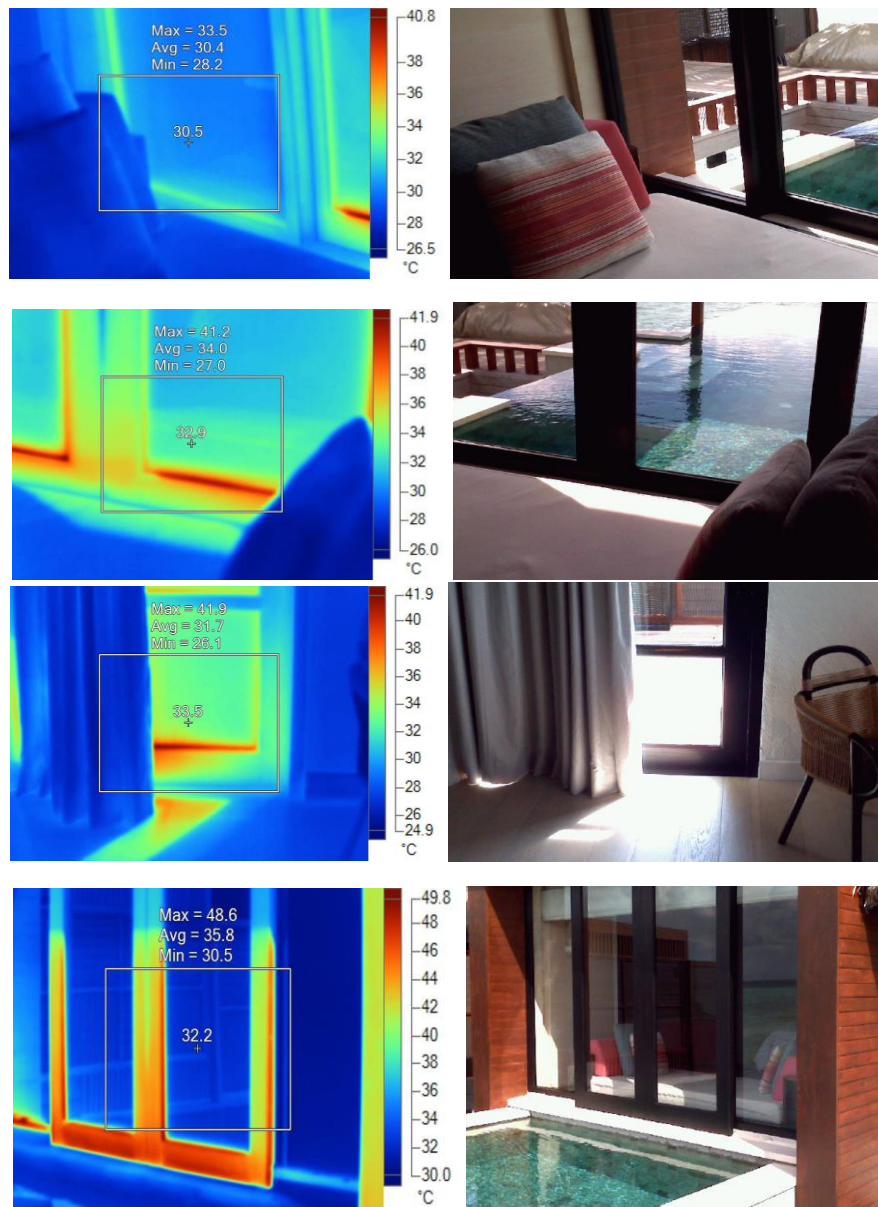


As can be seen in the thermal images above, the external temperatures of the roof are at 28°C while the interior temperatures are at around 24.4°C indicating that the roof has excellent thermal insulation properties.

- **Glazing:**

Another important feature of the building envelope is glazing. The villas at the resort feature large windows and doors, which allow for plenty of natural light and views of the surrounding landscape. The windows are made from single glazed glass, which are not the most optimal option used to reduce heat gain and keep the interiors cool. The windows are also designed to be operable, which allows for natural ventilation. The thermal images below were taken across the facility bungalows in November 2022.

Figure 7: Thermal Images (left), original Image (Right)



As seen in the above pictures, the glazing's temperature from the inside is very close to the external one ( 31.7°C versus 33.5°C respectively ) indicating a poor thermal performance of the glazing.

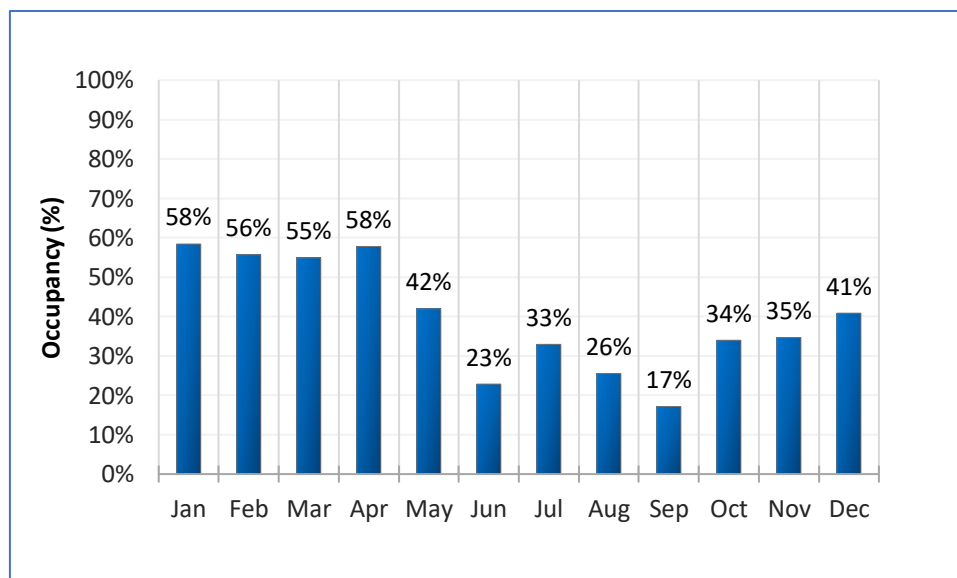
## 2.2. Occupancy

The following table depicts the occupancy rate variation of the resort during the baseline year extending from January 2022 until December 2022. As can be seen, the occupancy was fluctuating during the baseline period between a minimum of 17% in September and a maximum of 58% in January. The average occupancy rate throughout the baseline year is 40%.

Table 1: Occupancy Rates

| MONTH          | Days Per Month | Actual Room Nights | Available Room Nights | Occupancy Rate |
|----------------|----------------|--------------------|-----------------------|----------------|
| Unit           | Days           | RN                 | RN                    | %              |
| Jan-22         | 31             | 1,738              | 2,976                 | 58%            |
| Feb-22         | 28             | 1,497              | 2,688                 | 56%            |
| Mar-22         | 31             | 1,634              | 2,976                 | 55%            |
| Apr-22         | 30             | 1,664              | 2,880                 | 58%            |
| May-22         | 31             | 1,248              | 2,976                 | 42%            |
| Jun-22         | 30             | 658                | 2,880                 | 23%            |
| Jul-22         | 31             | 980                | 2,976                 | 33%            |
| Aug-22         | 31             | 759                | 2,976                 | 26%            |
| Sep-22         | 30             | 494                | 2,880                 | 17%            |
| Oct-22         | 31             | 1,011              | 2,976                 | 34%            |
| Nov-22         | 30             | 998                | 2,880                 | 35%            |
| Dec-22         | 31             | 1,216              | 2,976                 | 41%            |
| <b>TOTAL</b>   | <b>365</b>     | <b>13,897</b>      | <b>35,040</b>         | <b>40%</b>     |
| <b>Maximum</b> | -              | 1,738 [Jan]        | 2,976 [Jan]           | 58% [Jan]      |
| <b>Average</b> | 30             | 1,158              | 2,920                 | 40%            |
| <b>Minimum</b> | -              | 494 [Sep]          | 2,688 [Feb]           | 17% [Sep]      |

Figure 8: Occupancy Rates Variation



## 2.3. Kuda Huraa Climatic Conditions

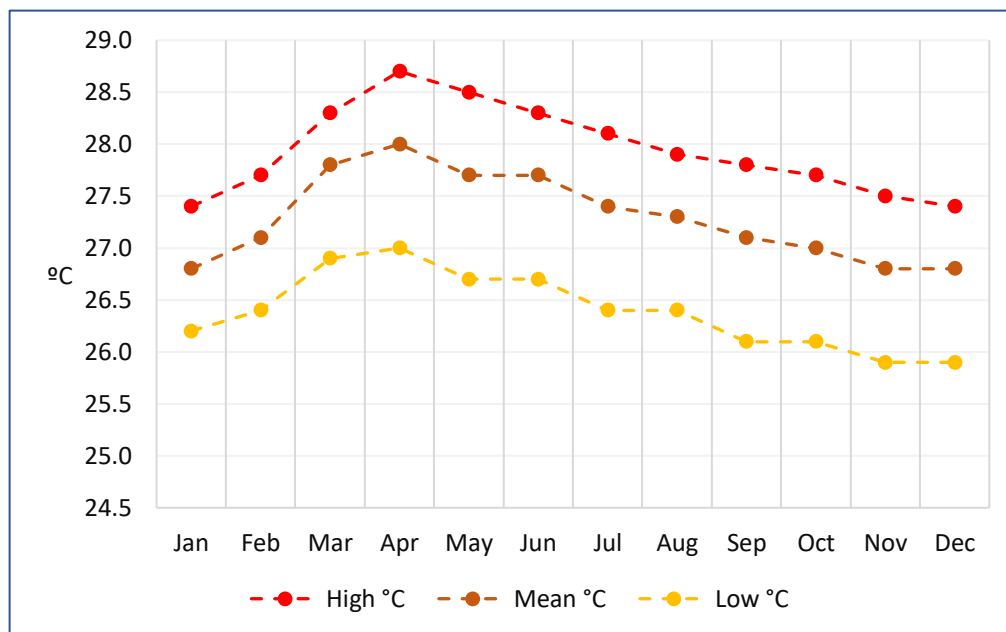
Maldives has a tropical climate characterized by significant rainfall in most months of the year. The climate in the Maldives is generally hot and humid, with temperatures ranging from 25°C to 31°C throughout the year. The country experiences two distinct seasons: the dry season (from December to April) and the wet season (from May to November). The wet season brings heavy rainfall and occasional thunderstorms, but overall, the weather remains warm and pleasant throughout the year. The following table depicts the weather data for Kuda Huraa, which will help in many of the energy analysis present in the following chapters.

Table 2: Maldives Climatic Data

| Month       | Jan  | Feb  | Mar  | Apr  | May  | Jun  | Jul  | Aug  | Sep  | Oct  | Nov  | Dec  |
|-------------|------|------|------|------|------|------|------|------|------|------|------|------|
| High °C     | 27.4 | 27.7 | 28.3 | 28.7 | 28.5 | 28.3 | 28.1 | 27.9 | 27.8 | 27.7 | 27.5 | 27.4 |
| Mean °C     | 26.8 | 27.1 | 27.8 | 28   | 27.7 | 27.7 | 27.4 | 27.3 | 27.1 | 27   | 26.8 | 26.8 |
| Low °C      | 26.2 | 26.4 | 26.9 | 27   | 26.7 | 26.7 | 26.4 | 26.4 | 26.1 | 26.1 | 25.9 | 25.9 |
| Prec. mm    | 86   | 43   | 59   | 121  | 234  | 140  | 148  | 154  | 185  | 203  | 226  | 219  |
| RH %        | 77%  | 76%  | 76%  | 78%  | 82%  | 81%  | 80%  | 80%  | 81%  | 81%  | 81%  | 80%  |
| Sunshine H. | 8.5  | 8.7  | 8.7  | 8.7  | 8.7  | 9.2  | 9.2  | 9.3  | 9.1  | 8.8  | 8.4  | 8.3  |

Source: Climate-Data.org

Figure 9: Temperature Variations in Kuda Huraa



## Degree Days:

Degree days are essentially a simplified representation of outside air-temperature data. They are widely used in the energy industry for calculations relating to the effect of outside air temperature on building energy consumption.

"Heating degree days", or "HDD", are a measure of how much (in degrees), and for how long (in days), outside air temperature was lower than a specific "base temperature" (or "balance point"). They are used for calculations relating to the energy consumption required to heat or cool buildings.

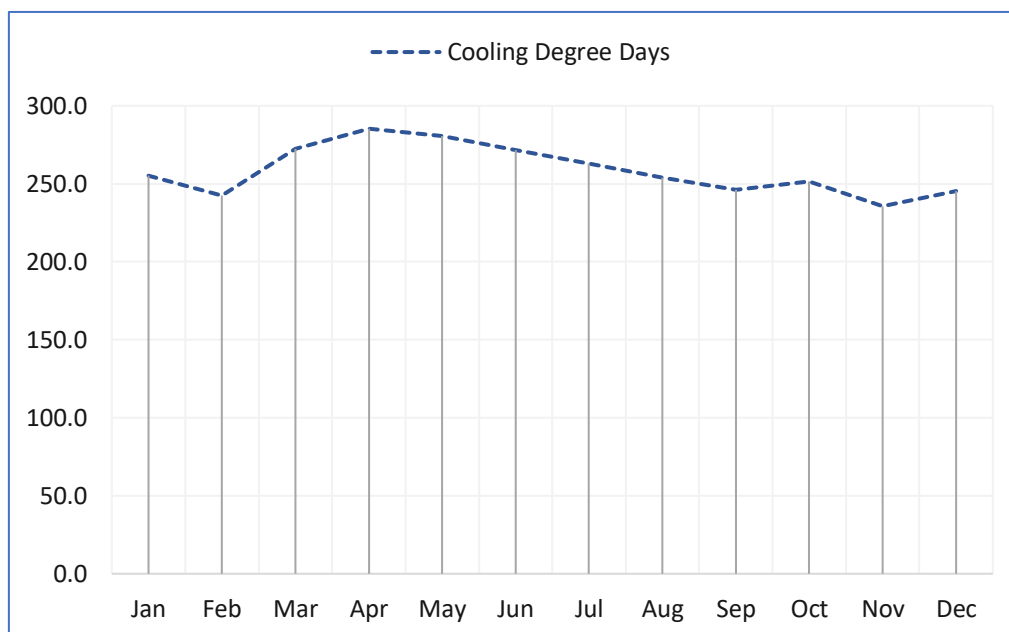
"Cooling degree days", or "CDD", are a measure of how much (in degrees), and for how long (in days), outside air temperature was higher than a specific base temperature. They are used for calculations relating to the energy consumption required to cool buildings.

"Heating degree days", or "HDD", are a measure of how much (in degrees), and for how long (in days), outside air temperature was lower than a specific base temperature. They are used for calculations relating to the energy consumption required to heat buildings.

In the present report, CDD was obtained for Kuda Huraa with a base temperature of 20.5°C. It was found that the maximum cooling degree day was 285 recorded in April, while the minimum degree day is 242 in February.

The below chart provides the monthly CDD Variations in Kuda Huraa across the baseline year of 2022.

Figure 10: CDD and HDD Variations in Kuda Huraa



## 2.4. Utilities: Energy Sources and Water

The utilities of FSRKH are electricity, Diesel, Water, and LPG. The following section depicts the accounting numbers for the utilities over the 12-month **baseline period** from **January 2022 until December 2022**.

### 2.4.1. Electricity (Fuel oil)

Electricity in FSRMKH is supplied through 2x1,000 KVA and 3x450 KVA diesel generators supplying electricity to the island. Moreover, electricity is supplied to Water Bungalows through 2x800KVA step-up/stepdown transformers. The electrical outputs of these diesel gensets are connected together via a synchronizing board, from which power lines are connected to the ATs and the MDBs.

Figure 11: Transformers room (Left), Total synchro board (Right)



Electricity rate is at an average of **\$0.317 /KWh** for the average of the baseline period and the total charges are at **\$2,409,965** representing **91.7%** of the total energy & water related utilities expenditures.

### 2.4.2. LPG

LPG is used to FSRMKH for Kitchen' operations. LPG rate is at **\$1.04/Kg** and the total charges are at **\$112,963** representing **4.3 %** of the total utilities expenditures.

Figure 12: LPG for kitchens with Gas detection system



### 2.4.3. Water

Water in FSRKH is provided through an on-site desalination plant that uses Reverse osmosis technology to convert seawater into fresh water in addition to and a water storage system. The total water charges are at **\$ 104,431 (excluding the electricity cost of RO which was already included in the electricity utility)** per year representing **4%** of the total utilities cost.

### 3. Utilities Accounting

The following sections detail the accounting and consumption numbers for utilities over the **baseline period extending from January until December 2022**. The annual consumption and related cost baseline are the aggregated consumption and associated costs for FSRMKH throughout the baseline period. This energy baseline serves as a benchmark consumption to assess the current energy performance along with the calculation of all the proposed Energy Efficiency Measures.

#### 3.1. Electricity (Fuel oil)

##### 3.1.1. Electricity (Fuel Oil) Consumption and Cost Breakdown

The following table shows the breakdown of the total energy consumption of FSRMKH for the baseline period, as provided by the electricity bills. The total amount of electricity consumption is **7,581,911 KWh** for a total cost of **\$2,409,965** at an average tariff of **\$0.317/KWh**.

Table 3: Diesel Gensets Electrical Consumption and Cost Breakdown

| Month        | Consumption      |                    | Rate           | Energy Value      | Deviation  |
|--------------|------------------|--------------------|----------------|-------------------|------------|
| Unit         | KWh              | \$                 | \$/KWh         | MJ                | -          |
| Jan          | 714,310          | 138,576            | \$0.19         | 2,571,516         | 12%        |
| Feb          | 654,040          | 117,727            | \$0.18         | 2,354,544         | 3%         |
| Mar          | 729,060          | 141,438            | \$0.19         | 2,624,616         | 13%        |
| Apr          | 730,070          | 243,843            | \$0.33         | 2,628,252         | 13%        |
| May          | 710,031          | 257,457            | \$0.36         | 2,556,112         | 11%        |
| Jun          | 625,150          | 201,372            | \$0.32         | 2,250,540         | -1%        |
| Jul          | 636,450          | 250,125            | \$0.39         | 2,291,220         | 1%         |
| Aug          | 599,420          | 236,171            | \$0.39         | 2,157,912         | -5%        |
| Sep          | 531,680          | 219,052            | \$0.41         | 1,914,048         | -19%       |
| Oct          | 553,300          | 217,170            | \$0.39         | 1,991,880         | -14%       |
| Nov          | 542,200          | 203,867            | \$0.38         | 1,951,920         | -17%       |
| Dec          | 556,200          | 183,166            | \$0.33         | 2,002,320         | -14%       |
| <b>Total</b> | <b>7,581,911</b> | <b>\$2,409,965</b> | <b>\$0.318</b> | <b>27,294,880</b> | <b>-</b>   |
| Maximum      | 730,070 (Apr)    | 257,457 (May)      | 0.41 (Sep)     | 2,628,252 (Apr)   | 13% (Apr)  |
| Average      | 631,825.92       | \$200,830          | \$0.32         | 2,274,573         | -          |
| Minimum      | 531,680 (Sep)    | 117,727 (Feb)      | 0.18 (Feb)     | 1,914,048 (Sep)   | -19% (Sep) |

It is important to mention that electricity in FSRMKH is generated through diesel generators. The diesel quantity and cost were provided from the engineering department as per the following table, in addition to the generator's services records until the month of October 2022. The electricity cost is therefore obtained by adding the cost of diesel to the maintenance cost, and a certain operation cost / KWH.

Table 4: Diesel Consumption and Cost Breakdown

| Month        | Consumption      |                    | Rate          | Energy Value      | Deviation  |
|--------------|------------------|--------------------|---------------|-------------------|------------|
| Unit         | liter            | \$                 | \$/liter      | MJ                | -          |
| Jan          | 189,315          | \$133,107          | \$0.70        | 6,866,455         | 9%         |
| Feb          | 166,967          | \$106,024          | \$0.64        | 6,055,893         | -3%        |
| Mar          | 184,995          | \$130,070          | \$0.70        | 6,709,769         | 7%         |
| Apr          | 205,088          | \$236,118          | \$1.15        | 7,438,542         | 16%        |
| May          | 196,733          | \$226,499          | \$1.15        | 7,135,506         | 12%        |
| Jun          | 172,603          | \$233,083          | \$1.35        | 6,260,311         | 0%         |
| Jul          | 177,084          | \$236,319          | \$1.33        | 6,422,837         | 2%         |
| Aug          | 166,385          | \$217,182          | \$1.31        | 6,034,784         | -4%        |
| Sep          | 147,447          | \$192,448          | \$1.31        | 5,347,903         | -17%       |
| Oct          | 154,183          | \$201,240          | \$1.31        | 5,592,217         | -12%       |
| Nov          | 149,682          | \$193,434          | \$1.29        | 5,428,966         | -15%       |
| Dec          | 162,276          | \$209,709          | \$1.29        | 5,885,751         | -6%        |
| <b>Total</b> | <b>2,072,758</b> | <b>\$2,315,233</b> | <b>\$1.12</b> | <b>75,178,933</b> | <b>-</b>   |
| Maximum      | 205,088 (Apr)    | 236,319 (Jul)      | 1.3504 (Jun)  | 7,438,542 (Apr)   | 16% (Apr)  |
| Average      | 172,730          | \$192,936          | \$1.127       | 6,264,911         | -          |
| Minimum      | 147,447 (Sep)    | 106,024 (Feb)      | 0.6350 (Feb)  | 5,347,903 (Sep)   | -17% (Sep) |

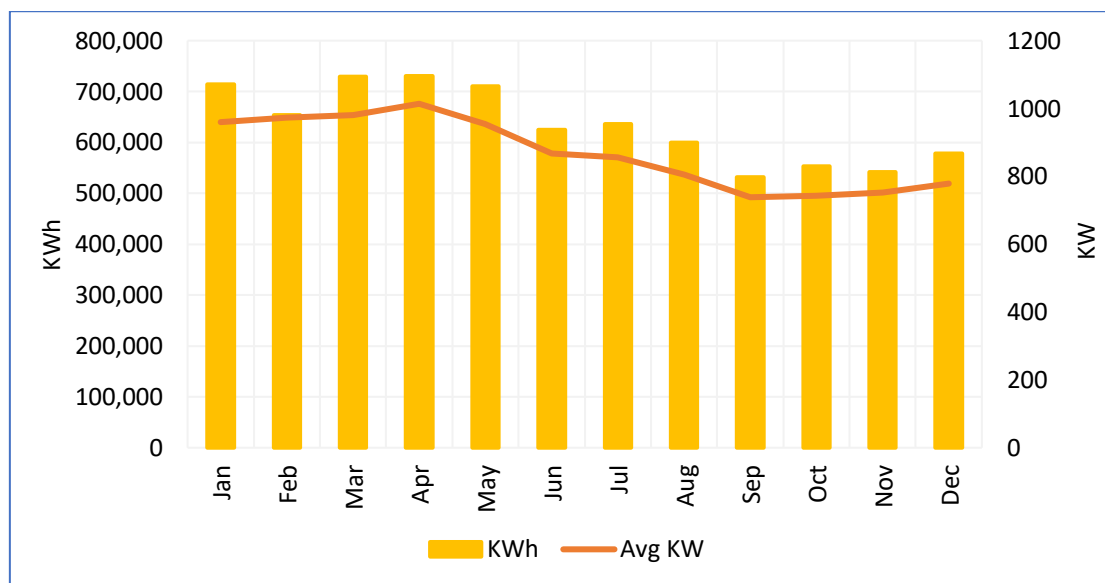
Table 5: Diesel Gensets maintenance and services costs

| Month        | Total Maintenance Cost (Gensets 1,2,3,4,5) |
|--------------|--------------------------------------------|
| Unit         | \$                                         |
| Jan          | 8,623                                      |
| Feb          | 10,294                                     |
| Mar          | 7,732                                      |
| Apr          | 8,402                                      |
| May          | 7,100                                      |
| Jun          | 3,122                                      |
| Jul          | 2,485                                      |
| Aug          | 2,400                                      |
| Sep          | 12,585                                     |
| Oct          | 14,371                                     |
| Nov          | 7,715                                      |
| Dec          | 20,115                                     |
| <b>Total</b> | <b>\$104,944</b>                           |

### 3.1.2. Load Profile

The below chart showcases the monthly electricity consumption during the baseline year along with the average electricity demand (in KW). It is seen that April carried the highest electricity consumption with about 15% higher than the average annual consumption while the lowest consumption was recorded in September with 16% lower than the annual average. KW Demand follows a similar path with an average of 1,014 KW in April and 738 KW in September.

Figure 13: Baseline Year - Electricity Monthly KWh and Average KW



The below figures show the electricity monthly consumption variation along with the occupancy rate and CDD during the baseline year. Electricity consumption seems to be highly correlated to both variables.

Figure 14: Electricity monthly consumption variation with occupancy rate

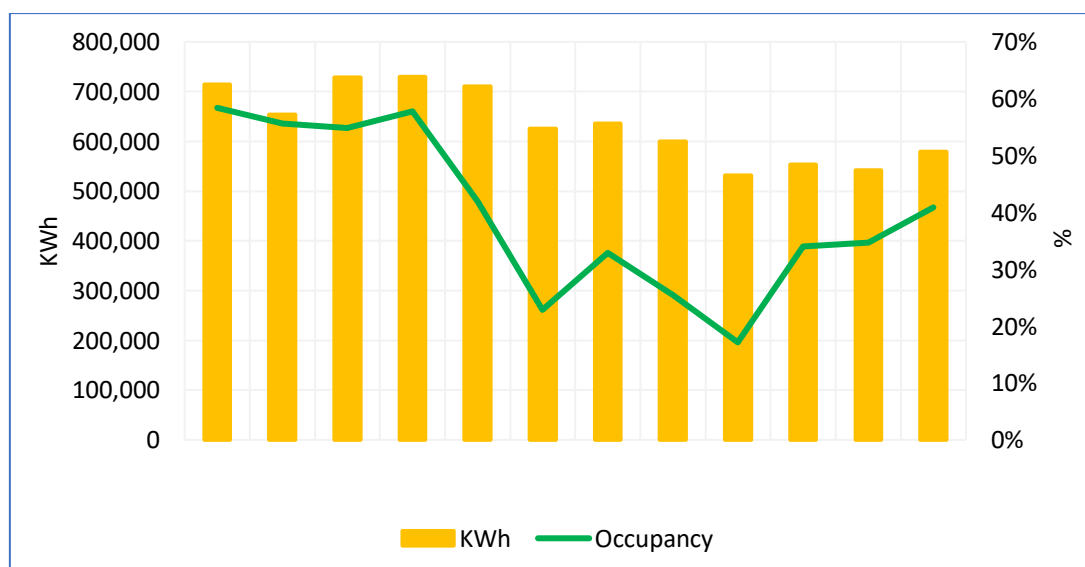
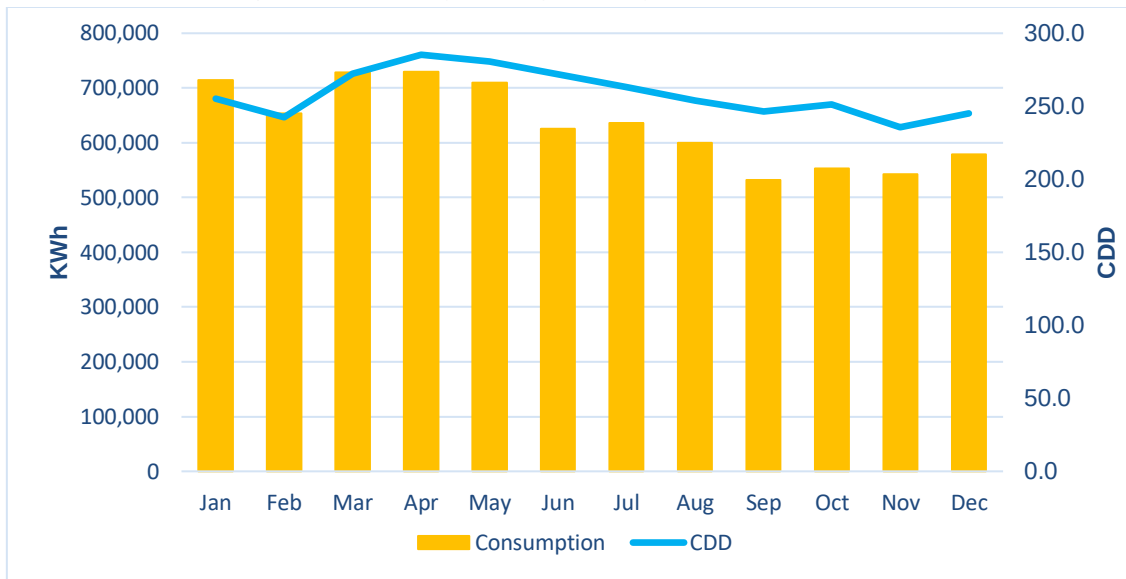


Figure 15: Electricity monthly consumption variation with CDD



### 3.1.3. Electricity Baseline – Regression Analysis

In statistical modeling, regression analysis is a set of statistical processes for estimating the relationship among variables. It focuses on the relationship between a dependent variable and one or more independent variables.

In the case of FSRMKH for a given month, a relationship exists between the energy consumption (the main parameter) and the following variables: the occupancy rate and the weather conditions (mathematically represented by the cooling degree days) Using the regression analysis, an empirical equation is modeled to generate projected energy consumption as a function of the above mentioned variables. The equation will be of the form:

$$KWh = A \times OCC + B \times CDD + C$$

OCC is the occupancy rate and CDD is the Cooling Degree Day

A, B and C are the coefficient to be calculated in the process of the regression analysis

In the present regression analysis, the historical variable parameters used for the modeling are found in the below table.

Table 6: Regression Analysis Variables

| Month | Occupancy   | CDD |
|-------|-------------|-----|
| Unit  | Room Nights | CDD |
| Jan   | 1,738       | 255 |
| Feb   | 1,497       | 242 |
| Mar   | 1,634       | 272 |
| Apr   | 1,664       | 285 |
| May   | 1,248       | 281 |
| Jun   | 658         | 272 |
| Jul   | 980         | 263 |
| Aug   | 759         | 254 |
| Sep   | 494         | 246 |
| Oct   | 1,011       | 251 |
| Nov   | 998         | 236 |
| Dec   | 1,216       | 245 |

The variables in the equation would then be:

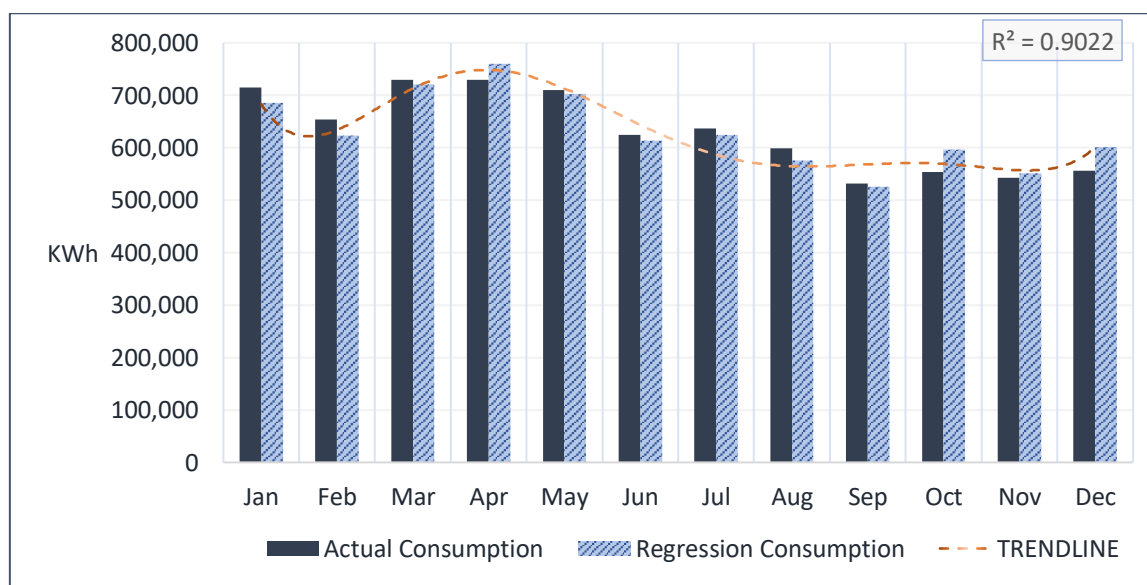
$$KWh = 108.6 \times OCC + 2,764 \times CDD - 208,782.9$$

Table 7: Electricity Projected Consumption, Residuals and variation

| Month        | Actual Consumption | Regression Consumption | Residuals | Deviation |
|--------------|--------------------|------------------------|-----------|-----------|
| Unit         | KWh                | KWh                    | KWh       | %         |
| Jan          | 714,310            | 685,303                | 29,007    | 4%        |
| Feb          | 654,040            | 623,882                | 30,158    | 5%        |
| Mar          | 729,060            | 721,276                | 7,784     | 1%        |
| Apr          | 730,070            | 760,609                | -30,539   | -4%       |
| May          | 710,031            | 702,433                | 7,598     | 1%        |
| Jun          | 625,150            | 613,747                | 11,403    | 2%        |
| Jul          | 636,450            | 624,672                | 11,778    | 2%        |
| Aug          | 599,420            | 575,789                | 23,631    | 4%        |
| Sep          | 531,680            | 525,859                | 5,821     | 1%        |
| Oct          | 553,300            | 596,111                | -42,811   | -8%       |
| Nov          | 542,200            | 551,023                | -8,823    | -2%       |
| Dec          | 556,200            | 601,206                | -45,006   | -8%       |
| <b>Total</b> | <b>7,581,911</b>   | <b>7,581,911</b>       | <b>0</b>  | <b>0%</b> |

The residuals are the difference between the actual and the projected energy consumption for the same values of independent variables. The variation is the percentage of difference between the actual and the projected Electricity Consumption. As can be seen, the annual regression shows an exact match (difference of 0), whereas on a monthly basis we find alternating variations mainly due to the change in occupancy, weather conditions and normalized days. The following chart gives a clear visibility on the regression model and the actual consumption for the baseline period.

Figure 16: Electrical Regression and Actual Energy Consumption



**R-squared** is a statistical measure linked to the regression model of how close the data are to the fitted regression line. It is also known as the coefficient of determination, which is the percentage of the response variable variation that is explained by a linear model (explained variation / total variation). R-squared is always between 0 and 100%: 0% indicates that the model explains none of the variability of the response data around its mean; 100% indicates that the model explains all the variability of the response data around its mean. In general, the higher the R-squared, the better the model fits the data. In the case of FSRMKH,  $R^2 = 0.9022$ , which means that the model is considered accurate.

## 3.2. LPG

### 3.2.1. LPG Baseline Consumption

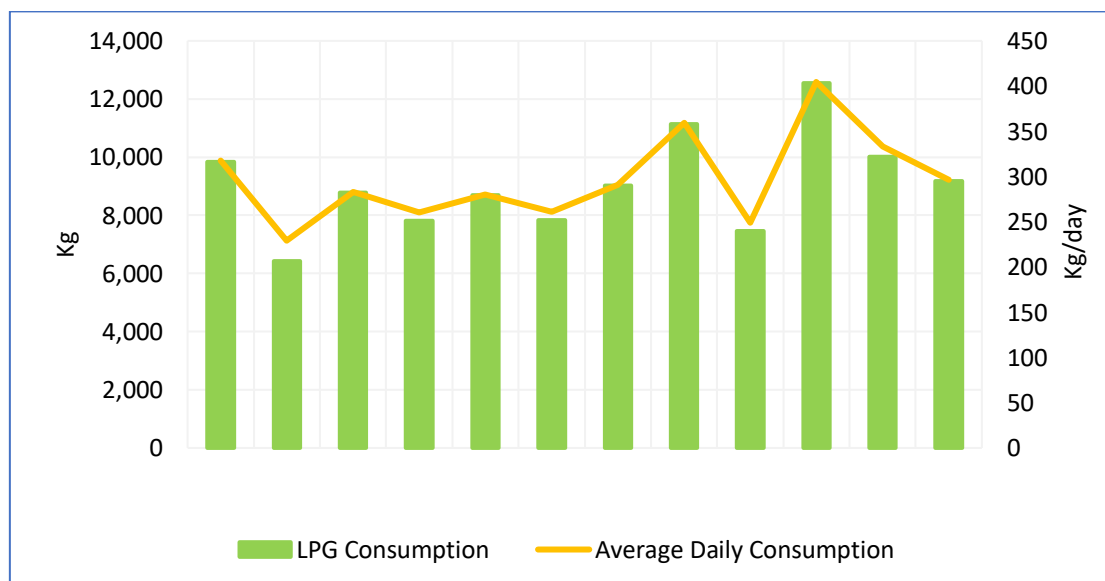
The following table shows the breakdown of the total LPG consumption of FSRMKH for the baseline period mainly used in the kitchens.

Table 8: LPG total consumption and cost

| Month        | LPG Consumption |                     | Rate          | Energy Value     | Deviation  |
|--------------|-----------------|---------------------|---------------|------------------|------------|
| Unit         | KG              | \$                  | \$/KG         | MJ               | -          |
| Jan          | 9,840           | \$9,840.00          | \$1.00        | 482,160          | 11%        |
| Feb          | 6,415           | \$5,773.50          | \$0.90        | 314,335          | 3%         |
| Mar          | 8,780           | \$8,780.00          | \$1.00        | 430,220          | 13%        |
| Apr          | 7,810           | \$7,810.00          | \$1.00        | 382,690          | 13%        |
| May          | 8,695           | \$10,434.00         | \$1.20        | 426,055          | 11%        |
| Jun          | 7,835           | \$10,969.00         | \$1.40        | 383,915          | -1%        |
| Jul          | 9,015           | \$9,015.00          | \$1.00        | 441,735          | 0%         |
| Aug          | 11,141          | \$11,141.27         | \$1.00        | 545,922          | -6%        |
| Sep          | 7,469           | \$7,469.00          | \$1.00        | 365,981          | -19%       |
| Oct          | 12,541          | \$12,541.00         | \$1.00        | 614,509          | -15%       |
| Nov          | 10,006          | \$10,005.93         | \$1.00        | 490,291          | -17%       |
| Dec          | 9,184           | \$9,184.43          | \$1.00        | 450,037          | -10%       |
| <b>Total</b> | <b>108,732</b>  | <b>\$112,963.13</b> | <b>\$1.04</b> | <b>5,327,850</b> | <b>-</b>   |
| Maximum      | 12,541 (Oct)    | \$12,541 (Oct)      | \$1.40 (Jun)  | 614,509 (Oct)    | 13% (Apr)  |
| Average      | 9,061           | \$9,413.59          | \$1.04        | 443,987          | -          |
| Minimum      | 6,415 (Feb)     | \$5,774 (Feb)       | \$0.90 (Feb)  | 314,335 (Feb)    | -19% (Sep) |

The below chart showcases the monthly gas consumption during the baseline year versus the average daily rate. It is seen that October carried the highest LPG consumption with about 38% higher than the average annual consumption while the lowest consumption was recorded in February with 29% lower than the annual average. Demand follows a similar path with an average of 405 KG in October and 229 KG in February.

Figure 17: Baseline Gas Consumption Variations vs average daily consumption



### 3.3. Water

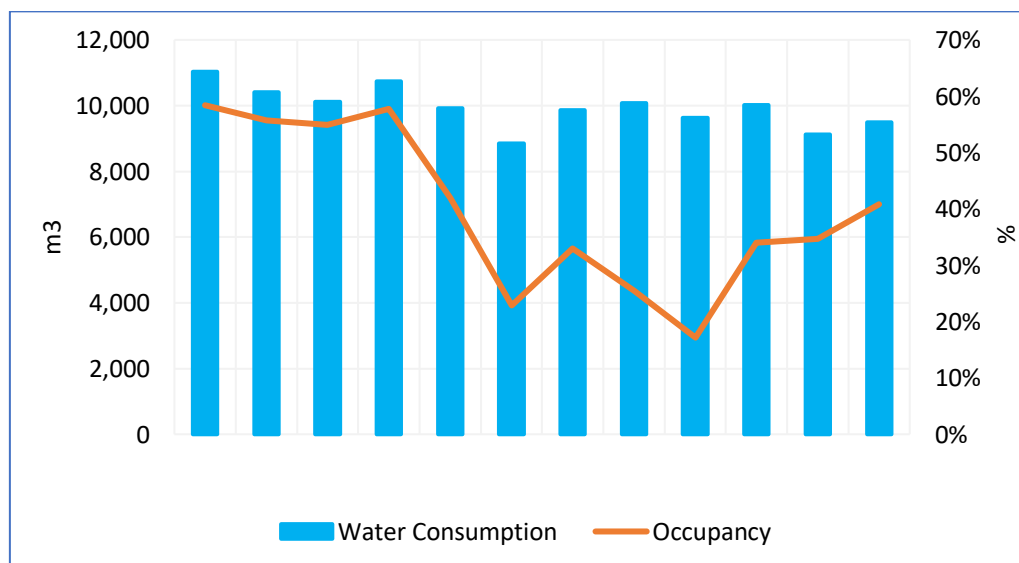
#### 3.3.1. Water Overall Consumption and Cost

The annual baseline water consumption and cost is shown in the following chart with a total of **119,279 m<sup>3</sup>**. It is important to mention that the cost presented is the cost of chemicals used in the RO system at a rate of **\$ 0.88 / m<sup>3</sup>** excluding the cost of electricity which was already accounted in the electricity utility cost. Accounting for the cost of electricity, the total cost of water becomes at **\$ 3.15/ m<sup>3</sup>**.

Table 9: Water Baseline Consumption and Cost

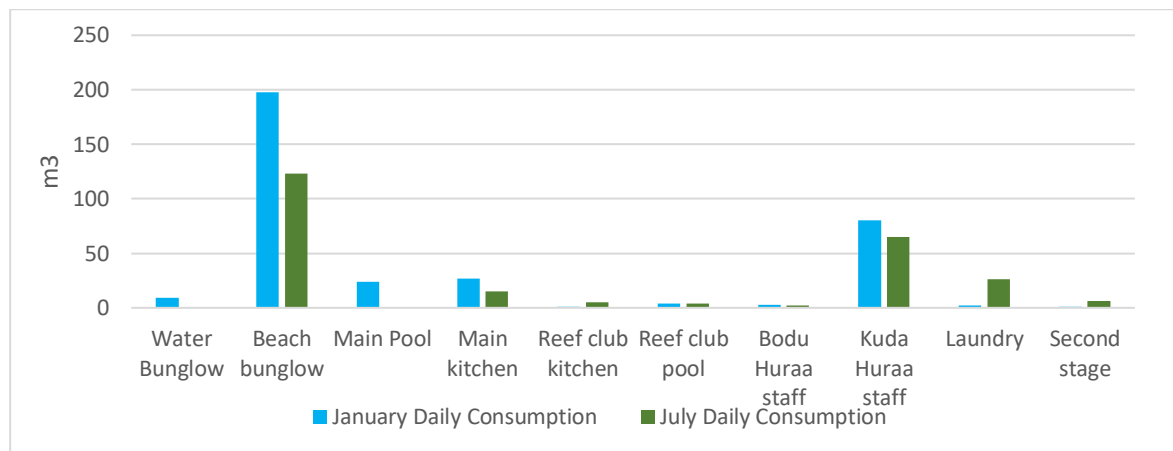
| Month        | Water Consumption |                  | Rate              | Deviation  |
|--------------|-------------------|------------------|-------------------|------------|
| Unit         | m <sup>3</sup>    | \$               | \$/m <sup>3</sup> | -          |
| Jan          | 11,040            | \$6,204          | \$0.56            | 10%        |
| Feb          | 10,412            | \$18,742         | \$1.80            | 5%         |
| Mar          | 10,115            | \$5,685          | \$0.56            | 2%         |
| Apr          | 10,731            | \$4,217          | \$0.39            | 7%         |
| May          | 9,918             | \$7,934          | \$0.80            | 0%         |
| Jun          | 8,848             | \$13,272         | \$1.50            | -12%       |
| Jul          | 9,861             | \$8,875          | \$0.90            | -1%        |
| Aug          | 10,074            | \$9,067          | \$0.90            | 1%         |
| Sep          | 9,637             | \$7,257          | \$0.75            | -3%        |
| Oct          | 10,023            | \$8,066          | \$0.80            | 1%         |
| Nov          | 9,122             | \$6,063          | \$0.66            | -9%        |
| Dec          | 9,498             | \$9,050          | \$0.95            | -5%        |
| <b>Total</b> | <b>119,279</b>    | <b>\$104,431</b> | <b>\$0.88</b>     | <b>-</b>   |
| Maximum      | 11,040 (Jan)      | 18,742 (Feb)     | 1.80 (Feb)        | 10% (Jan)  |
| Average      | 9,940             | \$8,703          | \$1               | -          |
| Minimum      | 8,848 (Jun)       | 4,217 (Apr)      | 0.39 (Apr)        | -12% (Jun) |

Figure 18: Water Baseline Consumption Variations



The graph below serves as a comparison between the daily water consumption breakdown during a typical wet month (July) versus a typical dry month (January) consumption breakdown.

Figure 19: Dry vs Wet daily typical consumption breakdown



To analyze further the water consumption of the resort, daily water consumption during one month with high occupancy rate-April 2022 (58%) and one month with low occupancy rate – September 2022 (17%) were considered. Water consumption is plotted in stacked bar chart for the following key end-users: staff accommodation, guest villas, main kitchen, reef club kitchen, pools, and laundry. It is important to mention that the water bungalow water meters were out of order from 4th April 2022 to 7th December 2022.

Figure 20: Daily Water Consumption vs occupancy (Room Nights) -April 2022

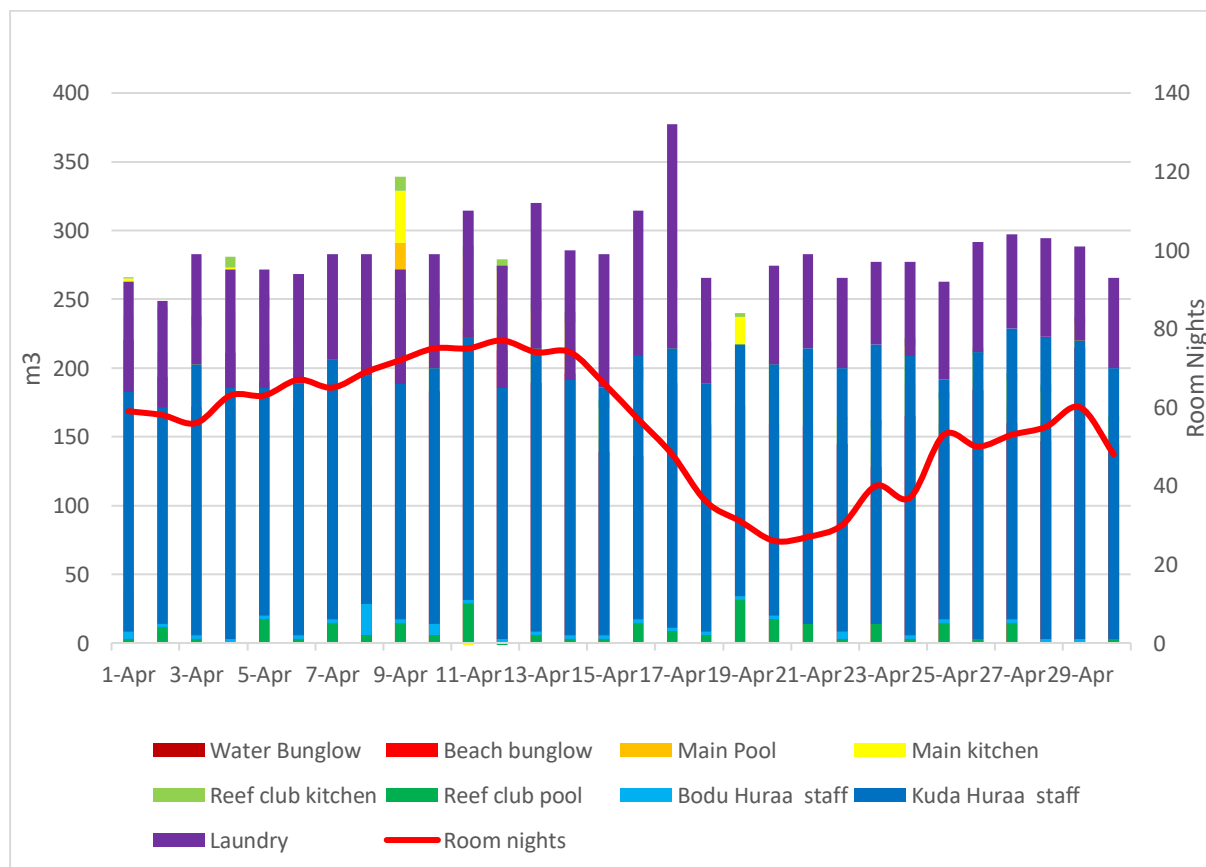
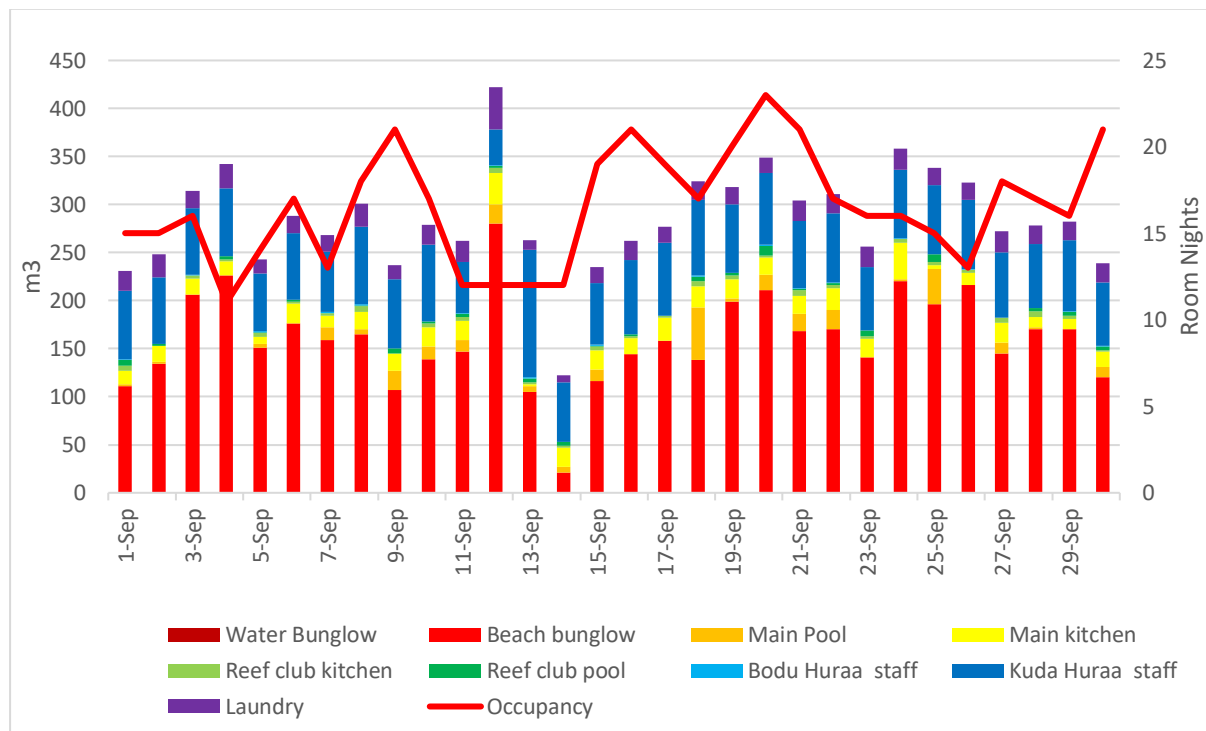


Figure 21: Daily Water Consumption - September 2022



- Total water consumption in April was 9,831 m<sup>3</sup> at an average of 36.4 m<sup>3</sup> /day while in September it is 8,546 m<sup>3</sup> with an average of 31.8 m<sup>3</sup> /day.
- The staff blocks' water consumption was seen to be at an average of 33.3 m<sup>3</sup> /day in April and 35.7 m<sup>3</sup> /day in September.
- Guestroom's bungalows have an average water consumption in April is 88.6 m<sup>3</sup> /day and 81 m<sup>3</sup> /day in September.
- Laundry water consumption is averaging at 28 m<sup>3</sup> /day in April while it is averaging at 19.5 m<sup>3</sup> /day in September.
- For the kitchens the average water consumption in April was seen to be 15.2m<sup>3</sup> /day and 10.5 m<sup>3</sup> /day in September.
- For the pools the average water consumption in April was seen to be 4.76 m<sup>3</sup> /day and 2.9 m<sup>3</sup> /day in September.

Overall the average water consumption between April and September has dropped by 12%. The main factor contributing to this variation is the occupancy rate variation whereas in April the occupancy rate was at its highest with 58% average while in September was at 17% only. Occupancy rate has an impact on water consumption in bungalows as well as in laundry and in F&B activities.

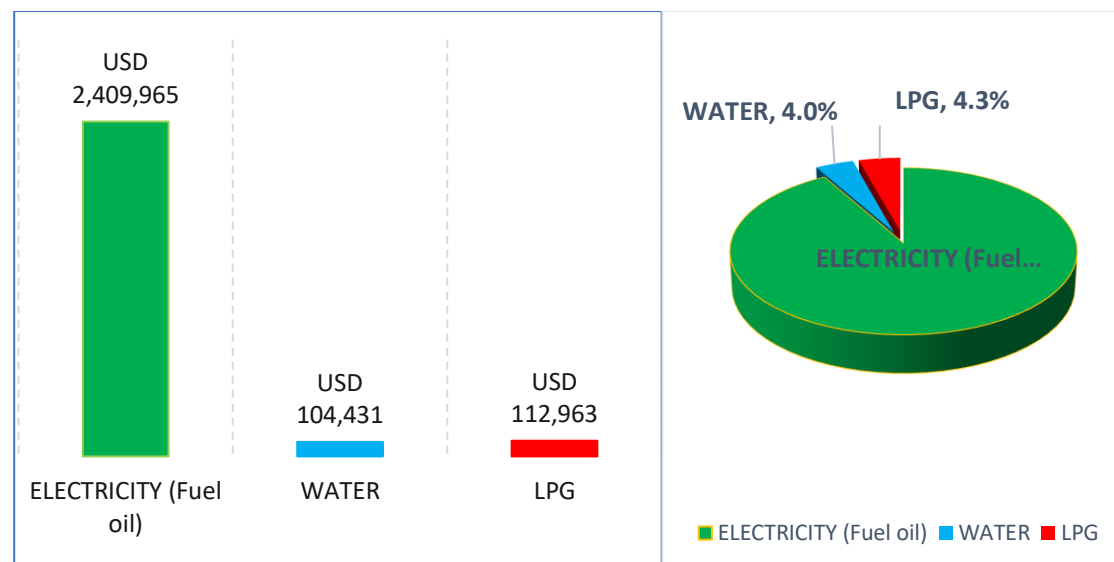
### 3.4. Total Utilities Cost

The below table provides the cost of each utility supply as compiled for the baseline year extending from January until December 2022.

Table 10: Annual Utilities Costs

| UTILITY    | ELECTRICITY<br>(Fuel Oil) | WATER              | LPG          | ALL UTILITIES |
|------------|---------------------------|--------------------|--------------|---------------|
| TOTAL COST | \$2,409,965               | \$104,431          | \$112,963    | \$2,627,359   |
| MAXIMUM    | 257,457 (May)             | 18,742 (Feb)       | 12,541 (Oct) |               |
| AVERAGE    | \$200,830                 | \$8,703            | \$9,414      | \$218,947     |
| MINIMUM    | 117,727 (Feb)             | 4,217 (Apr)        | 5,774 (Feb)  |               |
| PERCENTAGE | 91.7%                     | 4.0%               | 4.3%         | 100%          |
|            | USD/kWh                   | USD/m <sup>3</sup> | USD/kg       |               |
| Rates      | 0.32                      | 0.88               | 1.04         |               |

Figure 22: Utilities cost breakdown



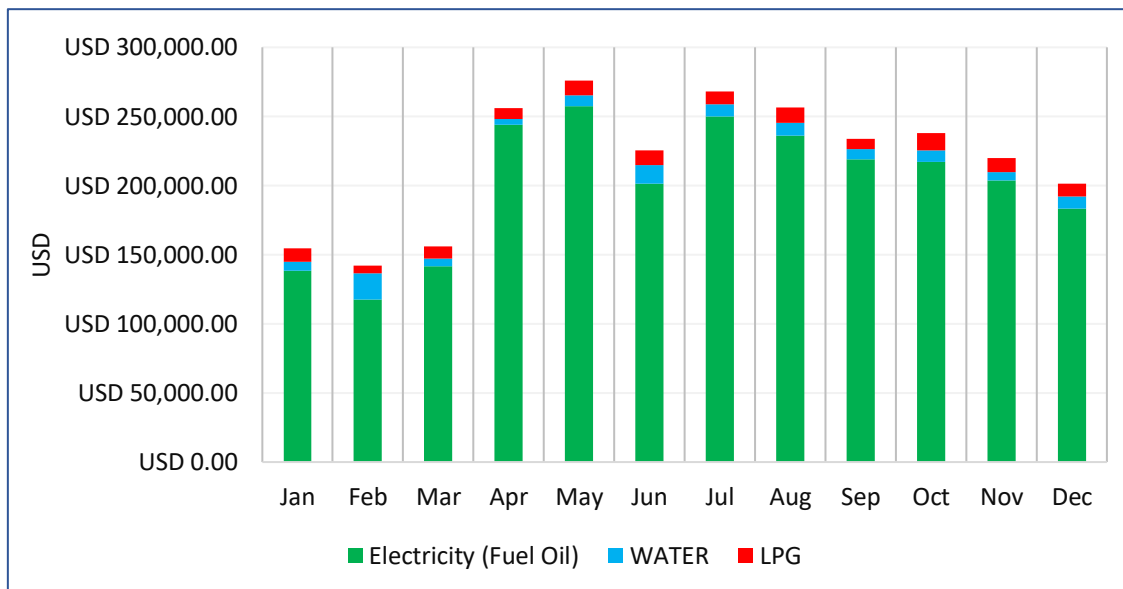
### 3.5. Utilities Cost Breakdown

The below table provides the monthly cost of each utility supply during for the baseline year.

Table 11: Utilities Cost Breakdown

| MONTH             | ELECTRICITY (Fuel Oil) | WATER               | LPG                 | TOTAL                 |
|-------------------|------------------------|---------------------|---------------------|-----------------------|
| Unit              | USD                    | USD                 | USD                 | USD                   |
| Jan               | \$138,576.14           | \$6,204.48          | \$9,840.00          | \$154,620.62          |
| Feb               | \$117,727.20           | \$18,741.60         | \$5,773.50          | \$142,242.30          |
| Mar               | \$141,437.64           | \$5,684.63          | \$8,780.00          | \$155,902.27          |
| Apr               | \$243,843.38           | \$4,217.28          | \$7,810.00          | \$255,870.66          |
| May               | \$257,457.32           | \$7,934.40          | \$10,434.00         | \$275,825.72          |
| Jun               | \$201,371.76           | \$13,272.00         | \$10,969.00         | \$225,612.76          |
| Jul               | \$250,124.85           | \$8,874.90          | \$9,015.00          | \$268,014.75          |
| Aug               | \$236,171.48           | \$9,066.60          | \$11,141.27         | \$256,379.35          |
| Sep               | \$219,052.16           | \$7,256.66          | \$7,469.00          | \$233,777.82          |
| Oct               | \$217,170.25           | \$8,065.51          | \$12,541.00         | \$237,776.76          |
| Nov               | \$203,867.20           | \$6,063.39          | \$10,005.93         | \$219,936.52          |
| Dec               | \$183,165.75           | \$9,049.69          | \$9,184.43          | \$201,399.87          |
| <b>Total</b>      | <b>\$2,409,965.13</b>  | <b>\$104,431.15</b> | <b>\$112,963.13</b> | <b>\$2,627,359.41</b> |
| Maximum           | \$257,457 (May)        | \$18,742 (Feb)      | \$12,541 (Oct)      | \$275,826 (May)       |
| Average           | \$200,830.43           | \$8,702.60          | \$9,413.59          | \$218,946.62          |
| Minimum           | \$117,727 (Feb)        | \$4,217 (Apr)       | \$5,774 (Feb)       | \$142,242 (Feb)       |
| <b>Percentage</b> | <b>92%</b>             | <b>4%</b>           | <b>4%</b>           | <b>100%</b>           |

Figure 23: Monthly utility cost breakdown



### 3.6. Energy & Carbon Footprint

#### 3.6.1. Energy Use Intensity (EUI)

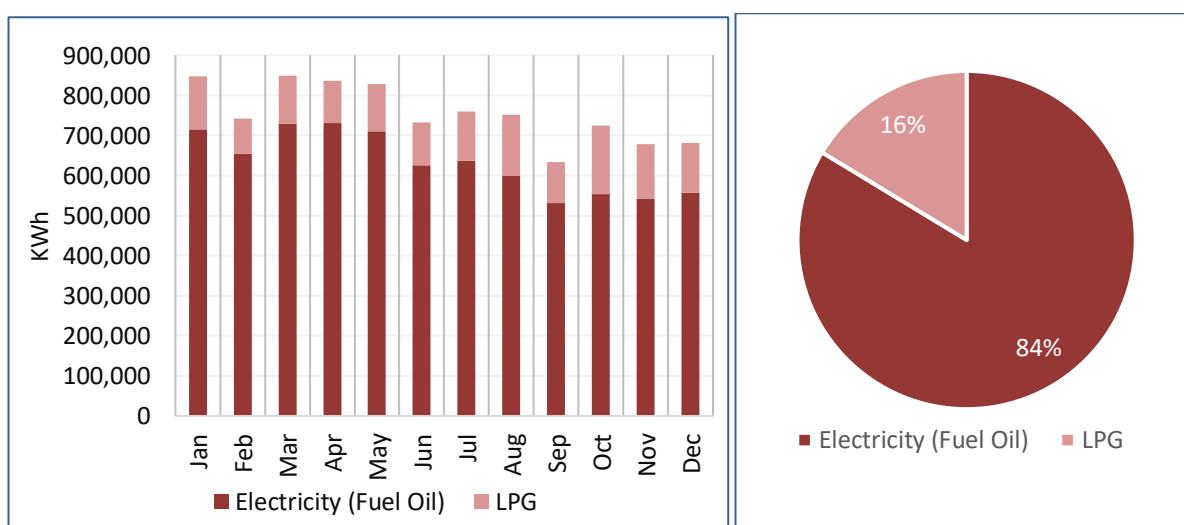
To be able to aggregate all the utilities' energy consumption, the energy consumption of all energy sources is converted to a single energy unit KWh. In the case of FSRMKH, the utility taken into consideration are electricity from fuel Oil and LPG consumptions.

Table 12: Total Utilities Energy

| MONTH             | Electricity (Fuel Oil) | LPG              | TOTAL            |
|-------------------|------------------------|------------------|------------------|
| Unit              | KWh                    | KWh              | KWh              |
| Jan               | 714,310                | 133,824          | 848,134          |
| Feb               | 654,040                | 87,244           | 741,284          |
| Mar               | 729,060                | 119,408          | 848,468          |
| Apr               | 730,070                | 106,216          | 836,286          |
| May               | 710,031                | 118,252          | 828,283          |
| Jun               | 625,150                | 106,556          | 731,706          |
| Jul               | 636,450                | 122,604          | 759,054          |
| Aug               | 599,420                | 151,521          | 750,941          |
| Sep               | 531,680                | 101,578          | 633,258          |
| Oct               | 553,300                | 170,558          | 723,858          |
| Nov               | 542,200                | 136,081          | 678,281          |
| Dec               | 556,200                | 124,908          | 681,108          |
| <b>Total</b>      | <b>7,581,911</b>       | <b>1,478,750</b> | <b>9,060,661</b> |
| Maximum           | 730,070 (Apr)          | 6,290,770 (Aug)  | 6,290,770 (Aug)  |
| Average           | 631,826                | 4,359,017        | 4,359,018        |
| Minimum           | 531,680 (Sep)          | 3,094,348 (Feb)  | 3,094,348 (Feb)  |
| <b>Percentage</b> | <b>84%</b>             | <b>16%</b>       | <b>100%</b>      |

The Energy Use Intensity is an indicator of the building's energy use as function of its size. EUI is obtained by dividing the total energy consumption (KWh) /year of the facility by the total conditioned area in m<sup>2</sup>. The total built up area in FSRMKH is 32,745 m<sup>2</sup> of which 12,976 m<sup>2</sup> is conditioned . Therefore the total EUI is **698.2 KWh/m<sup>2</sup>** .

Figure 24: Energy consumption variation and breakdown



### 3.6.2. Carbon Emissions Footprint

The carbon emissions calculation is done by multiplying the consumption of each utility by its corresponding emissions factor. The electricity emissions factor varies between countries based on the energy mix used. The CO<sub>2</sub> equivalent of 1 Liter of diesel oil used in diesel generators for electricity in Maldives is around 0.26832 kg CO<sub>2</sub>eq. While the CO<sub>2</sub> equivalent of 1 kg of LPG in Maldives is around 0.22774 Kg CO<sub>2</sub>eq. The **total annual carbon emissions** in FSRMKH are at **580,925 KgCO<sub>2</sub>**.

Table 13: FSHF Carbon Emissions Monthly Breakdown

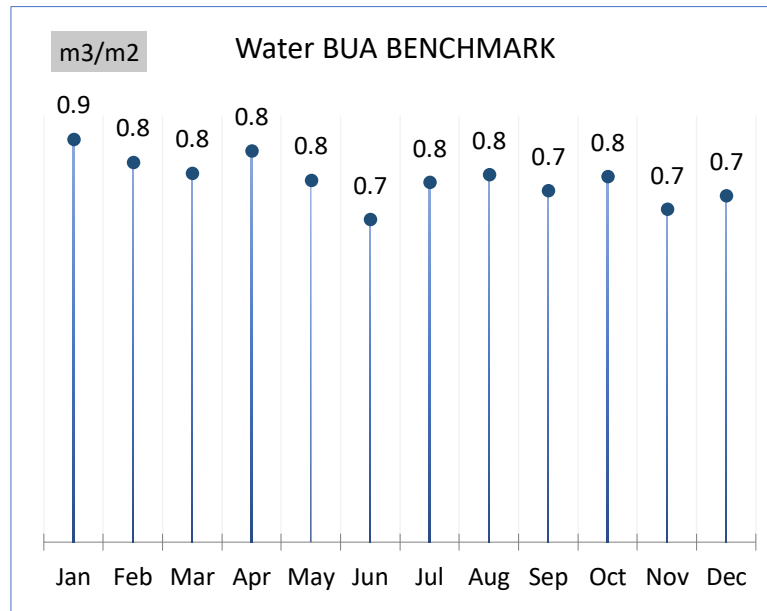
| MONTH             | Electricity (Fuel Oil) | LPG               | TOTAL             |
|-------------------|------------------------|-------------------|-------------------|
| Unit              | KgCO <sub>2</sub>      | KgCO <sub>3</sub> | KgCO <sub>2</sub> |
| Jan               | 50,797                 | 2,241             | 53,038            |
| Feb               | 44,801                 | 1,461             | 46,262            |
| Mar               | 49,638                 | 2,000             | 51,637            |
| Apr               | 55,029                 | 1,779             | 56,808            |
| May               | 52,787                 | 1,980             | 54,768            |
| Jun               | 46,313                 | 1,784             | 48,097            |
| Jul               | 47,515                 | 2,053             | 49,568            |
| Aug               | 44,644                 | 2,537             | 47,182            |
| Sep               | 39,563                 | 1,701             | 41,264            |
| Oct               | 41,370                 | 2,856             | 44,226            |
| Nov               | 40,163                 | 2,279             | 42,441            |
| Dec               | 43,542                 | 2,092             | 45,634            |
| <b>Total</b>      | <b>556,162</b>         | <b>24,763</b>     | <b>580,925</b>    |
| Maximum           | 55,029 (Apr)           | 2,856 (Oct)       | 56,808 (Apr)      |
| Average           | 46,347                 | 2,064             | 48,410            |
| Minimum           | 39,563 (Sep)           | 1,461 (Feb)       | 41,264 (Sep)      |
| <b>Percentage</b> | <b>96%</b>             | <b>4%</b>         | <b>100%</b>       |

FSRMKH's annual carbon footprint is at **44.76 kgCO<sub>2</sub>/m<sup>2</sup>**.

### 3.6.3. Water Intensity Benchmarking

The water consumption benchmark adopted in FSRMKH is calculated for the water consumption relatively to the gross area ( $\text{m}^3/\text{m}^2$ ) for each month during the baseline period. The average rates are  $0.77 \text{ m}^3/\text{m}^2$ .

Figure 25: Water Benchmark in  $\text{m}^3/\text{m}^2$



### 3.6.1. CHSB – Cornell Sustainability Hotel Benchmark – Real Consumption

The Cornell Sustainability Hotel Benchmark (CHSB) provides the most comprehensive benchmarking tool in the hospitality industry as it is based on a global database of more than 15000 properties across all regions and climate. The below table provides the key metrics of FSRMKH as compared to various levels of the CHSB for 8 Key Measures. The property has marks within the lower quartile in the Measures related to the carbon footprint, while the Water usage indicators fall between the lower quartile and the Mean, and the energy usage is between in the median and the upper quartile ranges.

Table 14: CHSB benchmark – using real consumption

| 2019 CALENDAR YEAR BENCHMARKS                                             |       |       |                |         |         |                |          |        |        |
|---------------------------------------------------------------------------|-------|-------|----------------|---------|---------|----------------|----------|--------|--------|
| MEASURE                                                                   | Count | Low   | Lower Quartile | Mean    | Median  | Upper Quartile | High     | SD     | FSMRKH |
| MEASURE 2: Hotel Carbon Footprint Per Room (kgCO <sub>2</sub> e)          | 78    | 2,623 | 11,642         | 28,266  | 23,203  | 36,861         | 184,774  | 26,314 | 2,780  |
| MEASURE 3: Hotel Carbon Footprint Per Occupied Room (kgCO <sub>2</sub> e) | 78    | 9.3   | 49.4           | 141.9   | 99.6    | 164.2          | 2,127.1  | 242    | 11     |
| MEASURE 4: Hotel Carbon Footprint Per Square Meter (kgCO <sub>2</sub> e)  | 78    | 19.2  | 77.3           | 163.8   | 152.5   | 220.6          | 468.1    | 108    | 45     |
| MEASURE 5: Hotel Energy Usage Per Occupied Room (kWh)                     | 78    | 60.2  | 127.7          | 288.5   | 221.6   | 354.3          | 3,698.4  | 410    | 138    |
| MEASURE 6: Hotel Energy Usage Per Square Meter (kWh)                      | 78    | 41.2  | 227.8          | 373.4   | 365.8   | 504.0          | 986.3    | 194    | 586    |
| MEASURE 8: Hotel Water Usage Per Occupied Room (L)                        | 52    | 532.6 | 1,080.5        | 2,580.1 | 1,847.8 | 3,734.4        | 14,408.6 | 2,224  | 2,158  |
| MEASURE 9: Hotel Water Usage Per Square Meter (L)                         | 61    | 443.8 | 1,746.6        | 3,380.1 | 2,585.4 | 4,379.7        | 12,104.0 | 2,432  | 3,643  |
| MEASURE 12: Hotel Energy From Renewables (%)                              | 79    | 0.0%  | 0.0%           | 0.8%    | 0.0%    | 0.0%           | 44.5%    | 0      | 0.0%   |

## 4. Electrical Energy Load Analysis & Energy Balance

### 4.1. Electrical Load Categories

A crucial milestone in energy audits is the buildup of the energy balance. This process starts with acquiring the load inventory – electrical end uses, and then running a complete simulation based on the operational schedule, load factors and behavioral statistics of each item in the inventory. The micro and macro energy consumption analysis of energy systems including lighting, air conditioning, ventilation and building envelope, serves the purpose of identifying where and how a facility uses energy. The breakdown of the energy consumption by end-use would be the energy balance, which will provide an important baseline for the roadmap in developing the Energy Efficiency Measures (EEMs) that would optimize the overall energy costs.

#### Information and Database Process

FSRMKH's Load inventory data is populated into an Energy Analysis System including assets name/category, actual electrical specification (KW) and operational schedule over 12 months (hours per day, week, month, or season). The second step is performing a total simulation and energy modeling of the facility with full bills reconciliation as per the baseline values.

In the present report, electrical loads were divided into the following categories:

#### 1- Lighting

This category corresponds to all types of existing lighting fixtures in the resort (LED and non-LED).

#### 2- HVAC Fans

This category corresponds to all extraction and fresh air fans responsible for the required circulation of air in the facility, in addition to kitchen hoods.

#### 3- Cooling

This category corresponds to all split units and VRVs installed across the facility.

#### 4- Motors

This category corresponds to all pumps (domestic, booster, and pool pumps), compressors, heat pumps and electric hot water tanks, in addition to the RO plants and Sewage Treatment Plant.

#### 5- Hot Water

This category corresponds to the electric heaters and heat pumps used for domestic hot water in guestrooms, staff accommodation, and SPA.

#### 6- Laundry

This category corresponds to all the laundry machines (washers, dryers, ironers...).

#### 7- Kitchen & Refrigeration

This category corresponds to all electricity driven Kitchen equipment (walk in cold rooms compressors, fridges, and other electric driven loads...).

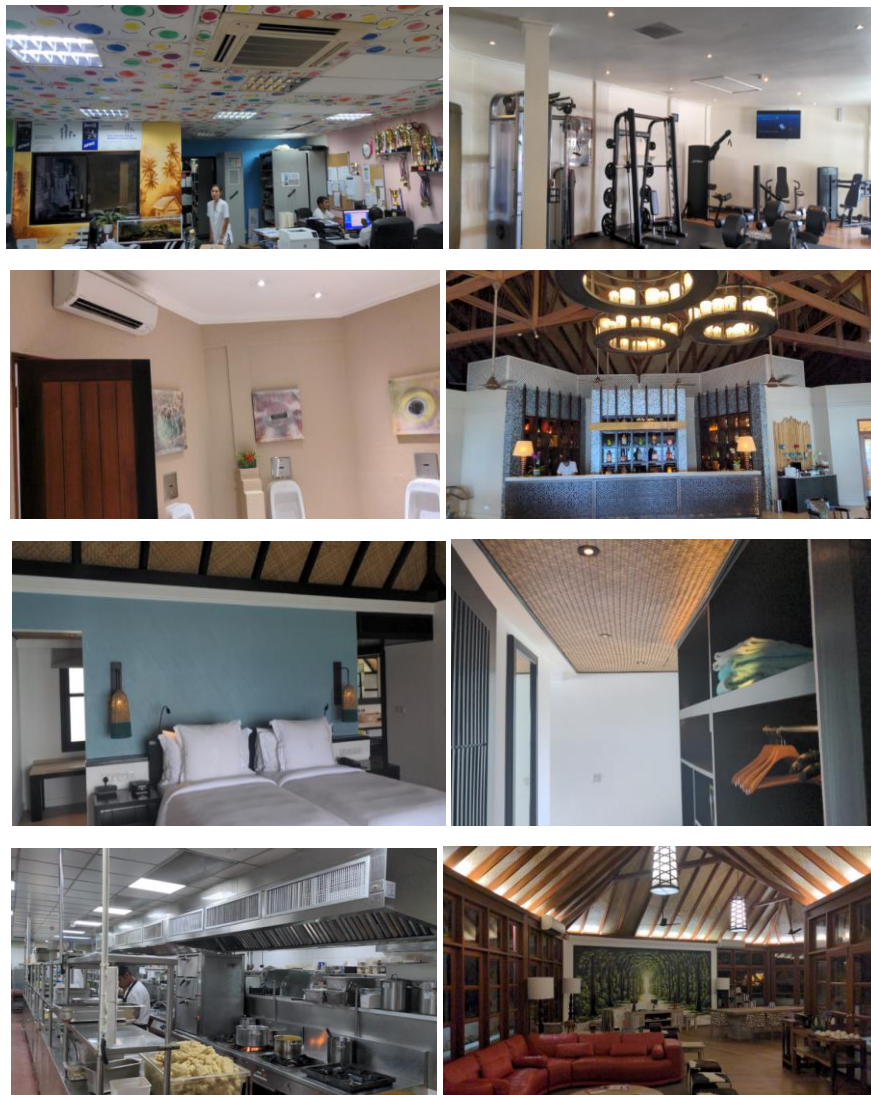
#### 8- Others

This category corresponds to all the remaining IT equipment and auxiliary equipment such as TVs in rooms, SPA and Gym equipment, in addition to pantry loads, and buggies...

### 4.1.1. Lighting

Lighting in hotels represent generally a crucial aesthetic element and has high load factors as the hours of operation of the common areas and the BOH are typically close to 24/7. In the case of FSMKH, existing lighting infrastructure is hybrid with primarily LED fixtures in the guest rooms' floors as well as the lobby and restaurants, while outdoor wall lights and entrance lights in guestrooms are still CFL type, pool lights are Halogen, Guestrooms bathrooms are incandescent, and some external lightings are metal halide. FSMKH has already started a LED retrofit campaign for lighting in the BOH, however there's still some areas with non-LED lighting fixtures. Lighting is fully operated manually across the resort. The below pictures show the lighting fixtures in different areas of the property.

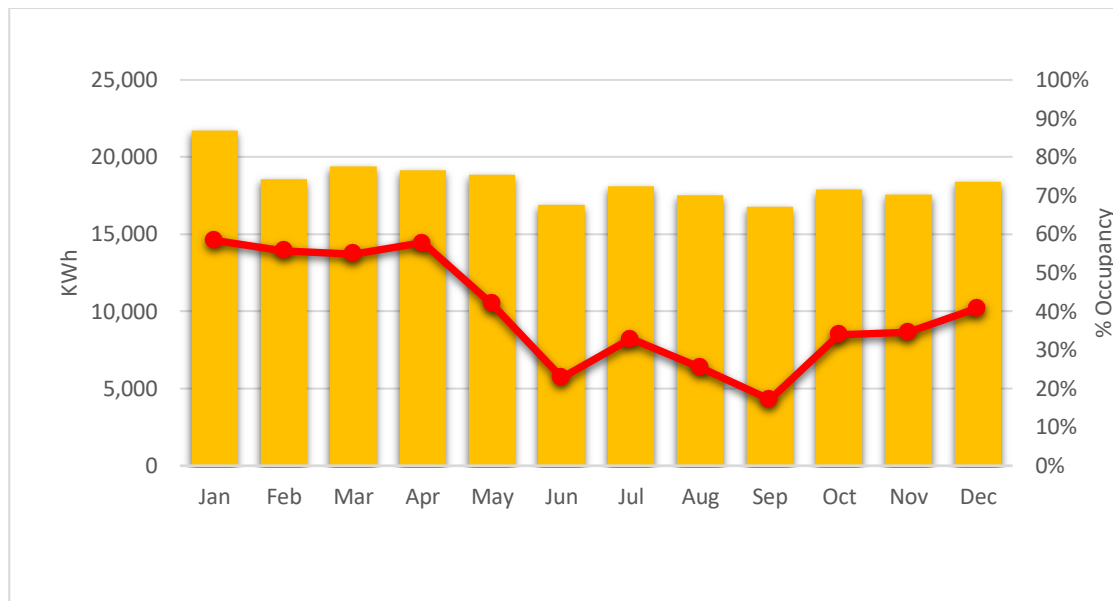
Figure 26: FSHF Lighting fixtures



- 1<sup>st</sup> Row: Offices LED T5 (left); and Gym area MR16 LED lighting
- 2<sup>nd</sup> Row: Public Toilet Lighting in Lobby (Left); FOH Lobby lighting (Right)
- 3<sup>rd</sup> Row: Guestroom LED Lighting
- 4<sup>th</sup> Row: kitchen Lighting (LED); Business center LED light (Right) – all LED.

Compiling all the lighting inventory of the property and simulating their operation according to the hours of usage and the bills reconciliation would yield to the following monthly energy consumption.

Figure 27: Lighting monthly energy consumption variation with occupancy



The total **lighting** category electrical consumption for the baseline period is **220,812 KWh**.

### 4.1.2. HVAC Fans

The HVAC Fans category in FSMRKH includes all extraction fans and fresh air fans in kitchens also known as kitchen hoods.

#### a. Ventilation and Extraction Fans

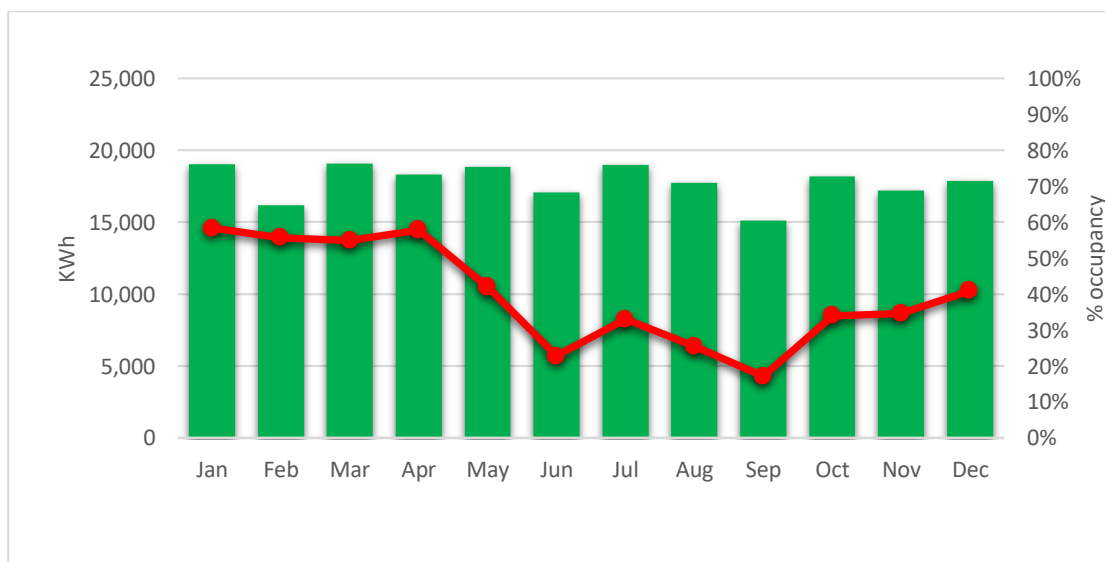
There are several ventilation and extractor fans in kitchens (Kitchen hoods) to ensure a good air exchange and odors extraction. Most fans operate on a 24/7 basis throughout the whole year. Only one Kitchen hood was equipped with a VFD for noise control. At the time of the field visit, the VFD was seen operating at a frequency of 29 Hz.

Figure 28: Reef Club kitchen - Kitchen hood



The total **HVAC Fans** category energy consumption for the baseline period is **213,553 KWh** with the monthly variations as per the below chart.

Figure 29: HVAC Fans Monthly Energy Consumption



### 4.1.3. Cooling

FSMRKH's space cooling needs are met by a huge number of split units and VRVs installed across the premises and mainly serving guestrooms, BOH, FOH and restaurants, shops, Kitchens, and Staff accommodations.

#### a. VRVs

FSMRKH has in total around 150 VRV units serving some of the guestrooms, Diving Center, GYM, Laundry, Main kitchen and reef club kitchen varying in load between 2.5 and 12 HP. Units are connected to Ecogen heat recovery systems which is being replaced gradually by heat pumps.

Figure 30: VRVs in guestrooms

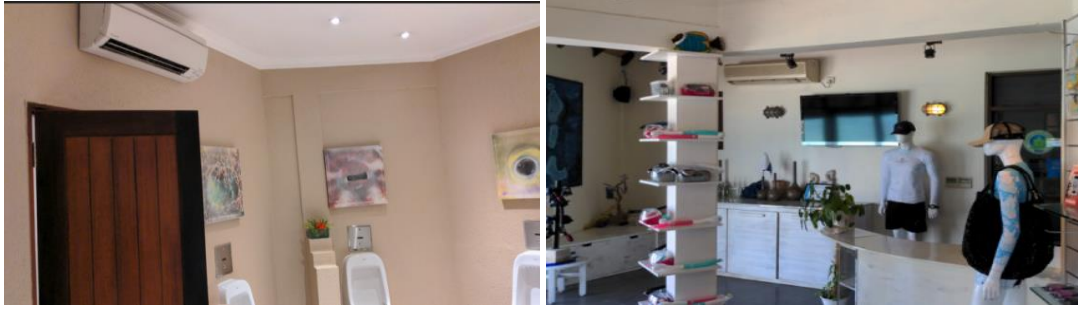


#### b. Split Units

In addition to VRVs, the facility cooling is met by over 245 split units serving a number of guestrooms, kids club, staff accommodations, BOH, IT and electrical rooms, resort shops, business center, lobby toilets, and SPA. Old Split units in guestrooms are currently under being retrofitted by new inverter split units.

Figure 31: BOH split Units (left), and Kitchen Split unit (right)





- 1<sup>st</sup> Row: BOH split Units (left), and Kitchen Split unit (right)
- 2<sup>nd</sup> Row: Toilet split units (Left), and shop split unit (right)

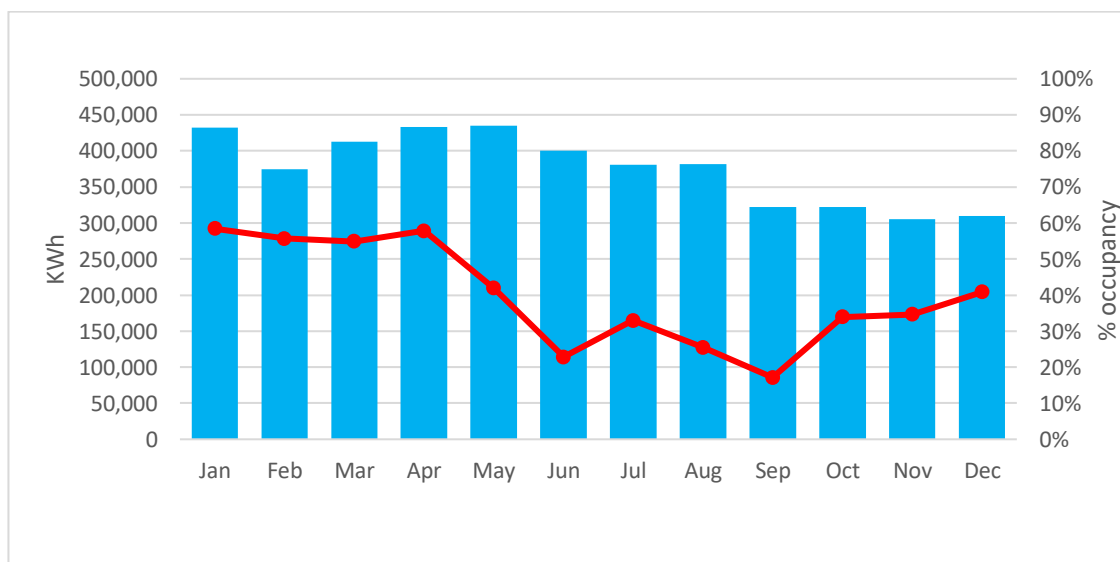
The table below shows the refrigerant type and charge for the VRF, and Split units installed in the resort.

Table 15: Split Units and VRVs Refrigerant Type and Charge

| Type       | Type / Model   | Refrigerant Type | Refrigerant Charge in kg |
|------------|----------------|------------------|--------------------------|
| VRF        | Dakin FXMQ-125 | R410A            | N/A                      |
| VRF        | Dakin FXCQ50M  | R410A            | N/A                      |
| VRF        | Dakin RXQ8AYM  | R410A            | 5.9                      |
| Split Unit | N/A            | N/A              | N/A                      |

Following the overall electricity simulation performed for all the assets and the bills reconciliation, the below chart provides the monthly electricity consumption for the cooling end user category for the baseline year which was seen to be around **4,509,046 KWh** annually.

Figure 32: Cooling Monthly Energy Consumption



#### 4.1.4. Motors

The Motors' end use category includes all pumps – Booster pumps, Pool pumps, in addition to RO plants and Sewage Treatment Plant. This category comprises also all compressors.

Some of the key installed and operating loads are here below:

- 96x 1.12 KW guestrooms pool pumps

Figure 33: Guestroom pool pumps



- 4x5.5 KW Booster pumps for domestic cold water 3 duty 1 standby with a pressure maintained at 2.7 bar.

Figure 34: Domestic Water Booster pumps



- 5x1.48KW main swimming pool pumps 3 duty 2 standby.

Figure 35: Main pool pumps



- 6x 2.94 KW Reef Club swimming pool pumps 3 duty 3 standby.

Figure 36: Reef Club Pool Pumps



- STPs: 3x80KW Sewage Treatment Plants for the treatment of sewage affluent from FSRMKH before being discharged to the ocean or for irrigation purposes.
- STP includes 2x16KW blower pumps.
- Each STP treats 80 m<sup>3</sup> of sewage water per day and include equalization pump which move the sewage from equalization tank to the aeration tank; Mix Liqueur Transfer Pump which moves the liqueur from aeration tank to clarifier tank; Sludge return pump which moves the sludge from Clarifier tank to the aeration tank, Supernatant Transfer Pumps moving the water from the Clarifier to tank to the Holding Tank, and finally the Displacement pump which take the product after filtration and UV to irrigation or to the Ocean .

Figure 37: STP Blower Pump



- 4X Desalination Plants (RO)

FSRKH has 4 desalination plants where sea water comes to one main tank for sedimentation then moves to a second tank from which pumps are pushing water to each RO plant.

Figure 38: RO plants



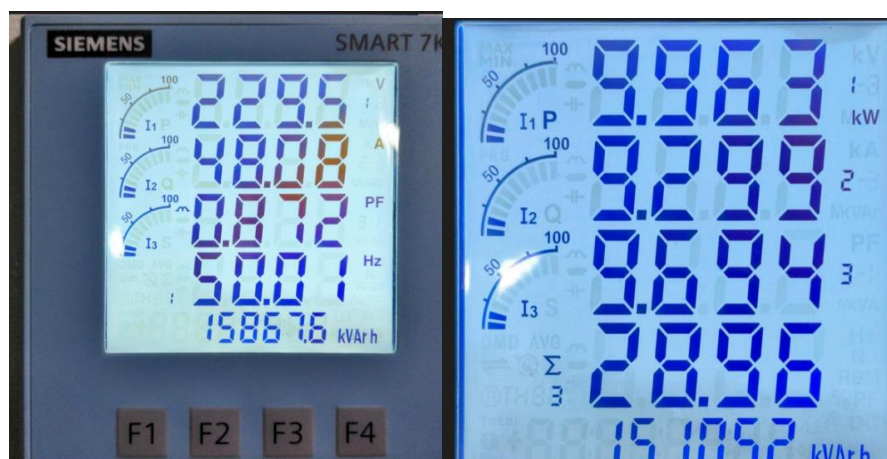
RO1: 80 KW RO plant with no VFDs. The pump was seen to be consuming 22.7KW at the time of the field audit in November 2022. The RO has a capacity of 80 m3/day.

Figure 39: RO1 panel



RO 2 : 56 KW RO plant. The plant has been newly installed , with 2 VFDs driven pumps running at 3.8 and 18.3 KW .The total power was seen on the panel at 48 KW.

Figure 40: RO2 panel



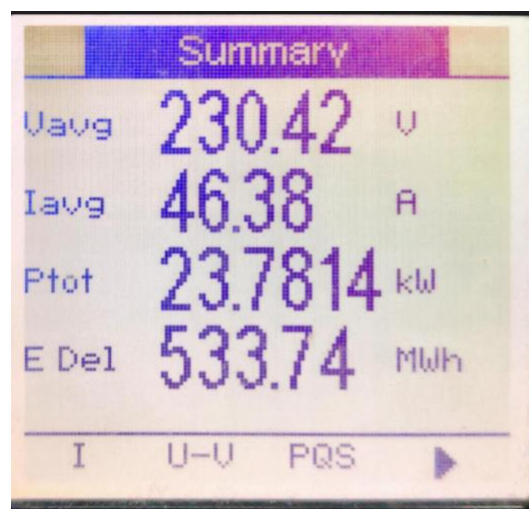
RO 3 :80 KW RO plant . No VFDs installed. The plant includes 1x30KW HP in addition to 1x5.5KW and 2x3KW pumps.

Figure 41: RO3 panel



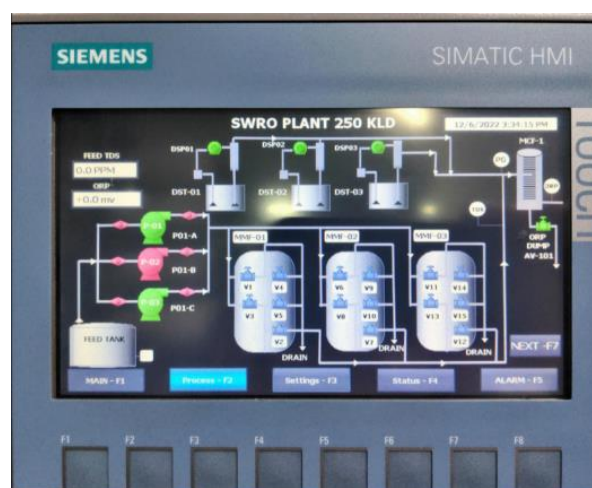
RO 4: 50 KW RO plant. No VFDs installed.

Figure 42: RO4 Panel



FSRMKH has also a second stage treatment plant for drinking water with 50 m<sup>3</sup>/day.

Figure 43:SWRO snapshot

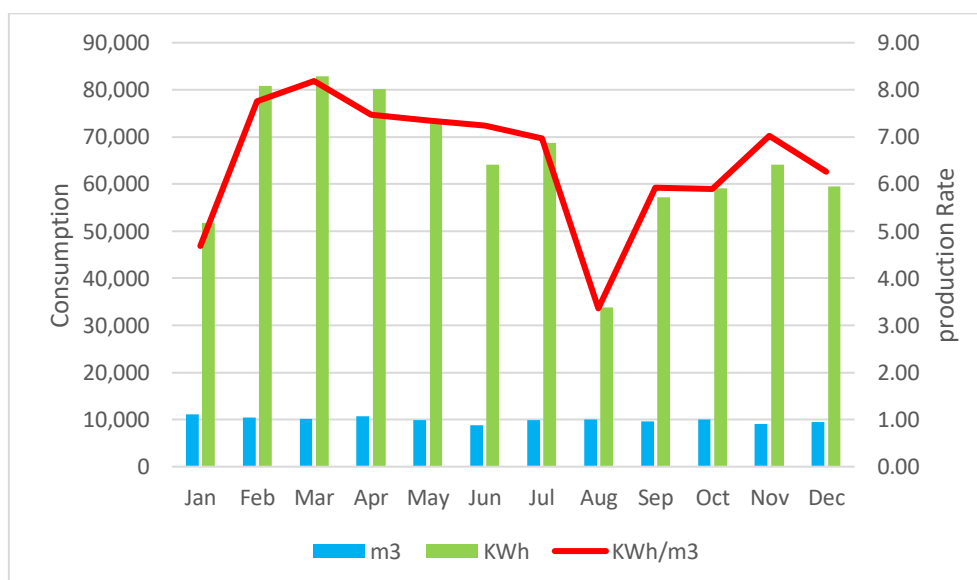


As mentioned in the water utilities accounting section, the RO plant produced during the baseline year of 2022 was at 119,279 m<sup>3</sup> out of which 2,258 m<sup>3</sup> was subject to the secondary water treatment for drinking water.

Table 16: RO Plant production and electricity consumption

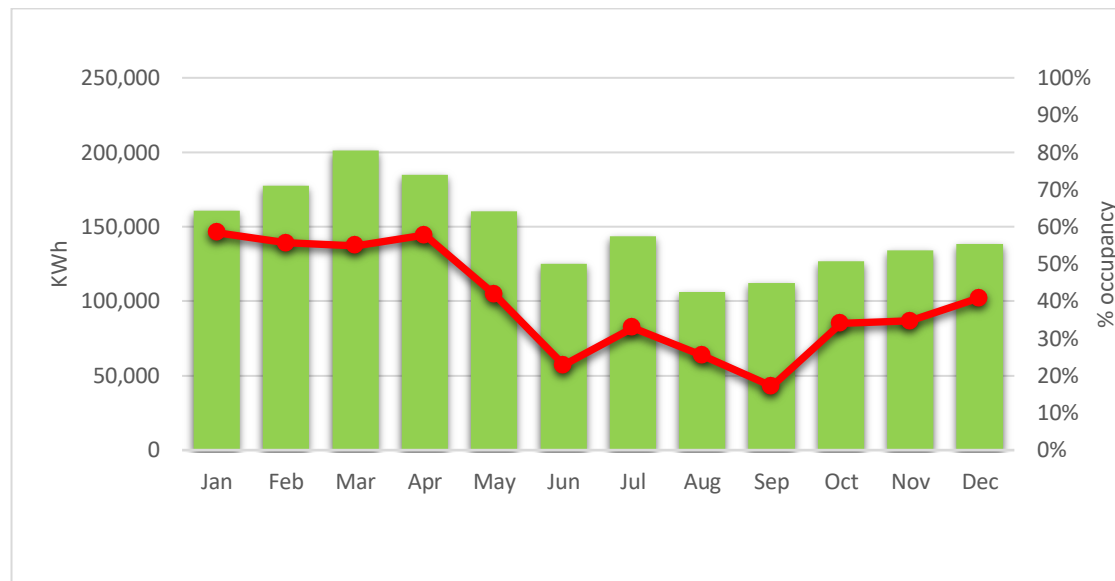
| Date         | RO production (Ro 1 : 4) | Second stage   | Electricity Consumption | Production Rate    |
|--------------|--------------------------|----------------|-------------------------|--------------------|
| Unit         | m <sup>3</sup>           | m <sup>3</sup> | KWh                     | KWh/m <sup>3</sup> |
| Jan          | 11,040                   | 242            | 51758.2944              | 4.69               |
| Feb          | 10,412                   | 189            | 80774.4                 | 7.76               |
| Mar          | 10,115                   | 230            | 82807.2                 | 8.19               |
| Apr          | 10,731                   | 179            | 80136                   | 7.47               |
| May          | 9,918                    | 169            | 72912                   | 7.35               |
| Jun          | 8,848                    | 158            | 64080                   | 7.24               |
| Jul          | 9,861                    | 186            | 68696                   | 6.97               |
| Aug          | 10,074                   | 180            | 33852                   | 3.36               |
| Sep          | 9,637                    | 172            | 57120                   | 5.93               |
| Oct          | 10,023                   | 182            | 59024                   | 5.89               |
| Nov          | 9,122                    | 174            | 64080                   | 7.02               |
| Dec          | 9,498                    | 197            | 59520                   | 6.27               |
| <b>Total</b> | <b>119,279</b>           | <b>2,258</b>   | <b>774,760</b>          | <b>6.50</b>        |

Moreover, the KWh consumption of the 4 RO plants were logged during the months of January and August with 51,758 and 33,852 KWh respectively. The rates in a peak dry month is then around 4.68 kwh/m<sup>3</sup> versus 3.36 KWh/m<sup>3</sup> in wet month.



Following the energy simulation and bills reconciliation, the total **Motors** category energy consumption for the baseline period is **1,176,456 KWh** with the monthly variations as per the below chart.

Figure 44: Motors Monthly Energy Consumption



#### 4.1.5. Hot Water

The hot water system installed in guest rooms, staff accommodations, and various locations within the resort is comprised of electric calorifiers.

- 6x4.5 KW Electric calorifiers for SPA bathtubs.

Figure 45: calorifiers in SPA



- 1x11 KW hot water tank for laundry -330 liters



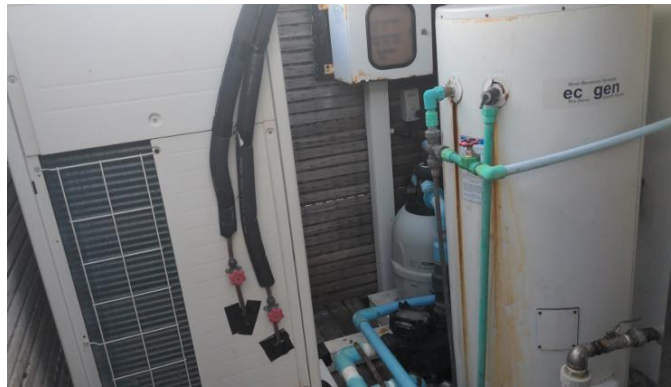
- 2x5.5 KW hot water tanks in BOH – 350Liters each

Figure 46: hot water tanks in BOH



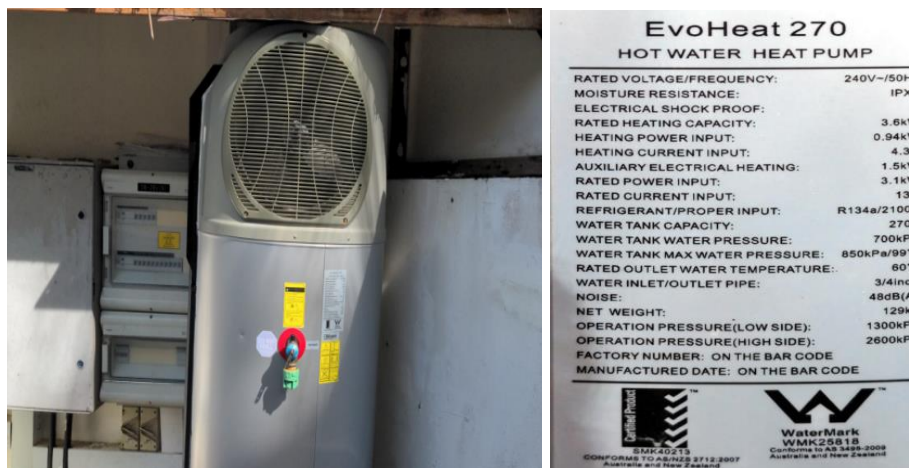
- 74x 4.8 KW Guestrooms Ecogen heat recovery units for hot water.

Figure 47: guestrooms hot water recovery system old

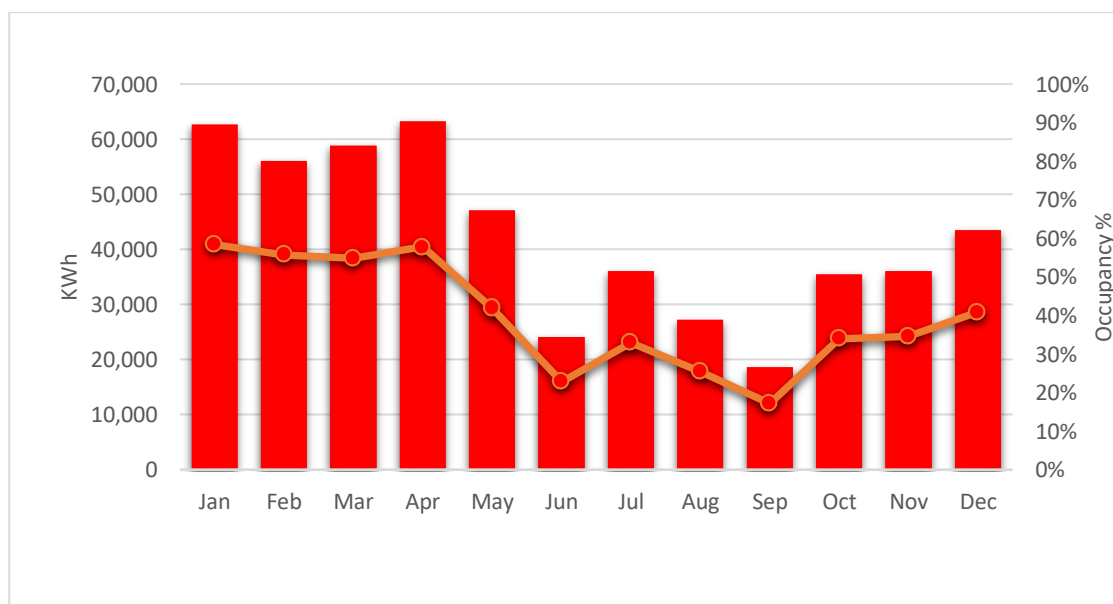


- 22 out of 96 guestrooms are retrofitted with 0.94 KW heat pumps for hot water production.

Figure 48: New heat pump (left), nameplate (right)



The total **Hot Water** category electrical consumption for the baseline period is **578,957 KWh** with the monthly variations as per the below chart.



#### 4.1.6. Kitchen Equipment

FSRMKH has 2 main kitchens: Reef Club kitchen and the Main kitchen . The kitchen loads include all cold rooms, single refrigeration units and miscellaneous kitchen equipment such as ovens, fryers etc...

At the time of the field visit, it was noticed that the kitchen loads have some working on gas while the vast majority working on 100% electricity and were kept ON even during unused times.

Figure 49: Main Kitchen



Figure 50: Cold Room (Left) with Compressor (Right)



Figure 51: Main Kitchen ice makers (left), oven Reef Club kitchen (Right)



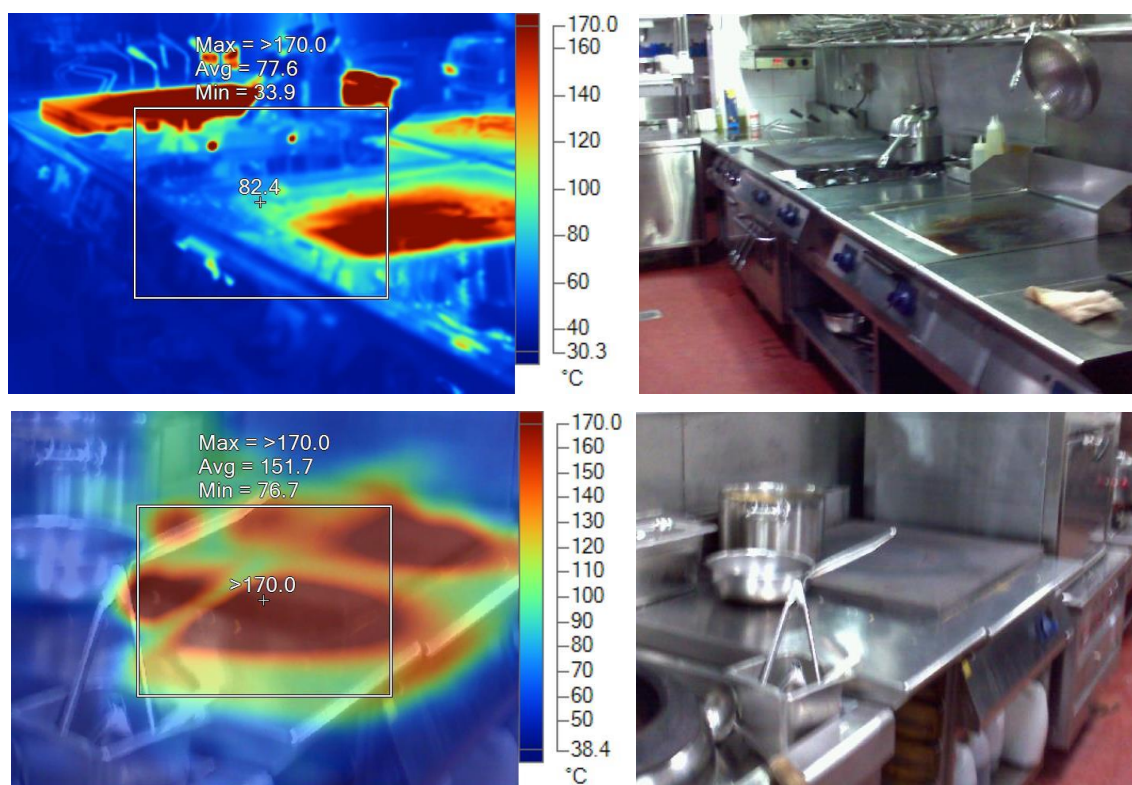
The table below shows the breakdown of loads in the kitchen and refrigeration category, as seen the refrigeration (cold room compressors and refrigerators) takes up to 91% of the installed load in kitchens and consume around 89% of the total electricity consumption allocated to the kitchens.

Table 17: kitchen load breakdown

| CATEGORY                  | SUB-CATEGORY      | LOAD |      | ENERGY |         |
|---------------------------|-------------------|------|------|--------|---------|
|                           |                   | Unit | KW   | %      | KWh     |
| KITCHEN AND REFRIGERATION | Refrigeration     |      | 99.1 | 91%    | 401,298 |
| KITCHEN AND REFRIGERATION | Kitchen equipment |      | 10.0 | 9%     | 50,780  |

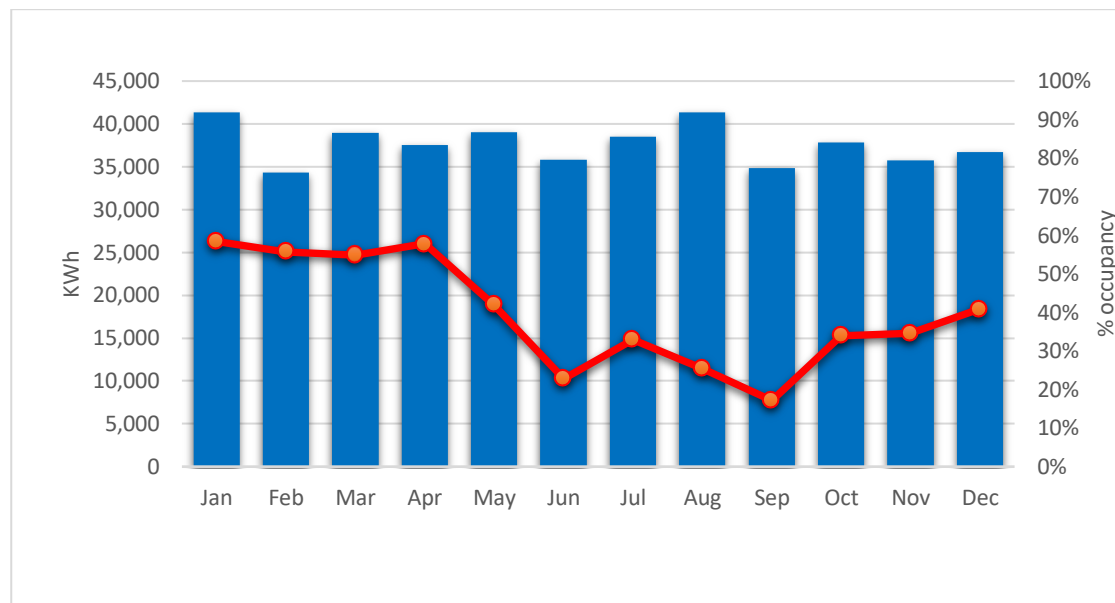
The following thermal pictures taken during the field visit in December 2022 show the non induction ovens turned on without being used, emitting heat to up 170°C.

Figure 52: Thermal picture oven



The total **Kitchen and Refrigeration load** category energy consumption for the baseline period is **452,078 KWh** with the monthly variations as per the below chart.

Figure 53: Kitchen Equipment Monthly Energy Consumption



#### 4.1.7. Others

This category comprises all the remaining electrical driven equipment mainly the low current equipment (IT loads, Telecom, CCTV) as well as electronic guestrooms appliances (such as TV and minibars), and other miscellaneous equipment such as buggy cars, gym machines, and miscellaneous low-load equipment.

Figure 54: Electrical Buggy Car (Left) pantry loads (Right)

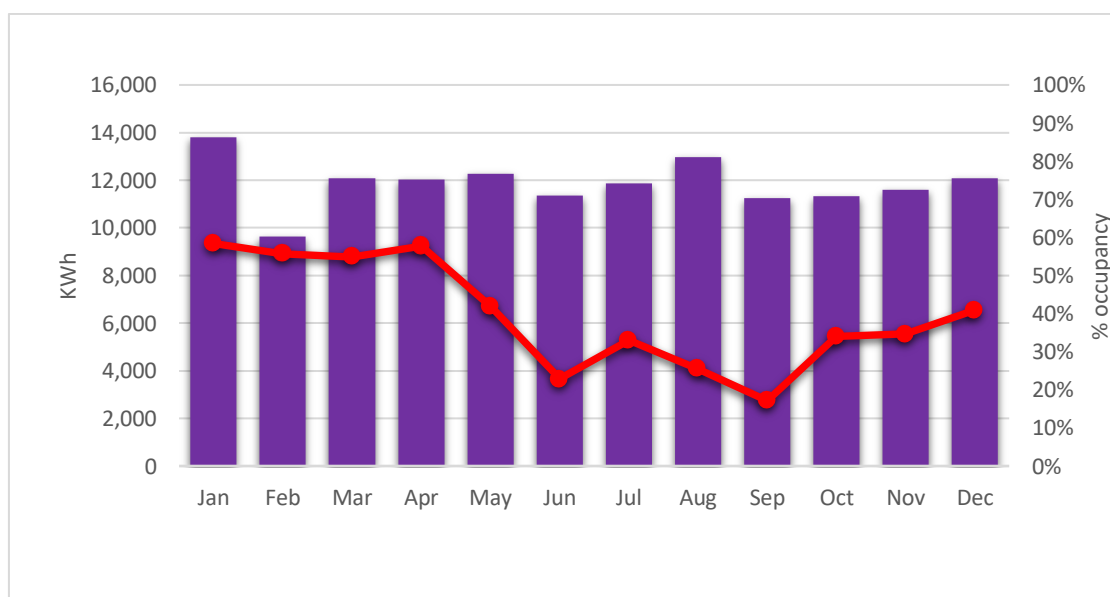


Figure 55: Gym Equipment



The total **Others** category energy consumption for the baseline period is **142,239 KWh** with the monthly variations found in the below chart.

Figure 56: Others Monthly Energy Consumption



#### 4.1.8. Laundry

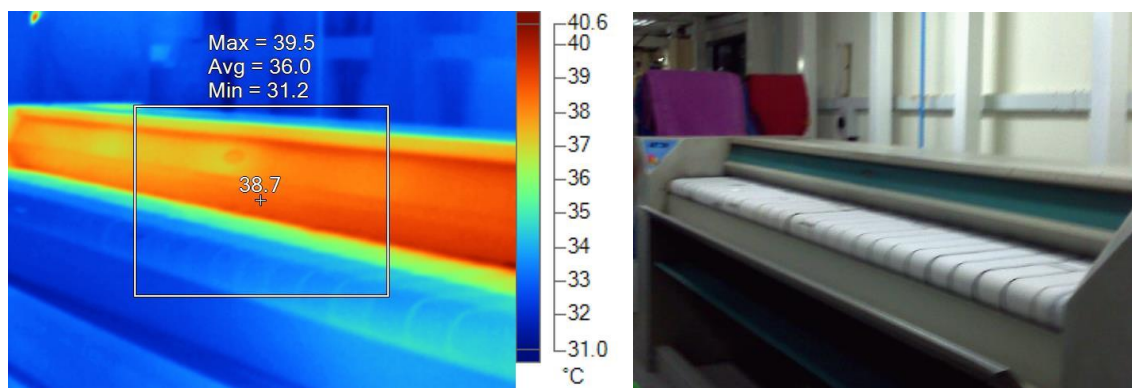
The laundry department in FSRMKH consists of 5 washing machines, 4 dryers (2 on electricity and 2 on LPG), 1 calendar unit and a set of ironing machines. The laundry working hours are from 6 am to 5 pm during low occupancy season and from 6 am to 11 pm when occupancy increases.

Figure 57: Laundry Loads



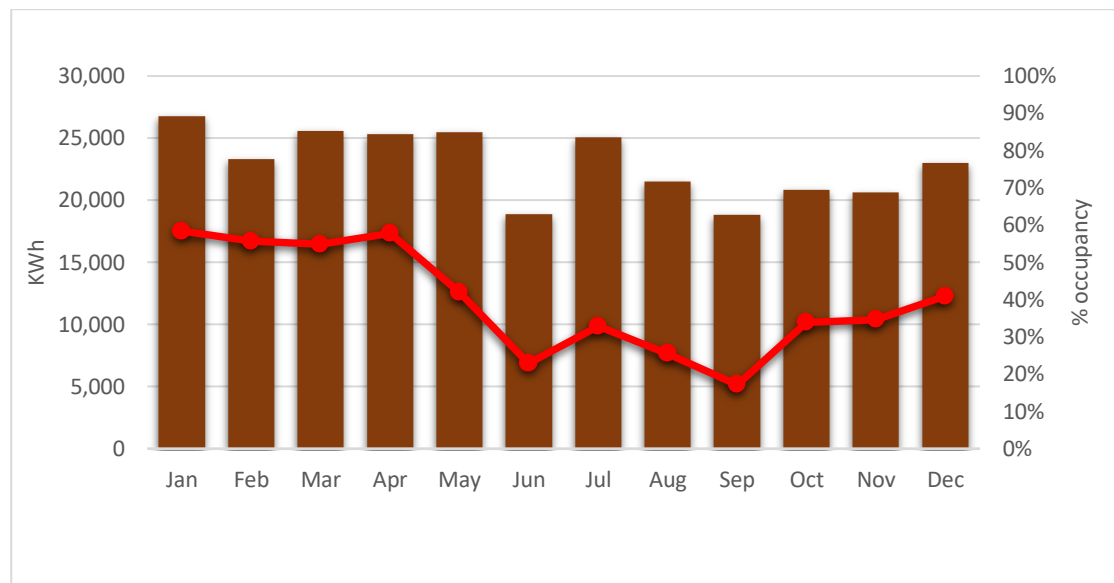
The thermal image below taken during the field visit shows the calendar unit with an average temperature of 36°C while beign on standby.

Figure 58: thermal image of calendar unit



The total **Laundry** category energy consumption for the baseline period is **275,193 KWh** with the monthly variations found in the below chart.

Figure 59: Laundry Monthly Energy Consumption



## 4.2. Energy Balance

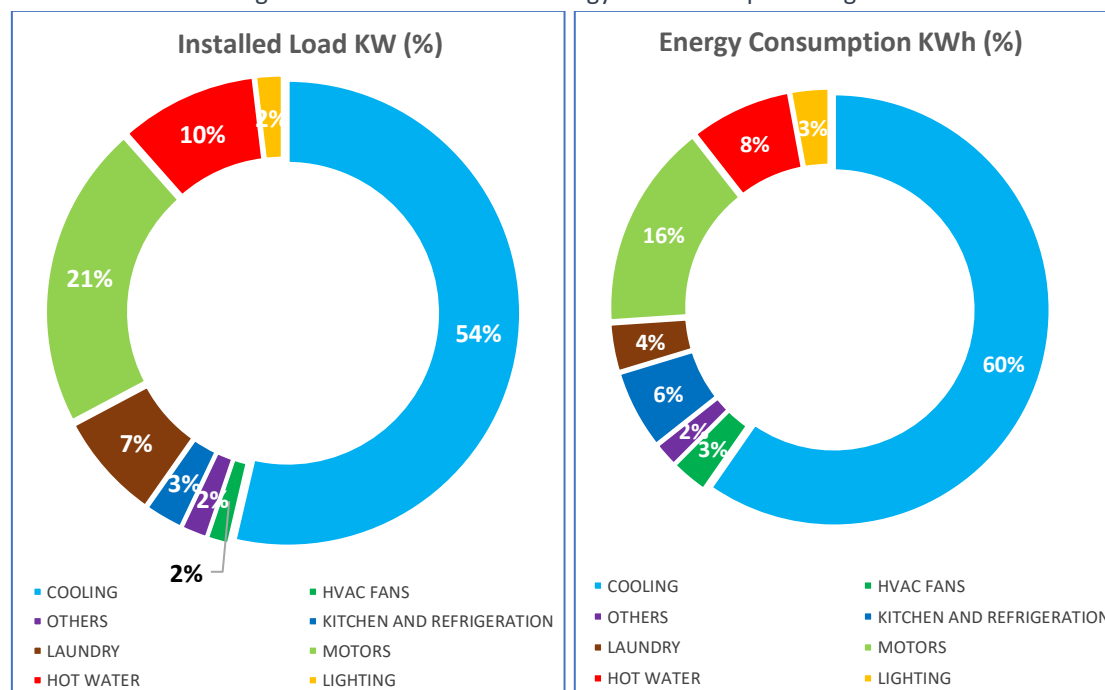
The Electrical Energy Balance provides the breakdown per chosen end-use category for both the installed electrical loads (KW) and the annual energy consumed (KWh). This is achieved following the energy simulation and bills reconciliation completion as per the load inventory, the related operations and the inspection done during the field energy audit.

The following table and figure provide the percentage shares of categories' total installed electrical power in KW and energy consumption in KWh.

Table 18: Electrical Load & Energy Consumption Breakdown by Category

| Category                | Installed Load                   |             | Energy Consumption                                  |                    |                     |             |
|-------------------------|----------------------------------|-------------|-----------------------------------------------------|--------------------|---------------------|-------------|
| Unit                    | KW                               | %           | KWh                                                 | \$                 | Kg. CO <sub>2</sub> | %           |
| COOLING                 | 2,165                            | 54%         | 4,522,533                                           | \$1,434,419        | 330,975             | 60%         |
| HVAC FANS               | 62                               | 2%          | 213,553                                             | \$67,696           | 15,620              | 3%          |
| OTHERS                  | 75                               | 2%          | 142,239                                             | \$45,090           | 10,404              | 2%          |
| KITCHEN & REFRIGERATION | 109                              | 3%          | 452,078                                             | \$143,309          | 33,067              | 6%          |
| LAUNDRY                 | 302                              | 7%          | 275,193                                             | \$87,236           | 20,129              | 4%          |
| MOTORS                  | 857                              | 21%         | 1,176,456                                           | \$372,937          | 86,051              | 16%         |
| HOT WATER               | 388                              | 10%         | 578,957                                             | \$183,529          | 42,347              | 8%          |
| LIGHTING                | 76                               | 2%          | 220,902                                             | \$70,026           | 16,158              | 3%          |
| <b>Total</b>            | <b>4,035</b>                     | <b>100%</b> | <b>7,581,911</b>                                    | <b>\$2,404,242</b> | <b>554,751</b>      | <b>100%</b> |
| Maximum                 | COOLING - 2,165 KW, 54% of total |             | COOLING - 4,522,533 KWh, \$ 1,433,643, 60% of total |                    |                     |             |
| Minimum                 | HVAC FANS - 62 KW, 2% of total   |             | OTHERS - 142,239 KWh, \$ 45,090, 2% of total        |                    |                     |             |

Figure 60: Electrical Load & Energy Breakdown percentages.



The cooling category shows the highest energy consumption at 60% of the total, followed by the Motors category at 16%. Similarly, the cooling category has the highest share of electricity-installed loads at 54% of the total followed by the motors category at 21%.

Figure 61: Annual energy Consumption per end use

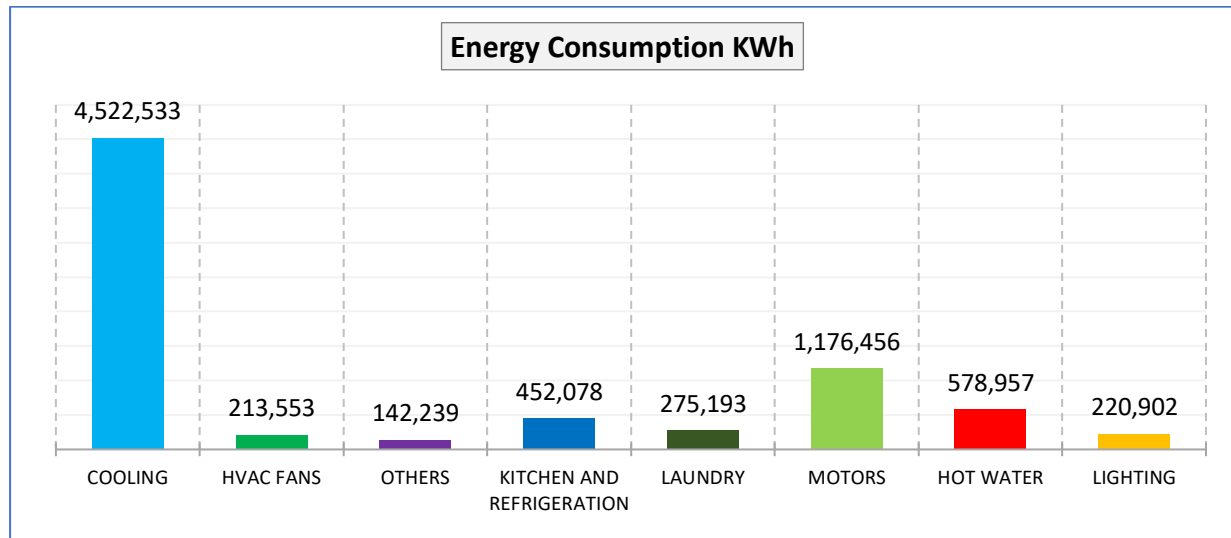
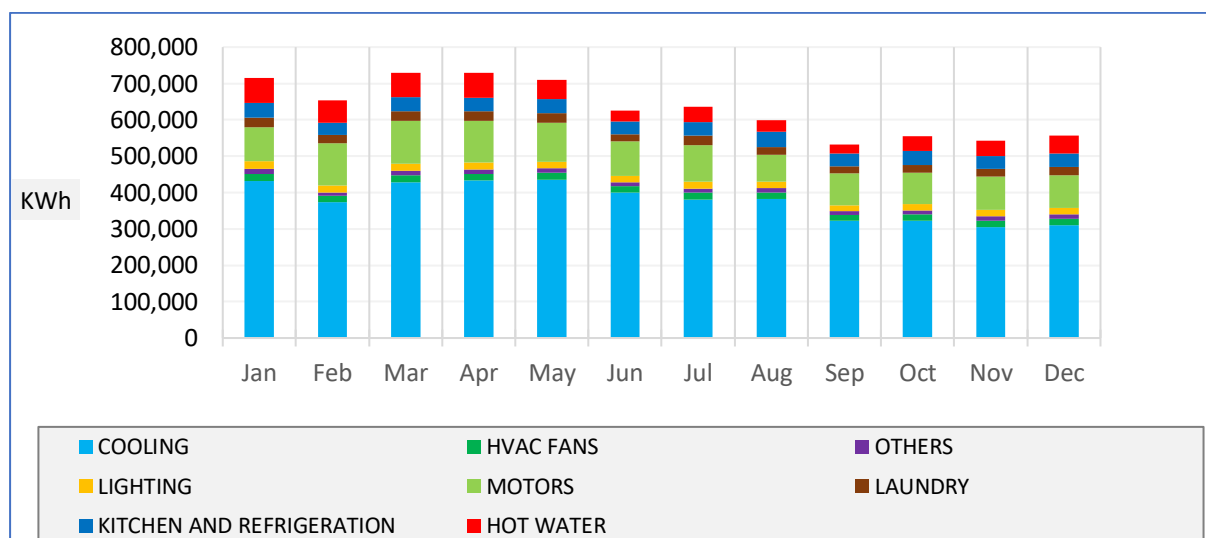


Figure 62: Monthly Electricity Consumption per end use



## 5. Water End Use Inventory

### 5.1. Water Inventory Categories

In a water audit, the water end use category refers to the various ways in which water is used within a building or facility. This category is crucial in helping identify where water is being used and how much water is being consumed in each end use category. The information is then used to determine where water conservation measures can be implemented to reduce overall water consumption and associated costs. There are typically several different water end use categories that are evaluated in a water audit. These categories can include:

FSRMKH water end use categories are based on the location as per the following:

- 1- **25 Guestrooms of the 100s series:** containing shower heads, toilets, water faucets and swimming pool.
- 2- **35 of the 200s series:** containing shower heads, toilets, water faucets and swimming pool.
- 3- **38 water villas:** containing shower heads, toilets, water faucets and swimming pool.
- 4- **Staff accommodation (E-hotel)**'s water end use category consumes water from their toilets containing water faucets and shower heads as well.
- 5- **Public area:** this end use category is made of water faucets, shower heads, toilets, and urinals in public restrooms, in addition to swimming pools (Main swimming pool, Kids club swimming pool, and Reef Club).
- 6- **Laundry machines and kitchens** (dishwashers).

The info is presented here below and is categorized by location, with the corresponding estimated water consumption in liters as per the datasheets provided by the FSMKH management team.

#### 5.1.1. Guestrooms 100s Series

| Location       | Type           | Inventory | Quantity | comments            |
|----------------|----------------|-----------|----------|---------------------|
| unit           |                |           | liters   |                     |
| GR 100s series | Faucets        | 52        | N/A      |                     |
|                | Shower Heads   | 54        | N/A      |                     |
|                | Toilets        | 27        | 135      | 5 liters per toilet |
|                | Swimming Pools | 25        | 667,500  | 26.7 m3 per pool    |

#### 5.1.2. Guestrooms 200s Series

| Location       | Type           | Inventory | Quantity  | comments            |
|----------------|----------------|-----------|-----------|---------------------|
| unit           |                |           | liters    |                     |
| GR 200s series | Faucets        | 69        | N/A       |                     |
|                | Shower Heads   | 67        | N/A       |                     |
|                | Toilets        | 35        | 175       | 5 liters per toilet |
|                | Swimming Pools | 33        | 1,279,410 | 38.77 m3 per pool   |

#### 5.1.3. Water Villas

| Location     | Type           | Inventory | Quantity  | comments            |
|--------------|----------------|-----------|-----------|---------------------|
| unit         |                |           | liters    |                     |
| Water Villas | Faucets        | 94        | N/A       |                     |
|              | Shower Heads   | 76        | N/A       |                     |
|              | Toilets        | 51        | 260       | 5 liters per toilet |
|              | Swimming Pools | 38        | 1,531,400 | 40.3 m3 per pool    |

#### 5.1.4. E-Hotel

| Location | Type         | Inventory | Quantity | comments            |
|----------|--------------|-----------|----------|---------------------|
| unit     |              |           | liters   |                     |
| E-Hotel  | Faucets      | 149       | N/A      |                     |
|          | Shower Heads | 150       | N/A      |                     |
|          | Toilets      | 150       | 750      | 5 liters per toilet |

**5.1.5. Public Areas**

| Location     | Type         | Inventory | Quantity  | comments               |
|--------------|--------------|-----------|-----------|------------------------|
| unit         |              |           | liters    |                        |
| Public Areas | Faucets      | 37        | N/A       |                        |
|              | Shower Heads | 22        | N/A       |                        |
|              | Toilets      | 45        | 225       | 5 liters per toilet    |
|              | Urinals      | 10        | 30        | 3 liters per one flush |
|              | Main Pool    | 1         | 1,500,000 | 1500 m3 per pool       |
|              | Reef Pool    | 1         | 750,000   | 750 m3 per pool        |
|              | Kids Club    | 1         | 7,500     | 7.5 m3 per pool        |

| Location | Type     | Inventory    | Quantity      | comments                     |
|----------|----------|--------------|---------------|------------------------------|
| unit     |          |              | liters /cycle |                              |
| Pools    | backwash | 1 time / day | 26,000        | Approximately 26 cubic meter |

**5.1.6. Laundry and Kitchens**

| Location             | Type                           | Inventory | Quantity      | comments                        |
|----------------------|--------------------------------|-----------|---------------|---------------------------------|
| unit                 |                                |           | liters /cycle |                                 |
| Laundry and Kitchens | Dishwashers                    | 3         | 12            | 4 liters per cycle.             |
|                      | Guest Laundry Washing Machines | 400       | 160000        | 400 Liter per machine per cycle |
|                      | Staff Laundry Washing Machines | 10        | 250           | 25 liter per machine per cycle  |

The Maldives has set a number of water standards and laws to ensure that the country's water resources are effectively managed and utilized. These standards and laws apply to water supply, wastewater, and water quality.

In terms of wastewater management, the Maldives has established a wastewater code that outlines the requirements for the construction, operation, and maintenance of wastewater treatment systems. The code also specifies the standards for the quality of treated wastewater that is discharged into the environment. As for the recommended flow rate for faucets, showers, and other water equipment, the Maldives has not established any specific standards yet.

However, international standards recommend that faucets have a maximum flow rate of 2.2 gallons per minute (GPM) or 8.3 liters per minute (LPM), while showerheads have a maximum flow rate of 2.5 GPM or 9.5 LPM. These standards are aimed at reducing water wastage and promoting water conservation.

As per the engineering department at FSMRKH, plumbing fixtures installed in FSMRKH are considered water efficient. Water efficient shower heads use 7.5 liters per minute or less with

maintaining a minimum of 275 Kilopascal (KPA); water efficient guest room faucets use a maximum of 5.7 liters (1.5 gallons) per minute; while Water efficient toilets include low-flow or dual flush, using 3 to 6 liters (0.8 to 1.6 gallons) per flush or less.

Toilets flushing in FSMKH is of a single flushing type , while water faucets are also based on manual operation and not infrared operated. During the field audit, the plumbing fixtures' flows in public restrooms in addition to guest changing rooms were measured. The measurement results of flow in faucets are indicated in the table below.

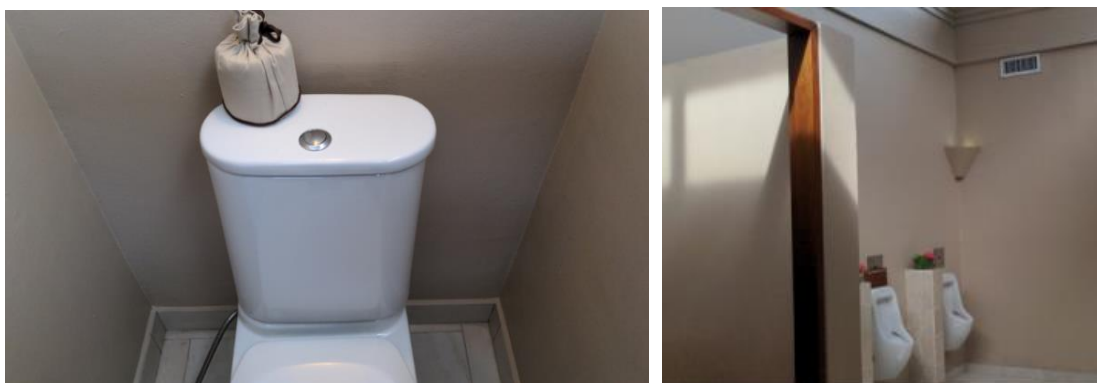
Table 19: Water flowmeter readings

| Unit                    | Time   | Quantity      | Flow  |
|-------------------------|--------|---------------|-------|
| seconds                 | Liters | Liters/Second |       |
| guestroom changing room | 20     | 2             | 0.099 |
| guestroom #108          | 13     | 2             | 0.150 |
| café Huraa male toilet  | 17     | 2             | 0.114 |
| guestroom #124          | 11     | 2             | 0.182 |
| guestroom #319          | 11     | 2             | 0.175 |

Figure 63: toilet's faucets public; (left) and guestrooms (right)



Figure 64: Restroom toilet (left), and urinary (right)



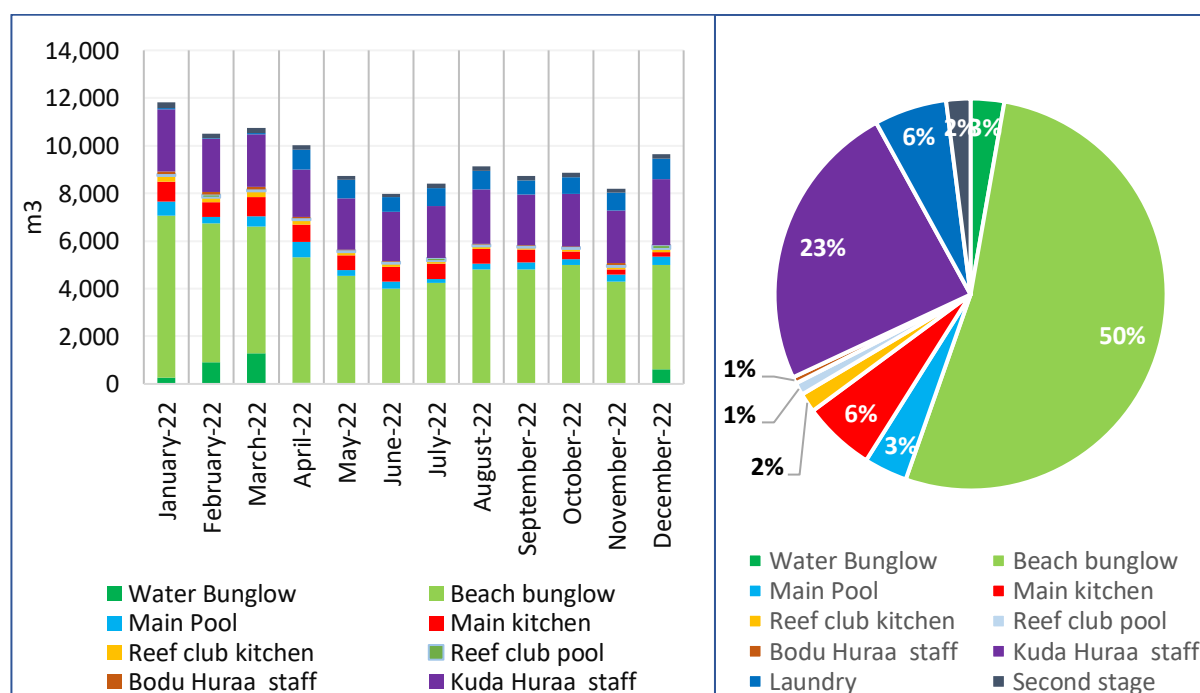
## 5.2. Water Balance

The table below presents the water consumption breakdown in FSRKH for the baseline year 2022 per end user. As can be noticed, the total metered consumption is 82% of the total RO production.

Table 20: Water consumption breakdown

| Date              | RO production (Ro 1 : 4) | Total Metered Consumption | Water Bungalow | Beach Bungalow | Main Pool    | Main kitchen | Reef club kitchen | Reef club pool | Bodu Huraa staff | Kuda Huraa staff | Laundry      | Second stage |
|-------------------|--------------------------|---------------------------|----------------|----------------|--------------|--------------|-------------------|----------------|------------------|------------------|--------------|--------------|
| Unit              | m3                       | m3                        | m3             | m3             | m3           | m3           | m3                | m3             | m3               | m3               | m3           | m3           |
| Jan               | 11,040                   | 5,448                     | 272            | 6,793          | 578          | 831          | 242               | 91             | 123              | 2,606            | 53           | 242          |
| Feb               | 10,412                   | 6,142                     | 901            | 5,835          | 277          | 624          | 189               | 131            | 92               | 2,239            | 32           | 189          |
| Mar               | 10,115                   | 6,275                     | 1,288          | 5,324          | 429          | 800          | 230               | 102            | 107              | 2,197            | 45           | 230          |
| Apr               | 10,731                   | 5,899                     | 48             | 5,270          | 638          | 721          | 196               | 90             | 37               | 1,989            | 841          | 179          |
| May               | 9,918                    | 5,577                     | 0              | 4,549          | 224          | 637          | 128               | 84             | 22               | 2,141            | 782          | 169          |
| Jun               | 8,848                    | 7,801                     | 1              | 4,002          | 305          | 615          | 118               | 87             | 19               | 2,071            | 612          | 158          |
| Jul               | 9,861                    | 10,047                    | 0              | 4,235          | 168          | 631          | 126               | 119            | 22               | 2,173            | 745          | 186          |
| Aug               | 10,074                   | 10,521                    | 0              | 4,818          | 220          | 638          | 108               | 65             | 22               | 2,301            | 784          | 180          |
| Sep               | 9,637                    | 9,924                     | 0              | 4,809          | 290          | 530          | 98                | 84             | 17               | 2,130            | 588          | 172          |
| Oct               | 10,023                   | 10,516                    | 0              | 4,984          | 263          | 302          | 109               | 82             | 37               | 2,194            | 701          | 182          |
| Nov               | 9,122                    | 9,067                     | 0              | 4,306          | 297          | 190          | 112               | 81             | 93               | 2,205            | 738          | 174          |
| Dec               | 9,498                    | 10,723                    | 614            | 4,392          | 341          | 181          | 144               | 145            | 23               | 2,764            | 838          | 197          |
| <b>Total</b>      | <b>119,279</b>           | <b>97,940</b>             | <b>3,124</b>   | <b>59,317</b>  | <b>4,030</b> | <b>6,700</b> | <b>1,800</b>      | <b>1,161</b>   | <b>614</b>       | <b>27,010</b>    | <b>6,759</b> | <b>2,258</b> |
| <b>Percentage</b> |                          | <b>82%</b>                | <b>3%</b>      | <b>50%</b>     | <b>3%</b>    | <b>6%</b>    | <b>2%</b>         | <b>1%</b>      | <b>1%</b>        | <b>23%</b>       | <b>6%</b>    | <b>2%</b>    |

Figure 65: Water consumption breakdown per end use category



As can be seen in the graphs above, the end user category that consumes water the most is the guestrooms (Beach Bungalows) with almost 50% of the total metered consumption, followed by the staff accommodations.

## 6. Indoor Air Quality data logging

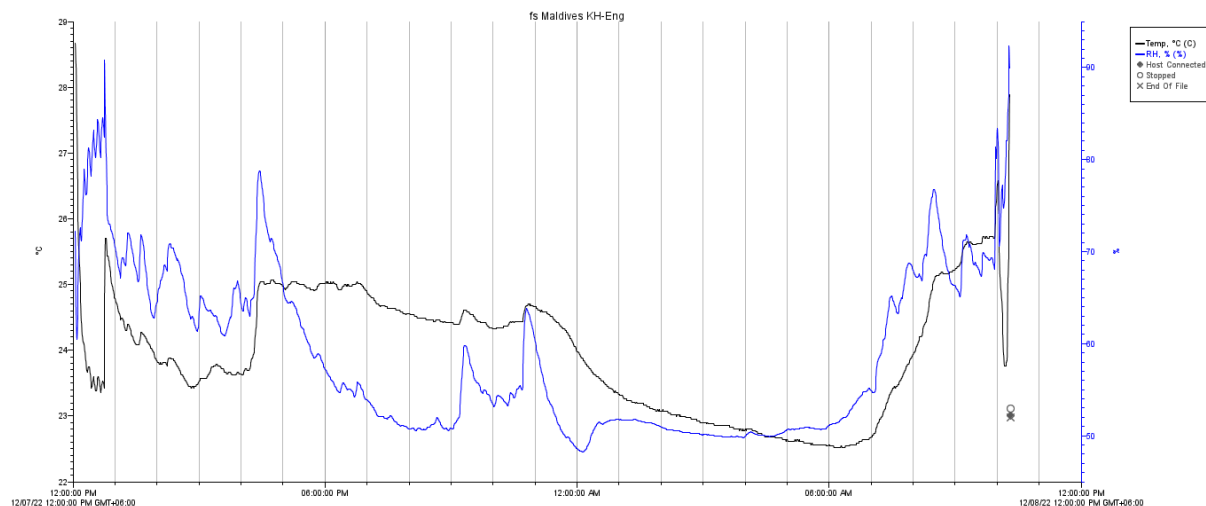
Indoor air quality (IAQ) focuses on airborne contaminants including thermal comfort quality and air quality, covering therefore temperature, humidity, CO<sub>2</sub>, CO and VOC (Volatile Organic Compounds) levels.

When it comes to standards, the main guidelines refer to standards promulgated by the American Society of Heating, Refrigerating and Air-conditioning Engineers Inc. (ASHRAE). These are found in the ASHRAE documents Ventilation for Acceptable Indoor Air Quality (ASHRAE 62-2001) and Thermal Environmental Conditions for Human Occupancy (ASHRAE 55-2004).

### Humidity and Temperature

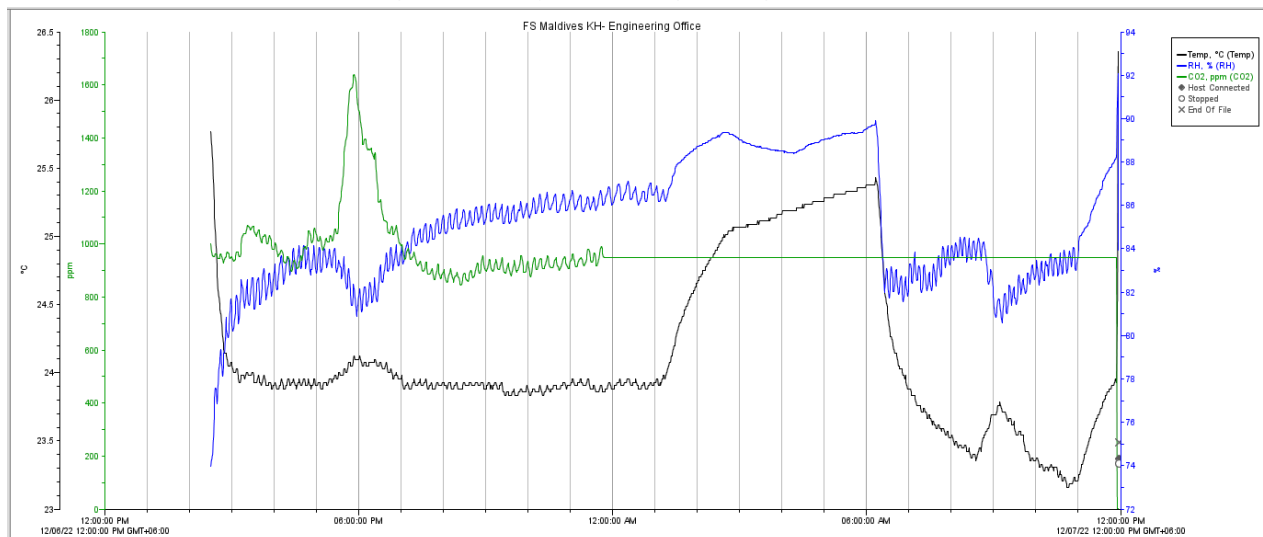
There is no “ideal” humidity level and temperature suitable for all building occupants. Many factors, such as personal activity and clothing may affect personal comfort. Acceptable relative humidity levels should range from 20 percent – 65 percent year-round. Levels less than 20 percent in the winter and greater than 65 percent in the summer should be considered off the standards. Elevated relative humidity can promote the growth of mold, bacteria, and dust mites, which can aggravate allergies and asthma. To achieve maximum occupant comfort, relative humidity should be maintained between 30% and 65%. The following are graphs, showing temperature and humidity plots in some spaces of FSHF. These measurements were taken during the field audit of FSRMKH in December 2022.

Figure 66: Temp/RH% at Housekeeping office



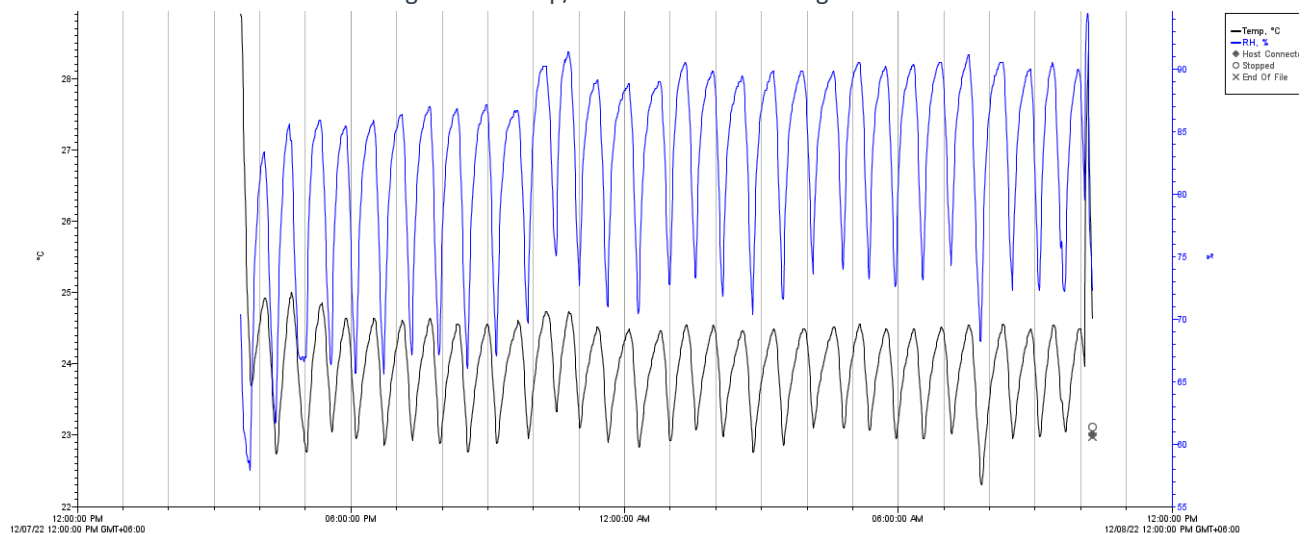
➔ The above chart shows that temperature varies between 22.5 and 28°C. RH is seen to be varying between 49% and 91%.

Figure 67: Temp/RH% at Engineering Office



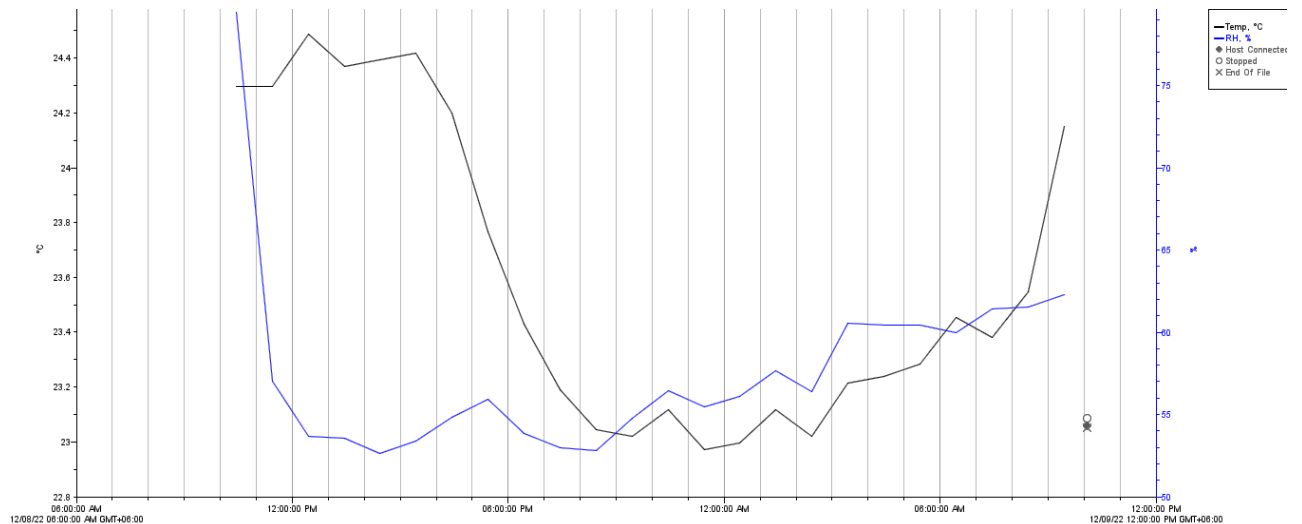
- ➔ Temperature remains constant at around 24 °C between 6 am and 12 am corresponding to the occupied time in these offices, and increases to around 26 °C at night. While RH varies between 80% and 88 % during occupied time and increases to 90% at night time. Similarly, CO2 levels increases during the occupancy window of the engineering offices to 1,700 ppm and decrease to around 1,000 ppm which indicates a poor ventilation of the space.

Figure 68: Temp/RH% at the Water Bungalow 318



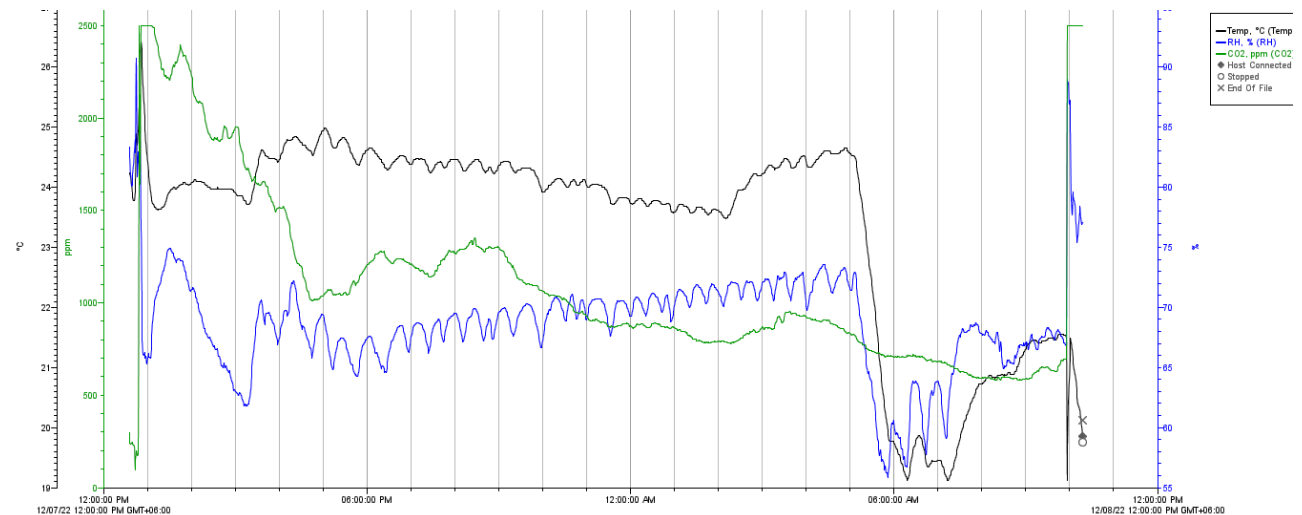
- ➔ Temperature is seen to fluctuate in a narrow band between 23 and 25°C and RH between 70% and 90% in the rented guestroom (Water Bungalow 318).

Figure 69:Temp/ RH% at the Business center



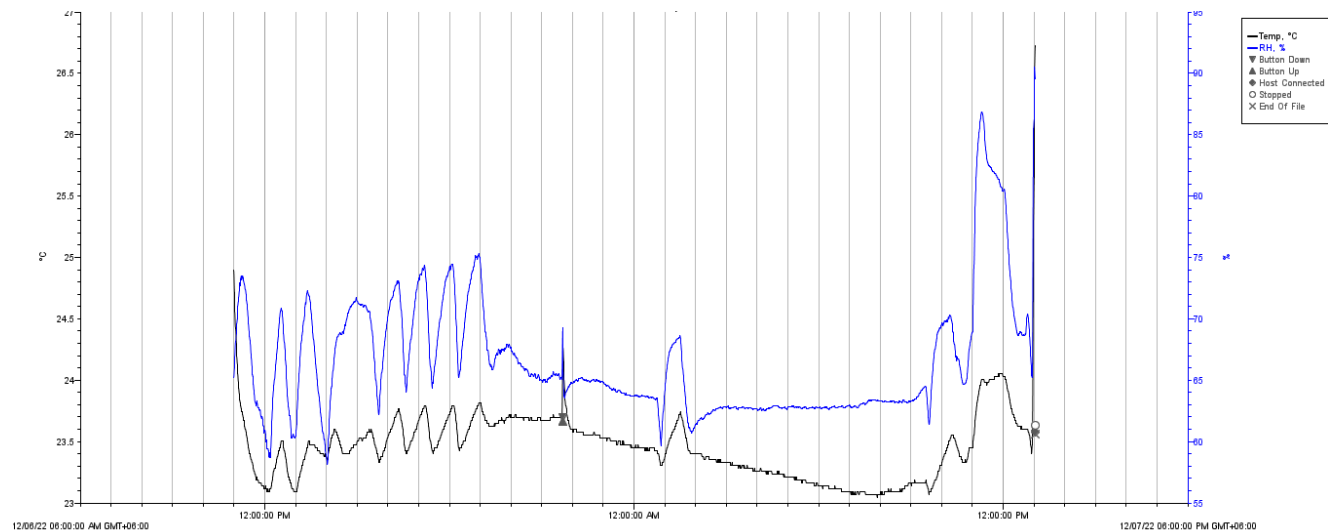
➔ Temperature is seen to fluctuate in a narrow band between 23 and 24.4°C and RH between 52% and 80% in the business center.

Figure 70: Temp/ RH% at the employee recreation



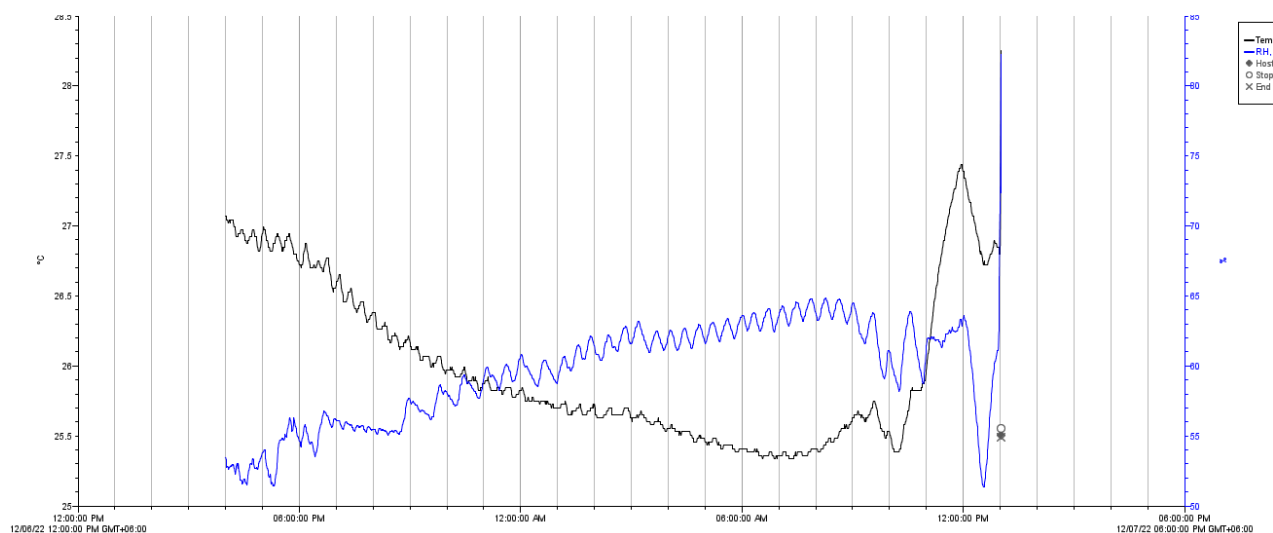
➔ Temperature is seen to stay constant 24°C between 6 pm and 6 am during Laundry non operation time, and then decreases to 21 °C once the laundry operation starts. CO2 concentration levels follow the same trendline and decreases from around 1500 ppm during laundry unoccupied times to around 950 ppm. RH levels vary in a short window around 70%.

Figure 71: Temp/ RH% at the gym



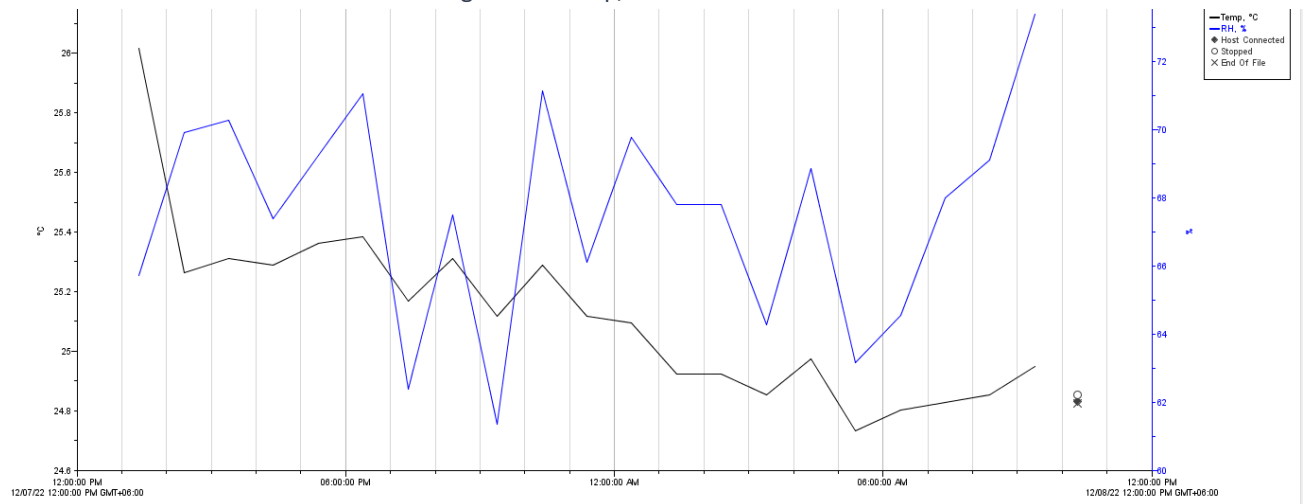
➔ Temperature and RH are seen to be constant at the Gym with an average of 23.5°C and 70 % respectively which indicates that the ventilation is continuously operating.

Figure 72:Temp/ RH% at the Kids Club



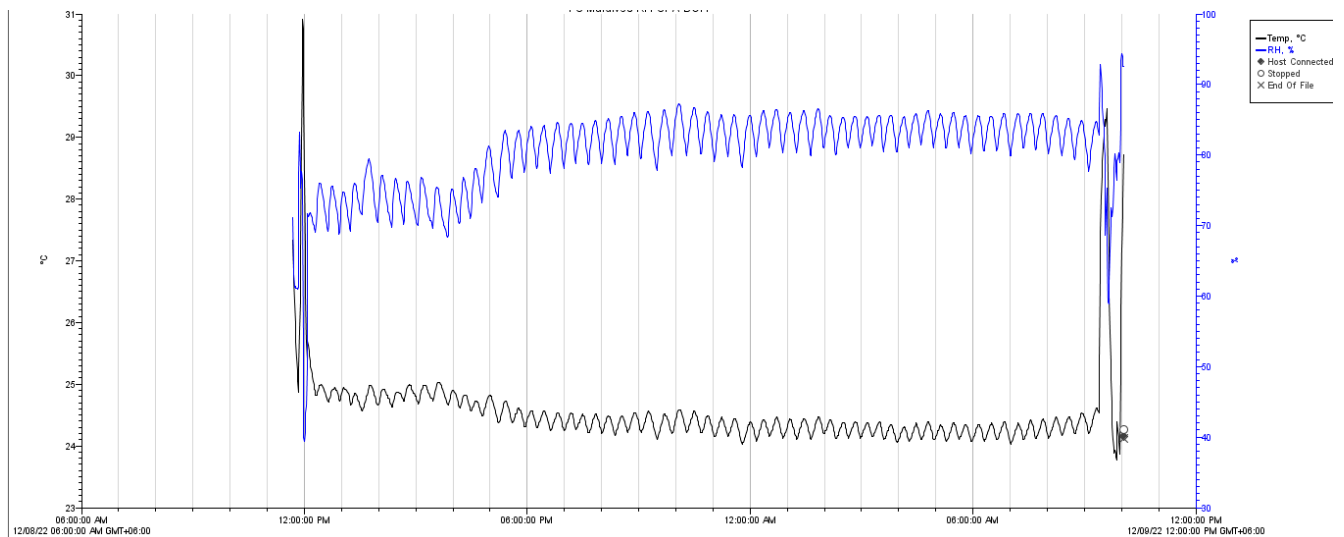
➔ Temperature decreases from 27°C to 25.5°C between 6 PM and 6AM , then increases to an average of 27.5°C at around 11 AM with a variation of RH between 70% and 54%.

Figure 73: Temp/ RH% at the Restroom



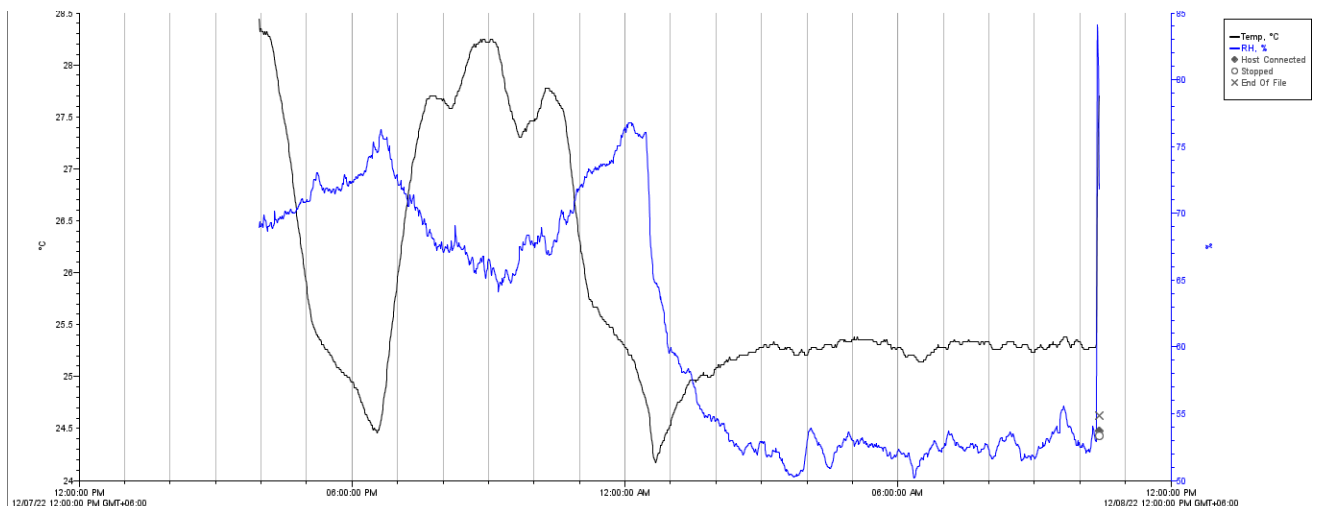
➔ Temperature is seen to be fluctuating at narrow range between 25.4 and 24.8°C and a RH at an average of 68% indicating that ventilation is continuously operating.

Figure 74:Temp/ RH% at the SPA BOH



➔ Temperature is seen to be constant at 24.8°C while RH also remains constant at 85% indicating that ventilation is continuously operating.

Figure 75:Temp/ RH% at the Reef Kitchen



- ➔ Temperature is seen to be fluctuating at 28 °C during the operation of the kitchen between 6 PM and 12AM and then decreases to 25°C between 12 AM and 6 AM during the unoccupied time of the kitchen. The RH follows a similar trend with an average of 70% between 6 PM and 12 AM versus 55% between 12 AM and 12 PM.

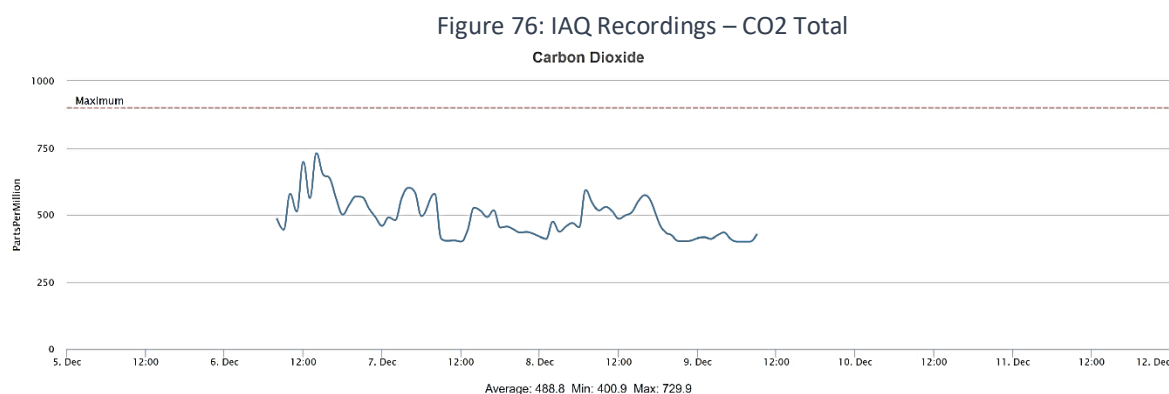
## IAQ recordings

An advanced IAQ recording unit - uHoo Aura was used across the property monitoring 13 different indoor environmental parameters among them – temperature, relative humidity, carbon dioxide, various particle sizes (PM10, PM4, PM2.5, PM1), carbon monoxide, formaldehyde, and volatile organic compounds (VOCs). The IAQ unit comes with a Virus Index which is an assessment of virus survival and transmission risk (such as the coronavirus) based on air quality.

Some key IAQ parameters information are found below:

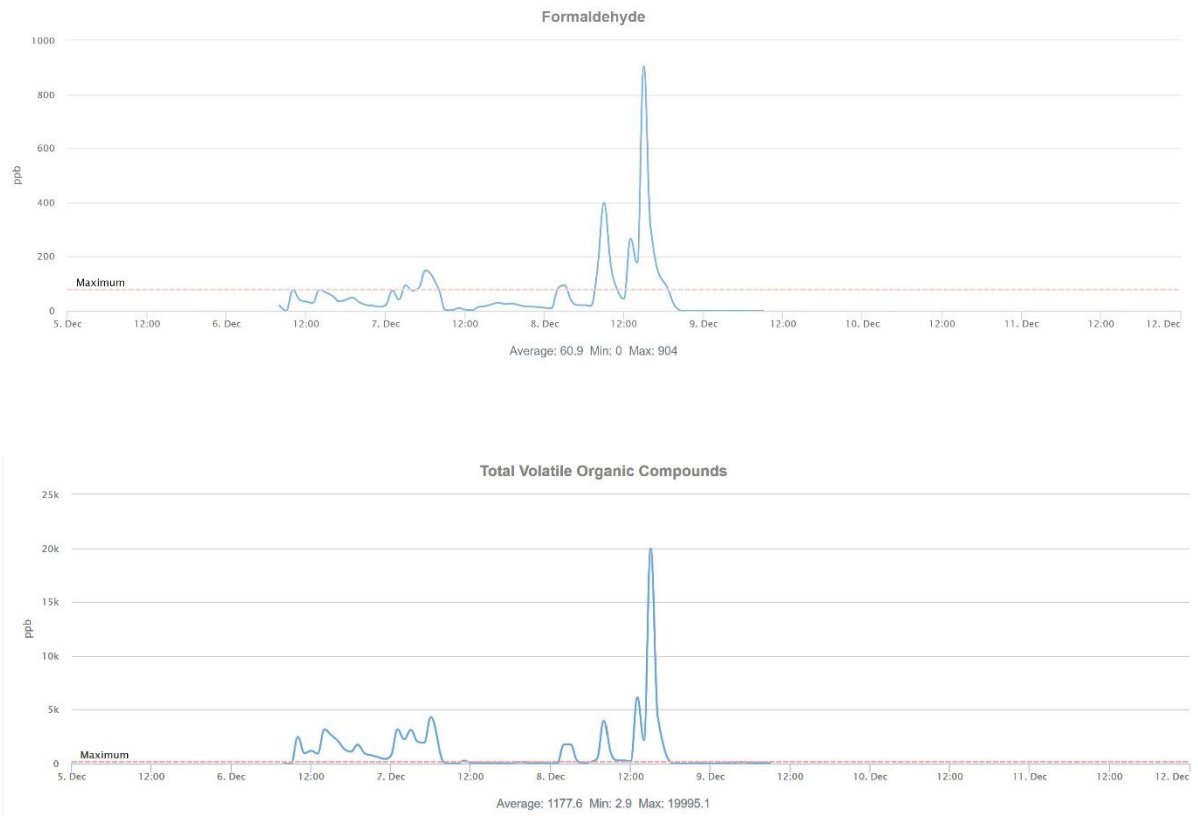
- **PM2.5:** Particulate matter is one of the most dangerous forms of pollution as the size of the particles is so small that they can get into the lungs causing numerous adverse effects. PM2.5, in particular, are particles which are 2.5 µm or less in diameter. Their threshold limit value is 25 µg/m<sup>3</sup>, based on 24-hour data.
- **CO<sub>2</sub>:** CO<sub>2</sub> is a natural compound in the air, with an average outdoor concentration of 300-400 ppm. Note that indoor levels are higher due to the confinement of indoor spaces.
- **TVOC:** The Total Volatile Organic Compound (TVOC) level is a measurement of the sum of all of the volatile organic compounds (VOC's) found in an air sample
- **Formaldehyde:** One of the most common VOCs – formaldehyde – can be emitted from numerous sources, such as furniture, incense burning, and cooking. Note that its threshold limit value is 0.1 ppm

The following set of IAQ recordings taken across different spaces of the Resort whereby all parameters were seen to be within OSHA and ASHRAE St 62.1 Standards for Ventilation and Indoor Air Quality. In that respect, the below charts covering CO<sub>2</sub> levels, Temperature, Formaldehyde, Particulate Matters, TVOCs, and Viruses.



- ➔ Index showcases levels around 500 ppm , well within the ASHRAE standards and confirming the adequate fresh air intake across the facility.

Figure 77:IAQ Recordings – Formaldehyde and TVOCs Total



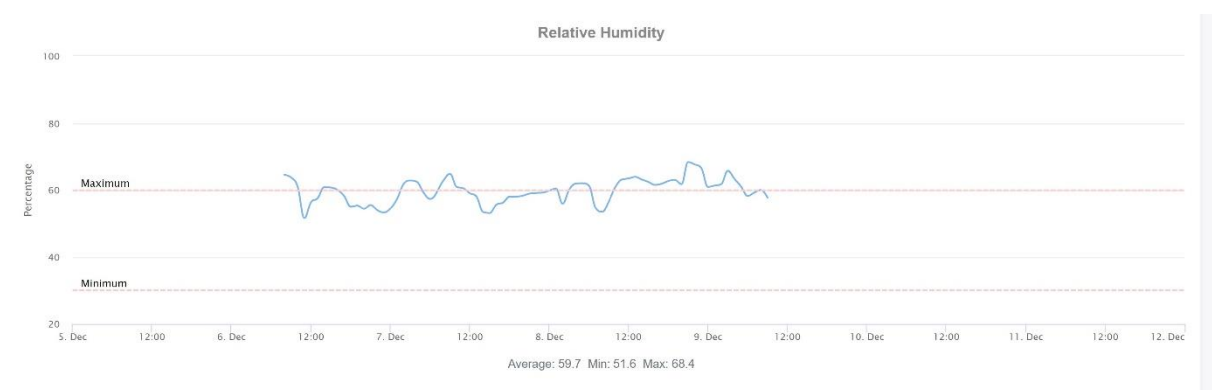
- ➔ The Formaldehyde concentration showed satisfactory levels across the period of measurement at except the sudden hike and drop as in this specific time a steamer for clothes was used.
- ➔ The TVOCs concentration showed higher levels across the period of measurement in addition to a sudden hike and drop as in this specific time a steamer for clothes was used.

Figure 78:IAQ Recordings – PM1,4,10 Total



➔ The Particulate matters concentration showed satisfactory limit across the reading period at concentration below 50  $\mu\text{g}/\text{m}^3$ , except the sudden hike and drop.

Figure 79:IAQ Recordings - RH Total

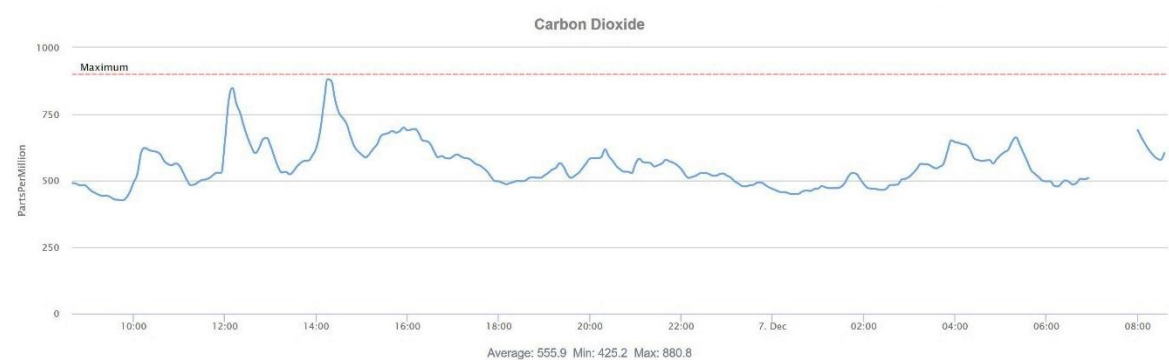


➔ RH was seen to be varying closely to the maximum levels reaching 60%.

## Canteen Case

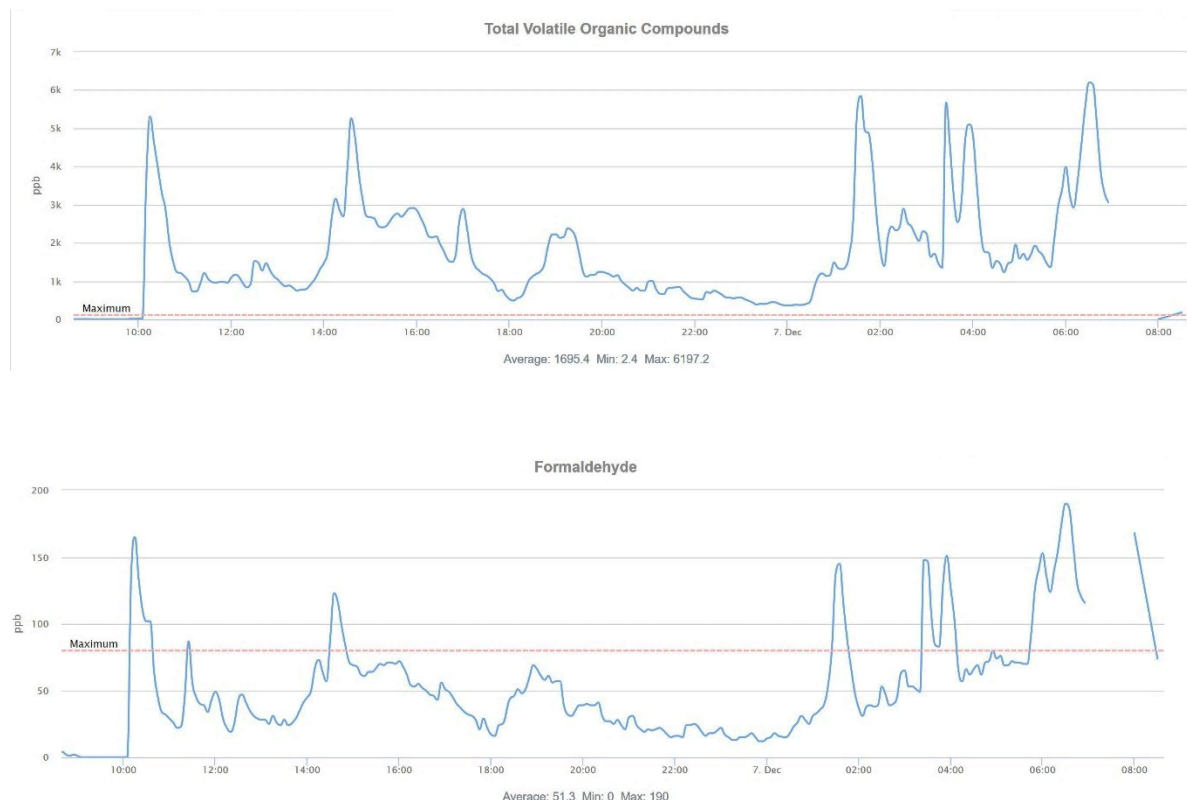
The following set of IAQ recordings were done in the staff canteen whereby all parameters were seen to be within OSHA and ASHRAE 62.1 Standards for Ventilation and Indoor Air Quality. In that respect, the below chart covering CO<sub>2</sub> showcases levels from 500 to maximum 800 ppm, well within the ASHRAE standards and confirming an adequate ventilation.

Figure 80: CO<sub>2</sub> levels - canteen



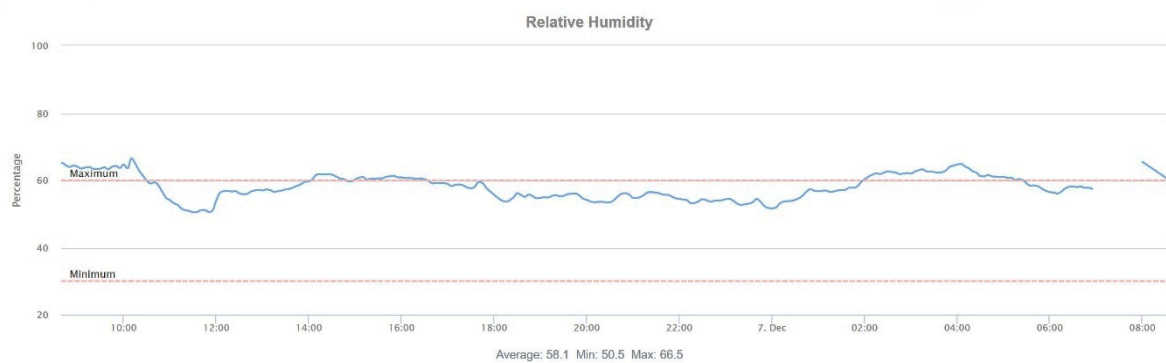
The TVOC showed high levels, exceeding the maximum and reaching 6,000 ppb. As for the Formaldehyde, the concentration was satisfactory most of the time except in sudden hikes which can be linked to the use of cleaning detergents.

Figure 81: TVOCs levels (up ) and Formaldehyde (down)



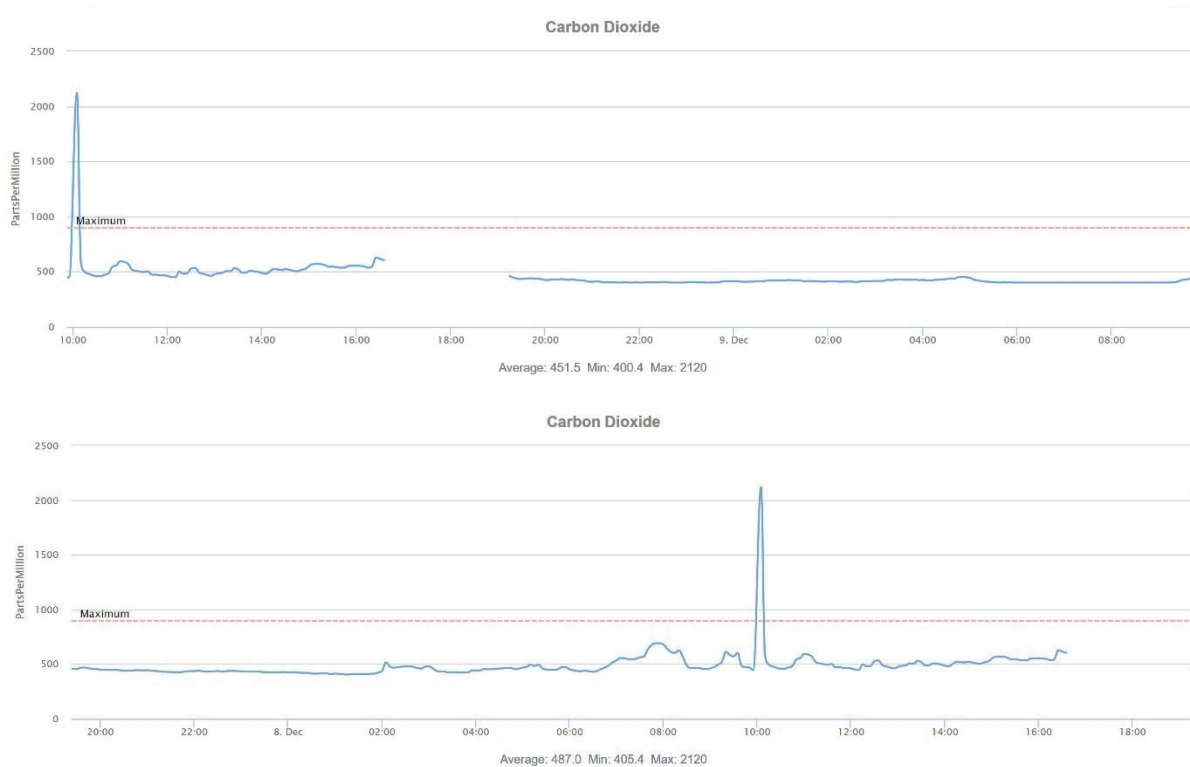
Relative humidity was seen to be satisfactory overall with an average of 60%.

Figure 82: RH %



### Guest Villa #124 Case

The following set of IAQ recordings were done in one of the changing rooms whereby all parameters were seen to be within OSHA and ASHRAE 62.1 Standards for Ventilation and Indoor Air Quality. In that respect, the below chart covering CO<sub>2</sub> showcases levels around 500 ppm, followed by a sudden hike and drop at around 10 am. However the levels are considered well within the ASHRAE standards and confirming an adequate ventilation.



The TVOC showed high levels, exceeding the maximum and reaching 50,000 ppb. As for the Formaldehyde, the concentration was satisfactory most of the time except in sudden hikes.

Figure 83: Formaldehyde levels

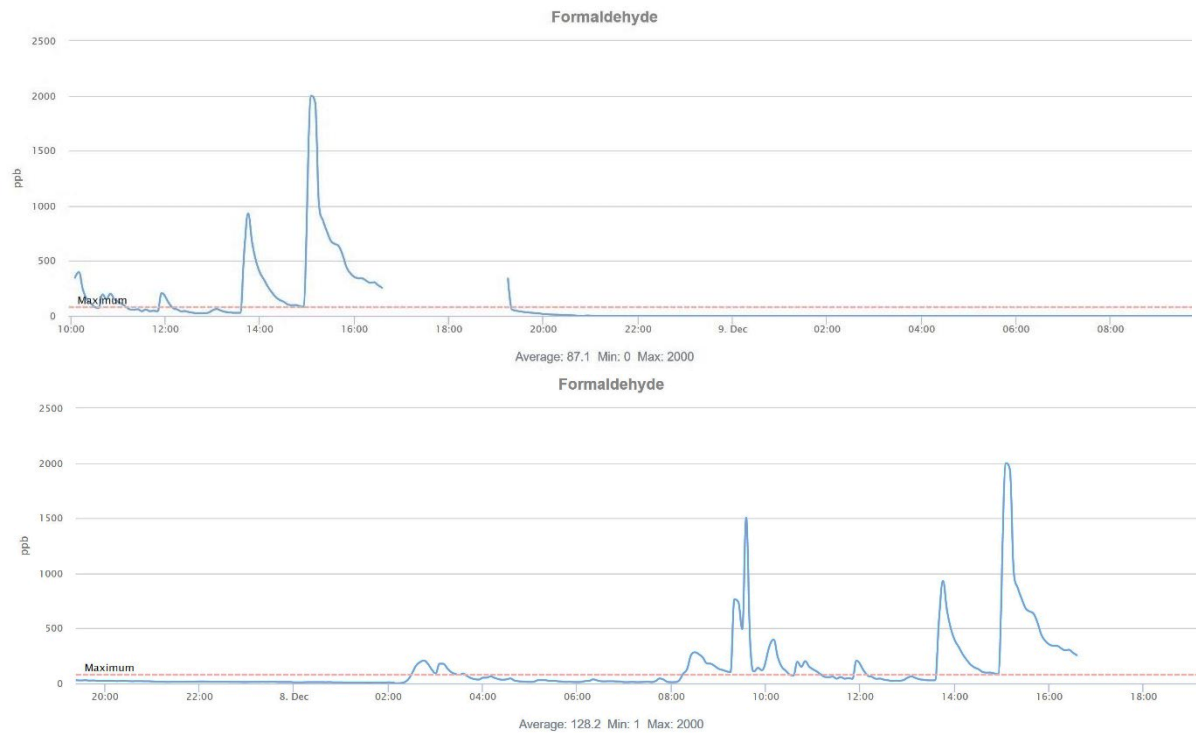
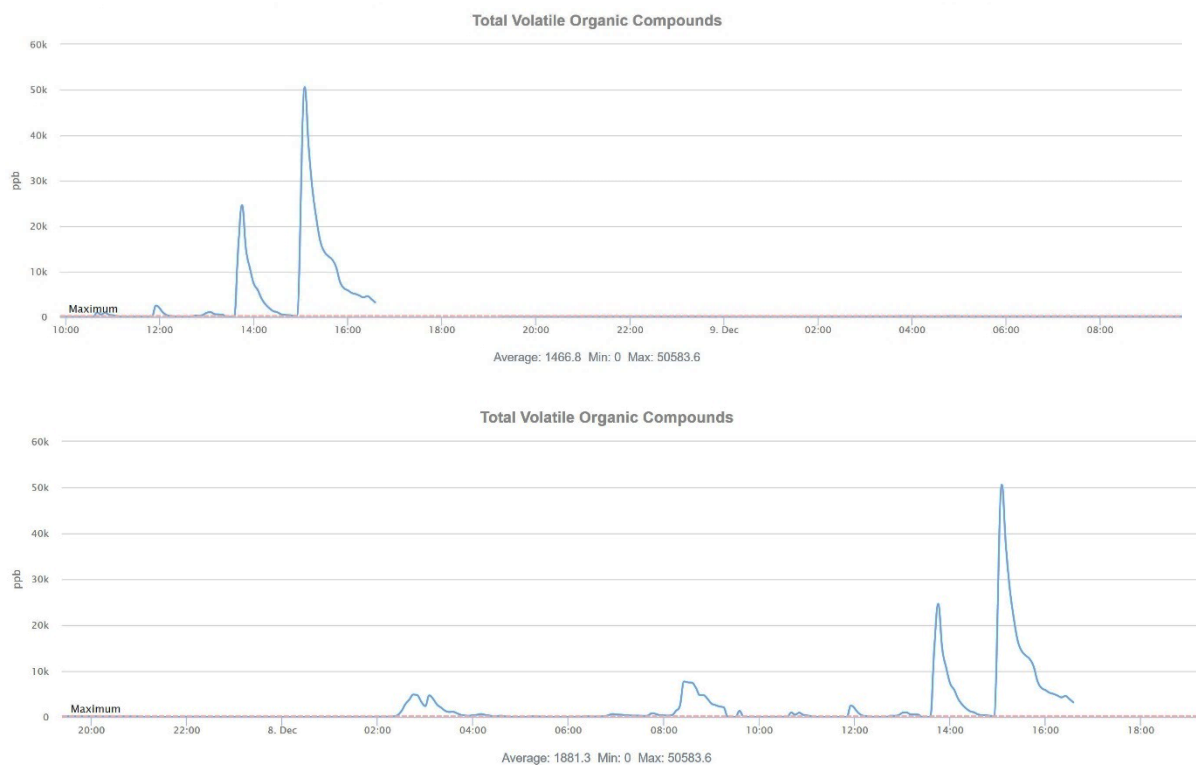


Figure 84: TVOCs levels



Temperature was seen to be varying within 22 °C and 26°C . On the other hand, RH was seen to be satisfactory at an average of 62%.

Figure 85: Temperature readings

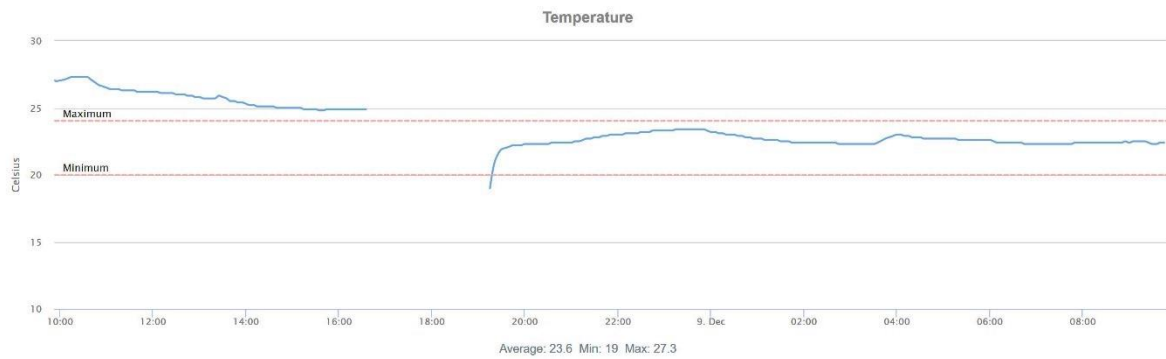
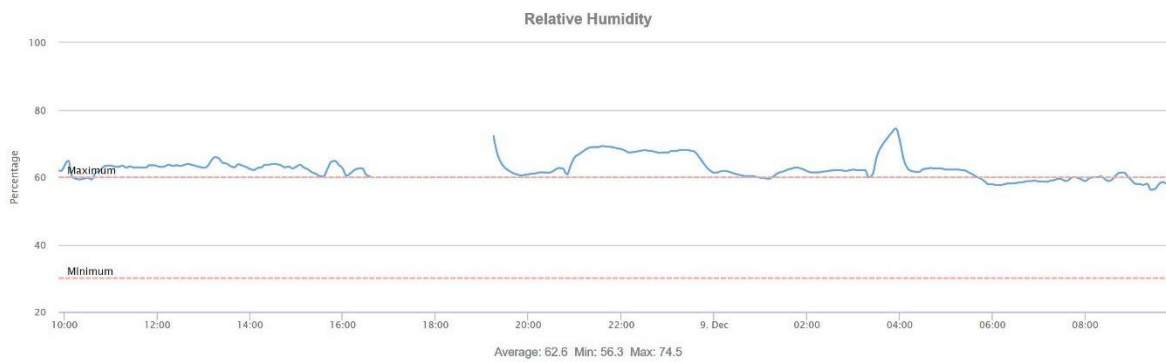


Figure 86: RH readings



## 7. Energy Efficiency Measures

The main objective of this section is to provide a roadmap that would include a number of Energy Efficiency Measure (EEM) which are all the actions that would reduce the utilities consumption. In this aim, EEG investigated a considerable number of energy efficiency solutions, which could yield to substantial energy cost and carbon footprint reductions in the event of their implementation. These opportunities include those who require very little or no CAPEX at all – Low Cost/No Cost – [LC/NC] as well as those needing capital expenditure [CAPEX]. The assessment of an EEM is its financial feasibility and technical viability. Each EEM is attributed with a color code which is a priority or urgency indicator for the EEM's implementation. The color codes adopted in this report along the proposed measures are found in the below tables:

Table 21: Energy Efficiency Measures Color Codes

| Color Code | Description                                                                       | Priority |
|------------|-----------------------------------------------------------------------------------|----------|
|            | Measures with very small CAPEX and excellent payback (<1 year) – Low Cost/No Cost | A        |
|            | Measures with good payback period (<3 years)– Immediate Implementation            | A        |
|            | Measures with midterm payback period (3 to 6 Years) – Strategic Implementation    | B        |
|            | Measures to be done during a planned retrofit/replacement                         | C        |
|            | Measures with high payback period (>6 years) – Upon CAPEX Availability            | D        |

Table 22: Energy Efficiency Measures

| CODE     | TYPE  | TITLE                                       |
|----------|-------|---------------------------------------------|
| EEM - 01 | LCNC  | Cooling Control of Split Units and VRVs     |
| EEM - 02 | LCNC  | Guest Villas Management in Unrented Mode    |
| EEM - 03 | CAPEX | Lighting Retrofit                           |
| EEM - 04 | CAPEX | Installation of Motion Detectors            |
| EEM - 05 | CAPEX | Installation of VFDs on the new STP blowers |
| EEM - 06 | CAPEX | Installation of VFDs on Kitchen Hoods       |
| EEM - 07 | CAPEX | Jacket Heat Recovery from Generators        |
| EEM - 08 | CAPEX | Installation of Metering System             |
| EEM - 09 | CAPEX | Photovoltaic Driven Electricity Generation  |

In the following pages, each EEM has been developed with the rationale, technical description/recommendations and the economical analysis. At the end of each EEM sub section, there is a summary table with all the key indicators as explained here below.

| EEM - XX                         |     |                 |
|----------------------------------|-----|-----------------|
| XXXX                             |     |                 |
| ENERGY SAVINGS                   |     |                 |
| ELECTRICITY                      | LPG | CO <sub>2</sub> |
| KWh                              | Kg  | Kg              |
|                                  |     | 0               |
| 0.0%                             |     | 0.0%            |
| ECONOMIC PERFORMANCE             |     |                 |
| Annual Cost Savings [USD]        |     |                 |
| Annual Share of Cost Savings [%] |     |                 |
| Implementation Cost [ \$]        |     |                 |
| Simple Payback [Years]           |     |                 |
| Net Present Value [ \$]          |     |                 |
| Internal Rate of Return [%]      |     |                 |

The first two lines include the EEM number and its title

The second box include the energy savings in each utility (per unit of measure) and the % savings from the same utility. Also, the CO<sub>2</sub> emissions reductions of the EEM is calculated

The third box include the Economical Performance with the Cost Savings , the % savings from all Utilities budget, the implementation cost, simple payback, IRR – Internal Rate of Return, as well as a Net Present Value calculated over 20 years.

## 7.1. EEM – 01: [LCNC] Cooling Control of Split Units and VRVs

### 7.1.1. Rationale

FSRMKH has the highest energy consumption and installed loads allocated to a huge number of split units and VRVs installed across the resort with zero control. The present EEM consists of implementing a control system on the split units/VRFs located in the BoH and other key areas.

### 7.1.2. Efficiency Action and Energy Savings

This EEM consists of managing all VRF systems and split units across the BOH area to ensure they are operated in an optimized way during night time.

The cooling demand in the Maldives is driven by both sensible and latent cooling with the latter being very important due to the high humidity.

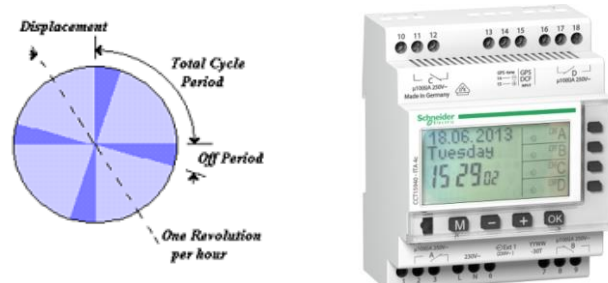
Latent cooling is the cooling that occurs when moisture is removed from the air. This type of cooling occurs when water vapor in the air is condensed into liquid water. As water vapor condenses, it releases heat, which causes the air to cool down.

Typically, both sensible and latent cooling occur simultaneously. For example, as warm air passes over a cold coil in an air conditioner, some of the moisture in the air will condense onto the coil, providing latent cooling, while the temperature of the air will also be reduced,

It is not possible to shut down the cooling units at night time if you want to keep the RH under adequate levels – even though they are still higher than standards in many areas across the property. To ensure providing an adequate minimum control of the latent cooling during night times while minimizing electricity usage, it is recommended to run all targeted split units/VRFs in a duty cycling mode which will operate them in a scheduled time ON and OFF across the hour between the idle hours.

The below figure showcases an hourly duty cycling example as well as one type of multi channel advanced scheduler that could be used for the application,

Figure 87: Example of duty cycle in an hour cycle (Left); multi channel advanced schedule controller (Right)

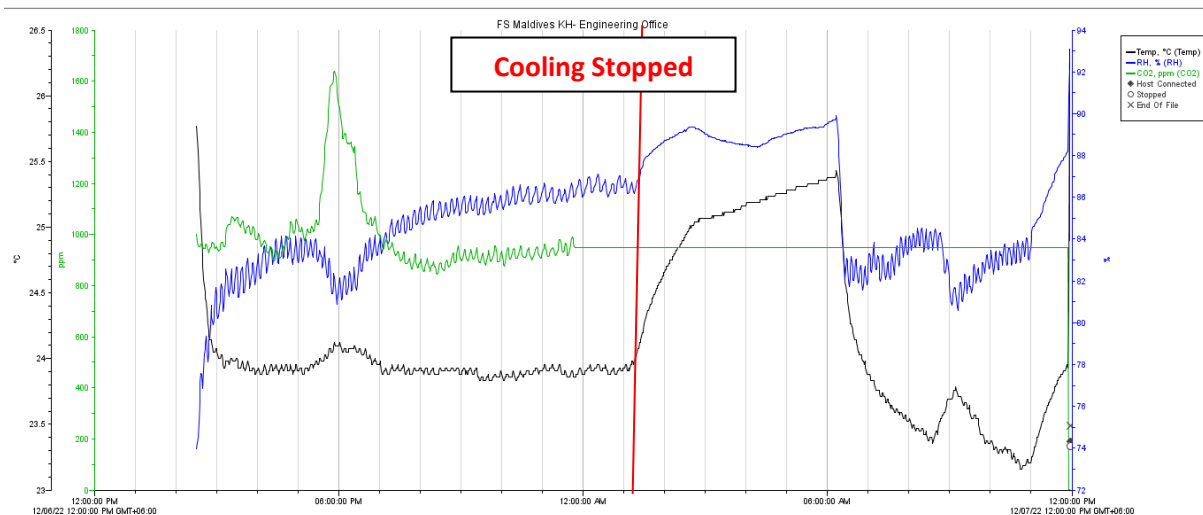


Based on the temp/RH recordings done during the field audit, the recommended operation during unoccupied times of offices /GYM/changing rooms/ pantries / etc. is to run the cooling units at the temperature of 23°C during 20 minutes time slots (20 minutes ON and then 20 minutes OFF) during the night -between 12hAM and 6hAM when the outside RH is the

highest. This temperature control will help control the RH and keep it within acceptable levels for IAQ.

The below chart is a test example showing the impact of stopping the A/C on the indoor RH %. As seen in the chart below, the RH has increased, but only from 88% to 90% then dropped back to 82% immediately after the A/C was put in action at around 6hAM. Accordingly the idea of duty cycling would be then to run the A/C for 50% of the time between 10hPM and 6hAM .

Figure 88: Engineering offices RH, Co2, and Temperature readings



The below table provides the key indicators of this EEM with the mentioned CAPEX revolving around the installation of the multi channels time scheduler units . Accordingly one such unit can control up to 4 different independent channels.

| EEM - 01                                |      |                 |
|-----------------------------------------|------|-----------------|
| Cooling Control of Split Units and VRVs |      |                 |
| ENERGY SAVINGS                          |      |                 |
| ELECTRICITY                             | LPG  | CO <sub>2</sub> |
| KWh                                     | Kg   | Kg              |
| 122,587                                 | 0    | 32,890          |
| 1.6%                                    | 0.0% | 1.6%            |
| ECONOMIC PERFORMANCE                    |      |                 |
| Annual Cost Savings [\$]                |      | USD 38,860.14   |
| Annual Share of Cost Savings [%]        |      | 1.5%            |
| Implementation Cost [\$]                |      | USD 18,000.00   |
| Simple Payback [Years]                  |      | 0.5             |
| Net Present Value [\$]                  |      | USD 487,490.24  |
| Internal Rate of Return [%]             |      | 216%            |

## 7.2. EEM – 02: [LCNC] Guest Villas Management in Unrented Mode ■

### 7.2.1. Rationale

Guest Villas' cooling represent a large energy end use and are working continuously across the year 24/7 even in unrented times to ensure minimum acceptable Indoor Air Quality. The present EEM focuses on optimizing the cooling systems during unrented scenario.

### 7.2.2. Efficiency Action and Energy Savings

This EEM consists of managing all VRF systems across the guestrooms to ensure they are operated in an optimized way during unrented periods.

The clear actions to be taken are the following:

- i. Ensure all split and VRF units are set at 26°C during unrented mode. This is to be enforced by the housekeeping and engineering departments.
- ii. Ensure housekeeping is closing the blackout curtains to reduce the heat gains and air infiltration. As can be seen in the thermal image below , there are high heat gains inside the villa during direct solar radiations. In addition, air infiltration minimization will reduce the humidity levels in the indoor areas.

Figure 89: unrented Guestroom with open curtains



Figure 90: Thermal image showing heat gains from glazing



The below table provides the key indicators of this EEM.

| EEM - 02                                 |      |                 |
|------------------------------------------|------|-----------------|
| Guest Villas Management in Unrented Mode |      |                 |
| ENERGY SAVINGS                           |      |                 |
| ELECTRICITY                              | LPG  | CO <sub>2</sub> |
| KWh                                      | Kg   | Kg              |
| 109,781                                  | 0    | 29,454          |
| 1.4%                                     | 0.0% | 1.4%            |
| ECONOMIC PERFORMANCE                     |      |                 |
| Annual Cost Savings [\$]                 |      | USD 34,800.59   |
| Annual Share of Cost Savings [%]         |      | 1.3%            |
| Implementation Cost [\$]                 |      | USD 0.00        |
| Simple Payback [Years]                   |      | -               |
| Net Present Value [\$]                   |      | USD 452,683.85  |
| Internal Rate of Return [%]              |      | -               |

## 7.3. EEM – 03: [CAPEX] Lighting Retrofit ■

### 7.3.1. Rationale

The majority of the lighting in FSRMKH is composed of LED lighting fixtures. However, there are still large quantities of non-LED lights specifically compact fluorescent, incandescent, metal haloid, and halogen in some areas such as BOH, Guestrooms, Lobby, Café Huraa, and outdoor coconut tree lighting.

Waiting for a failed bulb to be replaced by LED is not an effective way as lighting retrofit would provide a payback period below 18 months in most cases, especially in times of increased electricity tariffs.

It is therefore recommended to allocate a budget to retrofit ALL non-LED lights at once and optimize the related electricity consumption the earliest possible.

### 7.3.2. Efficiency Action and Energy Savings

At the present electricity tariffs, there is no point to wait for a lighting bulb/fixture to fail before retrofitting it with an LED equivalent. The process of retrofitting should be immediate to all non LED lighting fixtures as the ROI is very good. Even if in some areas the fit-out is not completed or initiated, as long as these areas are used , their lighting should be changed.

The field audit inventoried around 1,923 non LED lighting fixtures. The figures below show the potential retrofit areas.

Figure 91: Lighting Retrofit potential areas



The table below details the non Led fixtures which need to be retrofitted along with their locations, quantities, loads, energy consumptions and expected savings post retrofits.

Table 23: Lighting Retrofit BOQ and cost

| Existing Fixture Name         | Retrofit Fixture Name | Quantity     | Unit Price      | Total Price      |
|-------------------------------|-----------------------|--------------|-----------------|------------------|
| Unit                          | Unit                  | Fixtures     | \$              | \$               |
| CFL BULB 11 W                 | LED 5W                | 484          | \$ 10           | \$ 4,840         |
| CFL BULB 5W                   | LED 2.5W              | 107          | \$ 4            | \$ 428           |
| CFL BULB CLC 18W              | LED 9W                | 58           | \$ 20           | \$ 1,160         |
| INCANDESCENT E10 5W           | LED 2.5W              | 204          | \$ 4            | \$ 816           |
| HALOGEN 50W                   | LED 25W               | 32           | \$ 30           | \$ 960           |
| HALOGEN 35W                   | LED 18W               | 128          | \$ 35           | \$ 4,480         |
| INCANDESCENT PINPONG BULB 15W | LED 7W                | 128          | \$ 8            | \$ 1,024         |
| INCANDESCENT PINPONG BULB 40W | LED 20W               | 34           | \$ 40           | \$ 1,360         |
| METAL HALOID 36W              | LED 18W               | 85           | \$ 35           | \$ 2,975         |
| PAR38 80W                     | LED 40W               | 83           | \$ 80           | \$ 6,640         |
| T5 8W                         | LED 4W                | 312          | \$ 7            | \$ 2,184         |
| T5 14W                        | LED 7W                | 268          | \$ 12           | \$ 3,216         |
| <b>Total</b>                  |                       | <b>1,923</b> | <b>\$ 285.0</b> | <b>\$ 30,083</b> |

Table 24: Lighting Retrofit expected savings

| Type         | Quantity     | Existing Energy | Retrofit Energy | Savings       | Cost Savings        |
|--------------|--------------|-----------------|-----------------|---------------|---------------------|
| Unit         | Lightbulbs   | KWh             | KWh             | KWh           | \$                  |
| CFL          | 649          | 8,243           | 3,805           | 4,438         | \$ 1,406.85         |
| INCANDESCENT | 366          | 10,557          | 5,418           | 5,139         | \$ 1,629.06         |
| HALOGEN      | 160          | 17,105          | 8,214           | 8,891         | \$ 2,818.45         |
| METALHALOIDE | 85           | 13,593          | 6,796           | 6,797         | \$ 2,154.65         |
| PAR 38       | 83           | 29,682          | 14,841          | 14,841        | \$ 4,704.60         |
| T5           | 580          | 6,687           | 3,344           | 3,343         | \$ 1,059.73         |
| <b>Total</b> | <b>1,923</b> | <b>85,867</b>   | <b>42,418</b>   | <b>43,449</b> | <b>\$ 13,773.33</b> |

All of these retrofits are very easy to be installed and will provide quick energy savings topping the 43,000 KWh annually with no additional maintenance costs. The below table provides the key indicators of this EEM.

| EEM - 03                         |      |                 |
|----------------------------------|------|-----------------|
| Lighting Retrofit                |      |                 |
| ENERGY SAVINGS                   |      |                 |
| ELECTRICITY                      | LPG  | CO <sub>2</sub> |
| KWh                              | Kg   | Kg              |
| 43,449                           | 0    | 11,657          |
| 0.6%                             | 0.0% | 0.6%            |
| ECONOMIC PERFORMANCE             |      |                 |
| Annual Cost Savings [\$]         |      | USD 13,773.33   |
| Annual Share of Cost Savings [%] |      | 0.5%            |
| Implementation Cost [\$]         |      | USD 30,083.00   |
| Simple Payback [Years]           |      | 2.2             |
| Net Present Value [\$]           |      | USD 149,079.64  |
| Internal Rate of Return [%]      |      | 46%             |

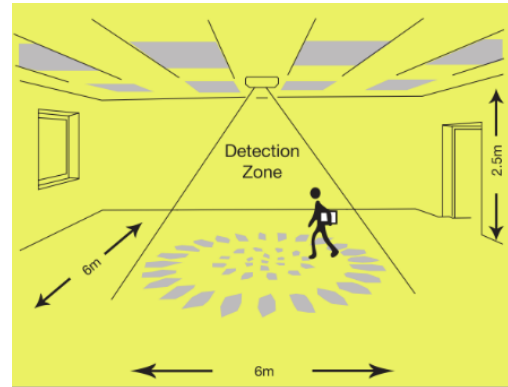
## 7.4. EEM – 04: [CAPEX] Lighting Control and Management

### 7.4.1. Rationale

Lighting Control's concept is based on managing lighting loads across various areas of a property through human manual control or through the introduction of specific technologies. It provides a major opportunity for any facility to reduce its overall lighting energy consumption. Lighting control examples are Time control, Daylight control and Occupancy control. The recommended control strategy in FSRMKH is motion detectors equipped with daylight controllers.

### 7.4.2. Efficiency Action and Energy Savings

To optimize the lighting control in FSRMKH, the strategy would be to install Motion Detectors or to enforce manual control in some areas. Motion detectors are devices that detect movement within their field of view. They work by sensing changes in the infrared radiation emitted by objects in their environment. This technology is commonly used in security systems, outdoor lighting, and automatic doors. The plan to include the following:



- i. Coordinate with F&B, Housekeeping and Engineering to ensure all rooms have their lighting controlled across the days.
- ii. Review potential Motion Detectors installations in: housekeeping pantries across the floors, and public bathrooms/ changing rooms.

In the case of FSRMKH, these motion detectors can be equipped with daylight controllers and can be used in public toilets, and changing rooms where lighting was seen to be operating at all times even when the space was vacant. A daylight controller (photocell) continuously measures the ambient light levels and is able to shut down /turn ON the lighting based on the day/night hours. This technology can be used to control all the lighting located in indoor spaces where daylight is important which helps in reducing the electrical energy consumption.

Figure 92: Motion Detector Potential





- Top Row: Lobby bathrooms, male and female
- Second Row: GYM changing room (left), pantries (Right)

The table below shows the potential MDs quantity and locations, in addition to the potential savings resulting.

Table 25: MD savings

| TYPE              | LOCATION               | LOAD        | QUANTITY  | TOTAL LOAD  | ANNUAL Energy Consumption | Number of MDs | Savings      | Cost Savings |
|-------------------|------------------------|-------------|-----------|-------------|---------------------------|---------------|--------------|--------------|
| UNIT              |                        | KW          |           | KW          | KWh                       | #             | KWh          | \$           |
| MR16 LED          | Changing Room - gym    | 0.01        | 30        | 0.20        | 1,708                     | 2             | 598          | 190          |
| MR16 LED          | Public toilet at lobby | 0.01        | 22        | 0.14        | 1,253                     | 2             | 439          | 139          |
| Fluorescent 2x36w | Pantries               | 0.07        | 10        | 3.52        | 10,698                    | 5             | 3,744        | 1,187        |
| LED 12W           | Pantries               | 0.01        | 10        | 3.52        | 12,845                    | 5             | 4,496        | 1,425        |
| <b>Total</b>      |                        | <b>0.10</b> | <b>72</b> | <b>7.38</b> | <b>26,504</b>             | <b>14</b>     | <b>9,276</b> | <b>2,941</b> |

The below table provides the key indicators of the EEM whereby the CAPEX represents the installation of 14 of Motion Detectors equipped with photosensors across the areas mentioned in the table above (public bathrooms, changing rooms, pantries).

| EEM - 04                         |      |                 |
|----------------------------------|------|-----------------|
| Installation of Motion Detectors |      |                 |
| ENERGY SAVINGS                   |      |                 |
| ELECTRICITY                      | LPG  | CO <sub>2</sub> |
| KWh                              | Kg   | Kg              |
| 9,276                            | 0    | 2,489           |
| 0.1%                             | 0.0% | 0.1%            |
| ECONOMIC PERFORMANCE             |      |                 |
| Annual Cost Savings [\$]         |      | USD 2,940.59    |
| Annual Share of Cost Savings [%] |      | 0.1%            |
| Implementation Cost [\$]         |      | USD 2,110.00    |
| Simple Payback [Years]           |      | 0.7             |
| Net Present Value [\$]           |      | USD 36,140.98   |
| Internal Rate of Return [%]      |      | 139%            |

## 7.5. EEM- 05: [CAPEX] VFD Installation on STP Blowers

### 7.5.1. Rationale

FSRMKH has 3x Sewage Treatment plants for treating water used for irrigation or before sending it back to the ocean, these STPs are equipped with newly installed 2x16KW blower pumps.

The present EEM is to install VFDs on these pumps which will allow implementing the proactive speed control bringing direct savings on their electricity consumption.

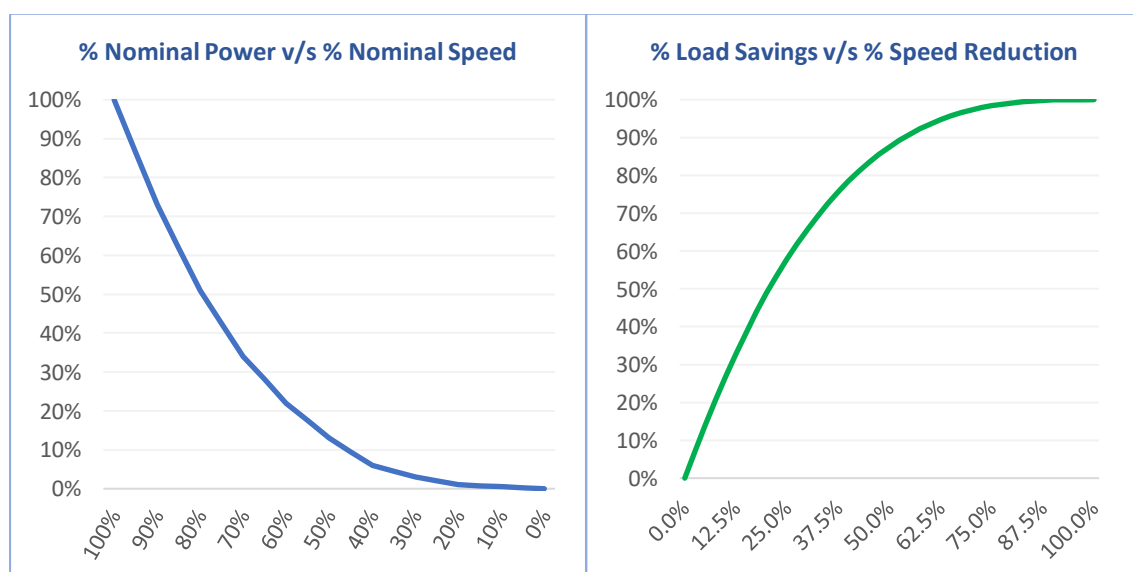
Figure 93:STP Blower Pump



### 7.5.2. Efficiency Actions and Energy savings

Obviously, the usage of VFDs will allow savings substantial energy used (if the speed reduction is well implemented in a PID/Automatic look or proactively changed manually or via a schedule time setting). A 10% speed reduction would lead to a 27% savings on the Power used. VFDs on pumps can be implemented on a pressure differential basis obtained from supply and return lines.

Figure 94: Power and Speed Variations



The variable parameter that would control the VFD on a blower in a sewage treatment plant is the airflow or the amount of air required in the aeration basin to maintain the dissolved oxygen (DO) concentration at a desired level.

The VFD controls the speed of the blower motor, which in turn controls the amount of air that is delivered to the aeration basin. By varying the speed of the motor, the VFD can adjust the airflow to match the changing demands of the plant, such as changes in influent flow rate, changes in organic loading, or changes in wastewater temperature.

To maintain the DO level in the aeration basin, the VFD will adjust the blower speed to increase or decrease the amount of air delivered to the basin. As the organic loading or flow rate increases, the VFD will increase the blower speed to deliver more air to the basin and maintain the DO level. Conversely, as the organic loading or flow rate decreases, the VFD will decrease the blower speed to reduce the amount of air delivered to the basin and maintain the DO level. The table below presents the potential savings obtained from the installation of VFDs on the blower pumps.

Table 26: VFD savings

| TYPE             | LOCATION       | LOAD | QUANTITY | TOTAL LOAD | Consumption | Overall Average Speed Reduction | Load/Energy Reduction | Savings | Cost Savings |
|------------------|----------------|------|----------|------------|-------------|---------------------------------|-----------------------|---------|--------------|
| unit             |                | KW   |          | KW         | KWh         | %                               | %                     | KWh     | USD          |
| STP BLOWER PUMPS | Admin Building | 16   | 2        | 32         | 42,048      | 15%                             | 40%                   | 16,819  | 5,332        |

Based on an average reduction of 15% of the speed of controlled pumps, the estimated energy savings can be calculated as per the below table.

| EEM - 05                                    |      |                 |
|---------------------------------------------|------|-----------------|
| Installation of VFDs on the new STP blowers |      |                 |
| ELECTRICITY                                 | LPG  | CO <sub>2</sub> |
| KWh                                         | Kg   | Kg              |
| 16,819                                      | 0    | 4,513           |
| 0.2%                                        | 0.0% | 0.2%            |
| ECONOMIC PERFORMANCE                        |      |                 |
| Annual Cost Savings [\$]                    |      | USD 5,331.69    |
| Annual Share of Cost Savings [%]            |      | 0.2%            |
| Implementation Cost [\$]                    |      | USD 10,200.00   |
| Simple Payback [Years]                      |      | 1.9             |
| Net Present Value [\$]                      |      | USD 59,154.24   |
| Internal Rate of Return [%]                 |      | 52%             |

## 7.6. EEM – 06: [CAPEX] Installation of VFDs on kitchen Hoods

### 7.6.1. Rationale

FSRMKH has 2 main kitchens (Reef Club kitchen, in addition to the Main kitchen) which have a number of kitchen hoods (fresh air and exhaust fans) of which only one has a VFD installed for noise control. This EEM consists of expanding the installation of VFDs on all the remaining kitchen hoods which will allow implementing the proactive speed control bringing direct savings on their electricity consumption.

### 7.6.2. Efficiency Action and Energy Savings

Obviously, the usage of VFDs will allow savings substantial energy used (if the speed reduction is well implemented in a PID/Automatic look or proactively changed manually or via a schedule time setting). A 10% speed reduction would lead to a 27% savings on the Power used.

VFDs (Variable Frequency Drives) on kitchen hoods work by controlling the speed of the exhaust fan motor in response to the cooking load. This means that the fan motor will only work as hard as it needs to, rather than constantly running at full speed.

When a kitchen hood equipped with a VFD is turned on, the VFD will detect the level of heat and smoke being generated by the cooking equipment. Based on this information, the VFD will adjust the fan motor's speed to create the ideal balance of air flow and suction to remove the smoke and fumes effectively.

Figure 95: Sensor on kitchen hood equipped with VFD



By adjusting the fan speed in response to the cooking load, VFDs can save energy by reducing the fan's power consumption when less air flow is required. This not only reduces energy costs but also prolongs the lifespan of the fan motor, reducing maintenance and replacement costs.

The table below shows the list of AHUs in the FSRMKH that need installation of VFDs, in addition to the savings.

Table 27: List of Kitchen Hoods and annual KWh consumption and potential savings

| NAME                | SUB CATEGORY | LOCATION            | LOAD      | QUANTITY  | TOTAL LOAD | ANNUAL         | Overall Average Speed Reduction | Load/Energy Reduction | Savings       | Cost Savings  |
|---------------------|--------------|---------------------|-----------|-----------|------------|----------------|---------------------------------|-----------------------|---------------|---------------|
| Unit                |              |                     | KW        |           | KW         | KWh            | %                               | %                     | KWh           | USD           |
| Exhaust (EF-RC/1)   | EF           | Reef Club           | 3.0       | 1         | 3          | 5,307          | 10%                             | 27%                   | 1,433         | 454           |
| Fresh Air(SF-RK/1)  | FAF          | Reef Club           | 1.4       | 1         | 1          | 2,459          | 10%                             | 27%                   | 664           | 210           |
| Exhaust (EF-MK/1)   | EF           | Main Kitchen        | 7.5       | 1         | 7          | 33,571         | 10%                             | 27%                   | 9,064         | 2,873         |
| Exhaust (EF-MK/2)   | EF           | Main Kitchen        | 7.5       | 1         | 7          | 33,536         | 10%                             | 27%                   | 9,055         | 2,870         |
| Exhaust (EF-MK/3)   | EF           | Main Kitchen        | 5.6       | 1         | 6          | 24,782         | 10%                             | 27%                   | 6,691         | 2,121         |
| Fresh Air (SF-MK/1) | FAF          | Main Kitchen        | 5.6       | 1         | 6          | 24,725         | 10%                             | 27%                   | 6,676         | 2,116         |
| Fresh Air (SF-MK/2) | FAF          | Main Kitchen        | 4.1       | 1         | 4          | 18,102         | 10%                             | 27%                   | 4,888         | 1,549         |
| Fresh Air (SF-MK/3) | EF           | Main Kitchen        | 4.0       | 1         | 4          | 17,660         | 10%                             | 27%                   | 4,768         | 1,512         |
| Exhaust (EF-BB/1)   | EF           | Barabaru Kitchen    | 7.5       | 1         | 7          | 19,369         | 10%                             | 27%                   | 5,230         | 1,658         |
| Exhaust (EF-KG/1)   | EF           | Kandu Grill Kitchen | 11.0      | 1         | 11         | 28,821         | 10%                             | 27%                   | 7,782         | 2,467         |
| <b>Total</b>        |              |                     | <b>57</b> | <b>10</b> | <b>57</b>  | <b>208,332</b> |                                 |                       | <b>56,250</b> | <b>17,831</b> |

The below Table shows the key indicators of this EEM.

| EEM - 06                              |      |                 |
|---------------------------------------|------|-----------------|
| Installation of VFDs on Kitchen Hoods |      |                 |
| ELECTRICITY                           | LPG  | CO <sub>2</sub> |
| KWh                                   | Kg   | Kg              |
| 56,250                                | 0    | 15,092          |
| 0.7%                                  | 0.0% | 0.7%            |
| ECONOMIC PERFORMANCE                  |      |                 |
| Annual Cost Savings [\$]              |      | USD 17,831.10   |
| Annual Share of Cost Savings [%]      |      | 0.7%            |
| Implementation Cost [\$]              |      | USD 22,490.00   |
| Simple Payback [Years]                |      | 1.3             |
| Net Present Value [\$]                |      | USD 209,455.76  |
| Internal Rate of Return [%]           |      | 79%             |

## 7.7. EEM – 07 [CAPEX]: Waste Heat Recovery for Hot Water Production (Jacket Heat Recovery) ■

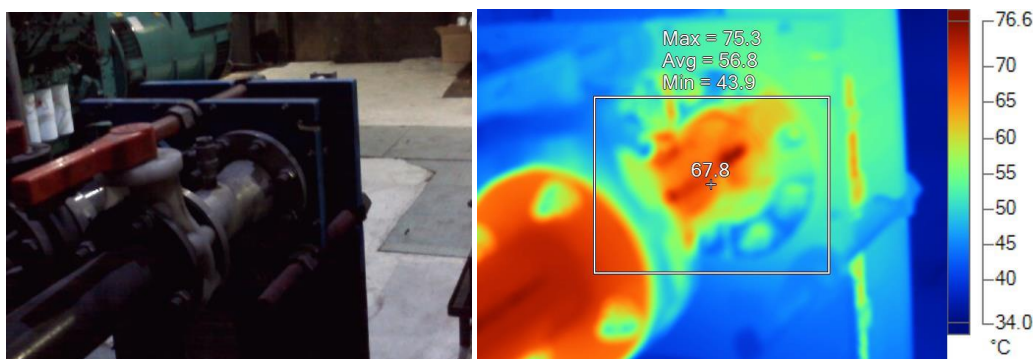
### 7.7.1. Rationale

As previously mentioned, FSRMKH has 5 diesel generators: 2x Cummins KTA 38 – G5 1000 kVA and 3x Cummins KTA 19 - G 3 450 kVA, working based on the following schedule and are operated based on a synchronization panel:

- 1x1000 KVA from 1AM to 6AM
- 1X1000 KVA plus 1X450 KVA from 6 AM to 8:30 AM
- 2X1000 KVA from 8:30 AM to 1AM.

In principle, diesel generators use only about 35% of their energy input (diesel) as this is turned into electricity production with the remaining 65% lost in between exhaust gases and jacket cooling. As shown in the thermal picture below, temperature at the heat exchanger is around 75 °C; therefore, there is a large opportunity for a Waste Heat Recovery project that would focus on recuperating wasted energy from the generators coolant and use it for hot water production.

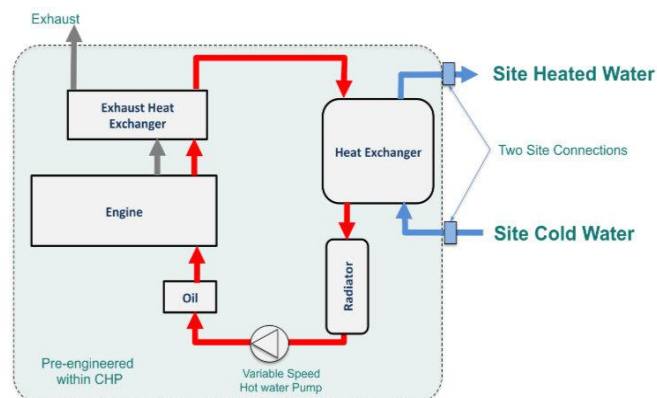
Figure 96: Genset room exhaust (left), and corresponding thermal image (right)



### 7.7.2. Proposed energy efficiency action

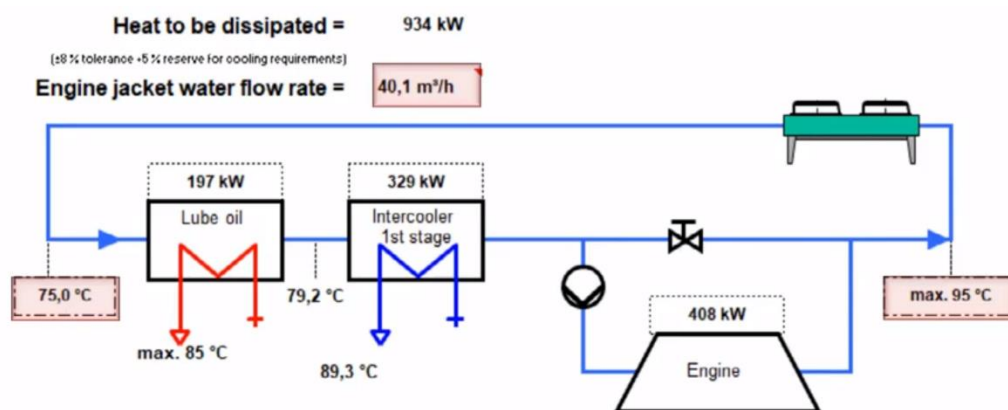
The simplest heat recovery application/method is to use the generators' cooling jackets/radiators for the heat recovery. The system will produce hot water that will be used in the production of hot water. The below figure demonstrates the discussed process.

Figure 97: Jacket Water Heat Recovery



As can be seen in the below figure, the below data is needed for the engine to be able to calculate the potential energy and fuel savings.

Figure 98: Data needed to calculate the jacket heat recovery savings (example of 1.5MW engine)



The heat rejected to the coolant jacket is found to be 539kW<sub>thermal</sub> and 247 KW<sub>thermal</sub> at full load (from generator data sheet) for the 1000 kVA engine and 450 KVA engines respectively. However, the heat rejected to the jacket water coolant is directly related to the load on the generator, therefore, the monthly heat rejected is multiplied by the monthly percentage load on the generator. Depending on the makeup water temperature, the targeted hot water temperature and the flow, a calculation of the savings can be done.

At the present time, FSRMKH is relying on electric hot water tanks for domestic hot water. The potential electric and thermal energy savings that can be achieved from the heat recovery system from the gensets are shown in the table below.

Table 28: WHR – 1,000 KVA gensets

| Month        | Gensets run time 1000 KVA | Heat Rejected to Jacket Water Coolant | Heat Recovered from Jacket Water Coolant | Electrical Energy Equivalence | Potential Electricity Savings |
|--------------|---------------------------|---------------------------------------|------------------------------------------|-------------------------------|-------------------------------|
| Unit         | Hours                     | kWh thermal                           | kwh thermal                              | kWhe                          | \$                            |
| January      | 1,544                     | 482,529                               | 313,644                                  | 313,644                       | \$ 99,425                     |
| February     | 1,168                     | 365,109                               | 237,321                                  | 237,321                       | \$ 75,231                     |
| March        | 1,019                     | 318,560                               | 207,064                                  | 207,064                       | \$ 65,639                     |
| April        | 1,817                     | 568,031                               | 369,220                                  | 369,220                       | \$ 117,043                    |
| May          | 1,004                     | 313,870                               | 204,016                                  | 204,016                       | \$ 64,673                     |
| June         | 1,284                     | 401,404                               | 260,913                                  | 260,913                       | \$ 82,709                     |
| July         | 1,002                     | 313,245                               | 203,609                                  | 203,609                       | \$ 64,544                     |
| August       | 1,018                     | 318,372                               | 206,942                                  | 206,942                       | \$ 65,601                     |
| September    | 1,283                     | 401,091                               | 260,709                                  | 260,709                       | \$ 82,645                     |
| October      | 1,330                     | 415,785                               | 270,260                                  | 270,260                       | \$ 85,672                     |
| November     | 1,513                     | 472,994                               | 307,446                                  | 307,446                       | \$ 97,460                     |
| December     | 1,008                     | 315,121                               | 204,829                                  | 204,829                       | \$ 64,931                     |
| <b>Total</b> | <b>14,990</b>             | <b>4,686,111</b>                      | <b>3,045,972</b>                         | <b>3,045,972</b>              | <b>\$ 965,573</b>             |

Table 29:WHR – 450 KVA gensets

| Month        | Gensets run time 450 KVA | Heat Rejected to Jacket Water Coolant | Heat Recovered from Jacket Water Coolant | Electrical Energy Equivalence | Potential Electricity Savings |
|--------------|--------------------------|---------------------------------------|------------------------------------------|-------------------------------|-------------------------------|
| Unit         | Hours                    | kWh thermal                           | kwh thermal                              | kWhe                          | \$                            |
| January      | 1,004                    | 131,434                               | 98,575                                   | 98,575                        | \$ 31,248                     |
| February     | 521                      | 68,256                                | 51,192                                   | 51,192                        | \$ 16,228                     |
| March        | 676                      | 88,495                                | 66,371                                   | 66,371                        | \$ 21,040                     |
| April        | 686                      | 89,804                                | 67,353                                   | 67,353                        | \$ 21,351                     |
| May          | 195                      | 25,462                                | 19,096                                   | 19,096                        | \$ 6,054                      |
| June         | 501                      | 65,586                                | 49,189                                   | 49,189                        | \$ 15,593                     |
| July         | 171                      | 22,320                                | 16,740                                   | 16,740                        | \$ 5,307                      |
| August       | 738                      | 96,612                                | 72,459                                   | 72,459                        | \$ 22,969                     |
| September    | 500                      | 65,507                                | 49,131                                   | 49,131                        | \$ 15,574                     |
| October      | 499                      | 65,376                                | 49,032                                   | 49,032                        | \$ 15,543                     |
| November     | 65                       | 8,509                                 | 6,382                                    | 6,382                         | \$ 2,023                      |
| December     | 529                      | 69,251                                | 51,939                                   | 51,939                        | \$ 16,465                     |
| <b>Total</b> | <b>6,085</b>             | <b>796,614</b>                        | <b>597,460</b>                           | <b>597,460</b>                | <b>\$ 189,395</b>             |

As can be seen in the tables above, the total potential recovered heat possible from the jacket water coolant is 3,643,432kWh<sub>th</sub> equivalent to 3,643,432 KWh electric which is much larger than the existing hot water demand .

The heat recovered would provide savings through removing all the electrical demands of all hot water tanks and potentially BOH services . As the potential production of heat is larger than the consumption , it can be considered that a 100% savings can be achieved on the targeted units and services.

The targeted equipment for the new hot water supply are as follows:

- All Electrical Hot Water units serving the staff accommodation, the main kitchen, the laundry , the canteen . Pushing further to areas such as the Gym and main pool is possible if the piping layout is easy.
- Washing machines in the laundry can have their water inlet supplied directly by hot water
- Dishwashers in the main kitchen can also have their water supply taken from the new hot water.

The heat recovery system will include the following components:

- One 4m<sup>3</sup> buffer tank (or 2x3m<sup>3</sup> tanks)
- One Heat Exchanger of the same size of the existing one for every generator used for the radiator cooling of the engines. The new heat exchanger will be installed in series with the existing one. Its primary side will be the radiator line while its secondary side will be the line connected to the above buffer tanks. The primary side line continues to the existing heat exchanger for further needed cooling.
- One pump with VFD circulating water between the buffer tank(s) and the new heat exchanger
- All pipes from the heat exchanger to the buffer tanks and from the buffer tanks to all the main cold supply line of the targeted hot water tanks or services
- Control system

The below table shows the key financial indicators for the above measure. The savings are equal to the current demand that will be covered minus the electrical consumption of the Hot water pump that will be added with the system. The mentioned CAPEX includes all the procurement and installation of the above components.

| EEM - 07                             |      |                 |
|--------------------------------------|------|-----------------|
| Jacket Heat Recovery from Generators |      |                 |
| ENERGY SAVINGS                       |      |                 |
| ELECTRICITY                          | LPG  | CO <sub>2</sub> |
| KWh                                  | Kg   | Kg              |
| 270,252                              | 0    | 72,509          |
| 3.6%                                 | 0.0% | 3.5%            |
| ECONOMIC PERFORMANCE                 |      |                 |
| Annual Cost Savings [\$]             |      | USD 85,669.86   |
| Annual Share of Cost Savings [%]     |      | 3.3%            |
| Implementation Cost [\$]             |      | USD 150,000     |
| Simple Payback [Years]               |      | 1.8             |
| Net Present Value [\$]               |      | USD 964,388.08  |
| Internal Rate of Return [%]          |      | 57%             |

## 7.8. EEM – 07 [CAPEX]: Energy Management/Building Management System ■

### 7.8.1. Rationale

A property of the size of FSRMKH along the exiting number of decentralized equipment should have as a priority a well-tuned, appropriately scaled Energy Management and Control System. At the present time the property has nearly no visibility on equipment's operation and virtually no proactive control which limits substantially the energy usage optimization possibilities. In addition, the key loads which are split units, VRF units, pool pumps and kitchen fans are all manually operated – or with fixed timers with potential human error and wastage. A small scale BMS will help FSRMKH management to accurately track and control all key loads in the premises for optimal energy usage.

### 7.8.2. EMS/BMS application

The facility has to plan for a small scale BMS/EMS that would focus on the following points:

- **Metering:** To connect all existing and future electrical and water meters to the new system which will allow automatic readings, trending and reporting. Currently the property is relying on manual readings of existing meters. Electrical meters that are currently without communication board to be gradually replaced with a Modbus connection for EMS/BMS connections.
- **Generators:** Connect the system to generators' synchro panel and operations to allow feeders reading , generators' loads, operating generators and all other key parameters to be visualized , recorded and analyzed .
- **RO Plant :** Connect the system to each RO line to monitor produced water and related KWh consumption for a continuous efficiency/performance check
- **STP:** Connect the key STP equipment status to the EMS/BMS
- **Café Huraa/Main Pool & Reef Club/Pool :** Connect the system to the two main pools pumps

The system can start small and then be expanded with time . The CAPEX needed for a system that would include the needed gateways/DDCs/Cabling and Platform would be around \$30,000.

The EEM doesn't provide direct energy savings but will allow a better energy analysis that would lead to optimize the energy efficiency of the property.

## 7.9. EEM – 08 [CAPEX]: Photovoltaic Driven Electricity Generation

### 7.9.1. Rationale

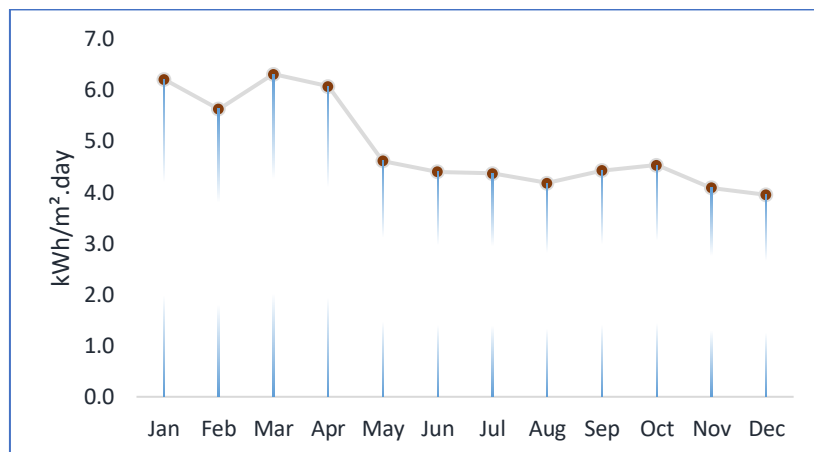
This EEM studies the potential areas in which solar panels can be installed. The recommended solution will be an online PV Power Plant without batteries whereby produced energy is used instantaneously by the property by integrating it into the present main electrical panels. Potential areas are delineated as per the figure below with an approximate area of 2,950 m<sup>2</sup>.

Figure 99: FSRMKH BOH's Roofs with delineated potential PV areas



The graph below shows the potential monthly solar radiation in Maldives in KWh/m<sup>2</sup>/day.

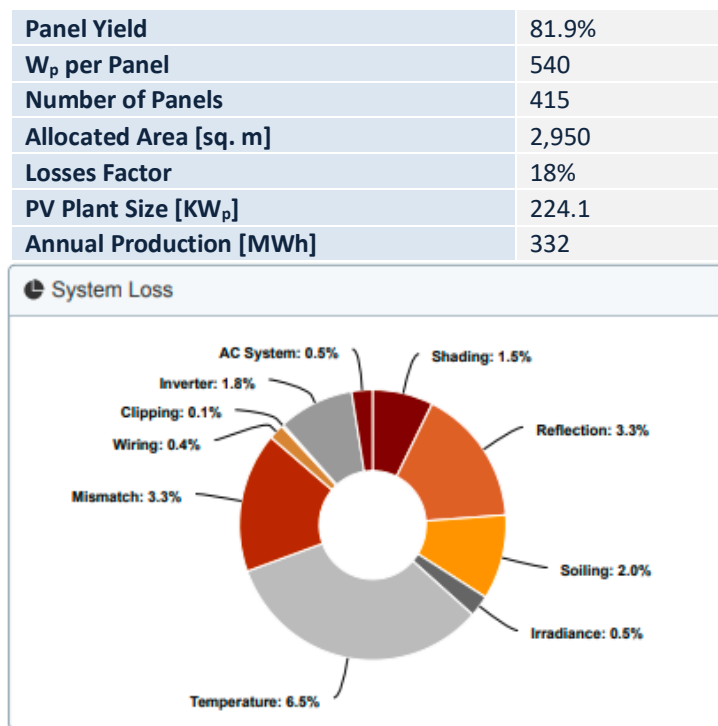
Figure 100: Monthly solar radiation – Kuda Huraa



### 7.9.2. Proposed energy efficiency action:

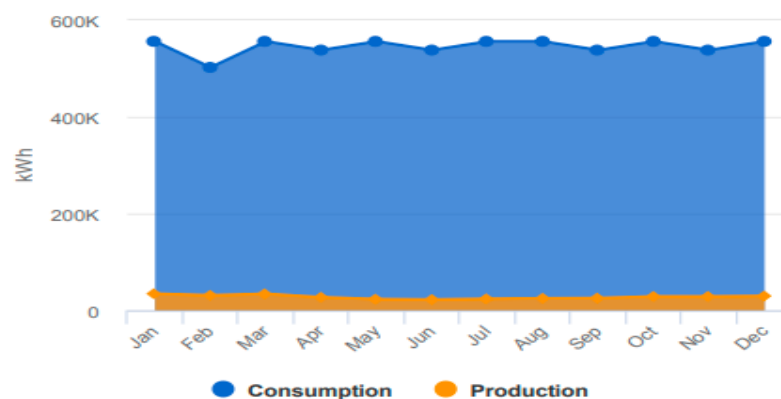
The present section covers the production of electricity via a PV plant to be installed on the roof of the available spaces mentioned earlier. The recommended solution will be an online PV Power Plant without batteries whereby produced energy is used instantaneously by the property by integrating it into the present main electrical panels. The key plant parameters from allocated panels, number of modules, performance ratio, losses, and overall plant size are shown in the following table and figure below.

Table 30: PV Plant Parameters (Up), and system loss (right)



The savings calculations are based on the available solar irradiance in Kuda Huraa along with the number of panels installed which would give the potential electricity produced (taking into consideration the efficiency of the panels). The below graph provides the total monthly electricity produced by the described PV plant presenting around 4.4% of the total consumption at FSMKH.

Figure 101: PV Electricity Production



The below table provides the total monthly electricity and cost savings by the described PV plant.

Table 31: Solar PV Energy Savings

| Month        | Actual Energy Consumption | PV Energy Produced | New Energy Consumption | Energy Cost Savings |             |
|--------------|---------------------------|--------------------|------------------------|---------------------|-------------|
| Unit         | kWh                       | kWh                | kWh                    | USD                 | %           |
| Jan          | 714,310                   | 34,325             | 679,985                | USD 10,915          | 4.8%        |
| Feb          | 654,040                   | 30,960             | 623,080                | USD 9,845           | 4.7%        |
| Mar          | 729,060                   | 34,057             | 695,003                | USD 10,830          | 4.7%        |
| Apr          | 730,070                   | 26,920             | 703,150                | USD 8,561           | 3.7%        |
| May          | 710,031                   | 23,090             | 686,941                | USD 7,343           | 3.3%        |
| Jun          | 625,150                   | 22,237             | 602,913                | USD 7,071           | 3.6%        |
| Jul          | 636,450                   | 23,707             | 612,743                | USD 7,539           | 3.7%        |
| Aug          | 599,420                   | 24,799             | 574,621                | USD 7,886           | 4.1%        |
| Sep          | 531,680                   | 25,177             | 506,503                | USD 8,006           | 4.7%        |
| Oct          | 553,300                   | 28,637             | 524,663                | USD 9,107           | 5.2%        |
| Nov          | 542,200                   | 28,578             | 513,622                | USD 9,088           | 5.3%        |
| Dec          | 556,200                   | 29,645             | 526,555                | USD 9,427           | 5.3%        |
| <b>Total</b> | <b>7,581,911</b>          | <b>332,132</b>     | <b>7,249,779</b>       | <b>USD 105,618</b>  | <b>4.4%</b> |
| Maximum      | 753,241 (Jul)             | 34,325 ( )         | 703,150 ( )            | \$10,915 ( )        | 5.3% ( )    |
| Average      | 666,458                   | 27,678             | 604,148                | \$8,801             | 4.4%        |
| Minimum      | 501,925 (Feb)             | 22,237 ( )         | 506,503 ( )            | \$7,071 ( )         | 3.3% ( )    |

### Needed Installation

The overall PV project includes the following parts/components:

- 415 PV modules each of 540 Wp of Efficiency > 81.9 % at STC, total system power of 224.1 KWp.
- 8 Inverters & Related Junction Boxes.
- Mounted Steel Structure and roofing system to support all panels
- Cabling and adequate accessories and cable trays.
- Provision of a room to install the inverters.

The table below details the key financial indicators of this EEM.

| EEM - 08                                   |      |                  |
|--------------------------------------------|------|------------------|
| Photovoltaic Driven Electricity Generation |      |                  |
| ENERGY SAVINGS                             |      |                  |
| ELECTRICITY                                | LPG  | CO <sub>2</sub>  |
| KWh                                        | Kg   | Kg               |
| 332,132                                    | 0    | <b>89,111</b>    |
| 4.4%                                       | 0.0% | <b>4.4%</b>      |
| ECONOMIC PERFORMANCE                       |      |                  |
| Annual Cost Savings [\$]                   |      | USD 105,285.84   |
| Annual Share of Cost Savings [%]           |      | 4.0%             |
| Implementation Cost [\$]                   |      | USD 246,510.0    |
| Simple Payback [Years]                     |      | 2.3              |
| Net Present Value [\$]                     |      | USD 1,123,041.57 |
| Internal Rate of Return [%]                |      | 43%              |

## 8. EEMs Summary Figures & Implementation Notes

### 8.1. EEMs Summary – CAPEX, Cost Savings & Payback Period

The following table provides the summary of the Energy Efficiency Measures (EEMs) with key indicators including CAPEX, % Savings from Total Utilities Cost and Payback period. As can be seen, if all EEMs are implemented, the total annual cost savings would be **\$304,493** at a CAPEX of **\$509,393** and a payback of **1.67 Years**.

Table 32: EEMs summary – CAPEX, Cost savings and Payback Period

| CODE             | EEM Name                                    | ANNUAL SAVINGS              |                     | COST                | PAYBACK     |
|------------------|---------------------------------------------|-----------------------------|---------------------|---------------------|-------------|
|                  |                                             | \$                          | KGs CO <sub>2</sub> | \$                  | Years       |
| -                | -                                           |                             |                     |                     |             |
| EEM - 01 [LCNC]  | Cooling Control of Split Units and VRVs     | \$38,860.14 (1.5%)          | 32,890              | \$18,000.00         | 0.46        |
| EEM - 02 [LCNC]  | Guest Villas Management in Unrented Mode    | \$34,800.59 (1.3%)          | 29,454              | \$0.00              | -           |
| EEM - 03 [CAPEX] | Lighting Retrofit                           | \$13,773.33 (0.5%)          | 11,657              | \$30,083.00         | 2.18        |
| EEM - 04 [CAPEX] | Installation of Motion Detectors            | \$2,940.59 (0.1%)           | 2,489               | \$2,110.00          | 0.72        |
| EEM - 05 [CAPEX] | Installation of VFDs on the new STP blowers | \$5,331.69 (0.2%)           | 4,513               | \$10,200.00         | 1.91        |
| EEM - 06 [CAPEX] | Installation of VFDs on Kitchen Hoods       | \$17,831.10 (0.7%)          | 15,092              | \$22,490.00         | 1.26        |
| EEM - 07 [CAPEX] | Jacket Heat Recovery from Generators        | \$85,669.86 (3.3%)          | 72,509              | \$150,000.00        | 1.75        |
| EEM - 08 [CAPEX] | Installation of Metering System             | \$0.00 (0.0%)               | 0                   | \$30,000.00         | 0.00        |
| EEM - 09 [CAPEX] | Photovoltaic Driven Electricity Generation  | \$105,285.84 (4.0%)         | 89,111              | \$246,510.00        | 2.34        |
|                  | <b>TOTAL *</b>                              | <b>\$304,493.14 (11.6%)</b> | <b>257,715</b>      | <b>\$509,393.00</b> | <b>1.67</b> |

## 8.2. EEMs Summary – Electricity (KWh) Savings

The below table provides the annual savings of every EEM and for each utility. Electricity savings are seen to be **960,546 KWh** per year representing **12.7%** of the actual electricity consumption.

Table 33:EEM Summary – Electricity (KWh) Savings

| CODE     | EEM Name                                    | ELECTRICITY          |
|----------|---------------------------------------------|----------------------|
| -        | -                                           | KWh                  |
| EEM - 01 | Cooling Control of Split Units and VRVs     | 122,587 (1.6%)       |
| EEM - 02 | Guest Villas Management in Unrented Mode    | 109,781 (1.4%)       |
| EEM - 03 | Lighting Retrofit                           | 43,449 (0.6%)        |
| EEM - 04 | Installation of Motion Detectors            | 9,276 (0.1%)         |
| EEM - 05 | Installation of VFDs on the new STP blowers | 16,819 (0.2%)        |
| EEM - 06 | Installation of VFDs on Kitchen Hoods       | 56,250 (0.7%)        |
| EEM - 07 | Jacket Heat Recovery from Generators        | 270,252 (3.6%)       |
| EEM - 08 | Installation of Metering System             | 0 (0.0%)             |
| EEM - 09 | Photovoltaic Driven Electricity Generation  | 332,132 (4.4%)       |
|          | <b>TOTAL</b>                                | <b>960,546 12.7%</b> |

## 9. Water Conservation Measures

The main objective of this section is to provide a roadmap that would include a number of Water Conservation Measure (WCM) which are all the actions that would reduce the water consumption. In this aim, EEG investigated a considerable number of water efficiency solutions, which could yield to substantial energy cost and water footprint reductions in the event of their implementation. These opportunities include those who require very little or no CAPEX at all – Low Cost/No Cost – [LC/NC] as well as those needing capital expenditure [CAPEX]. The assessment of an WCM is its financial feasibility and technical viability. Each WCM is attributed with a color code which is a priority or urgency indicator for the WCM's implementation. The color codes adopted in this report along the proposed measures are found in the below tables:

Table 34: Water Measures Color Codes

| Color Code | Description                                                                       | Priority |
|------------|-----------------------------------------------------------------------------------|----------|
|            | Measures with very small CAPEX and excellent payback (<1 year) – Low Cost/No Cost | A        |
|            | Measures with good payback period (<3 years)– Immediate Implementation            | A        |
|            | Measures with midterm payback period (3 to 6 Years) – Strategic Implementation    | B        |
|            | Measures to be done during a planned retrofit/replacement                         | C        |
|            | Measures with high payback period (>6 years) – Upon CAPEX Availability            | D        |

Table 35: Water Conservation Measures

| CODE     | TYPE  | TITLE                                                             |
|----------|-------|-------------------------------------------------------------------|
| EEM - 01 | CAPEX | Rainwater harvesting                                              |
| EEM - 02 | CAPEX | Installation of Water Leakage System                              |
| EEM - 03 | CAPEX | Change Aerators in Guestrooms and Public Toilet and Water Control |

## 9.1. WCM – 01: [CAPEX] Rainwater Harvesting

### 9.1.1. Rationale

FSRMKH has an overall water tariff of \$0.88/m<sup>3</sup>. On the other hand, FSRMKH located in Maldives which has a humid tropical monsoon climate with high precipitations exceeding 1,818 mm per year. On average, the island experiences around 140 to 150 rainy days per year. The driest month (February) has an average rainfall of less than 43mm, equivalent to 43L/m<sup>2</sup> while the wettest month is May at 234 L/m<sup>2</sup>. Recuperating part of the rainwater will help reduce the freshwater supplied from the RO plant and the related charges. The below chart provides the precipitation probability in Maldives.

Figure 102: Daily Chance of Precipitation in Maldives Kuda Huraa



### 9.1.2. Proposed water conservation action

Rainwater harvesting consists of collecting rainwater from roof surfaces (potential area delineated in the Figure below) and storing it for multipurpose usage since it is considered as pure fresh water. The area delineated below have a total of 2,850 m<sup>2</sup>.

Figure 103: Potential area for rain water harvesting



FSMRKH has already a complete drainage network including the rainwater drainage network. If the facility intends to implement rainwater harvesting project it will consist on connecting the rain water pipes to decentralized storage tanks and then pumping fresh water to a centralized tank from where rain water will be distributed to end users after a minimal filtration process.

Figure 104: Existing Rain water drainage piping



As mentioned, rainwater harvesting offers free fresh water ready to be used directly with minimal treatment, saving therefore are the fresh water volumes treated in the RO plant. Overall, and based on the average monthly rainfall, it is estimated to save 3,982 m<sup>3</sup> of water annually with the below table providing the key indicators of the project based on the water tariff at \$3.15/m<sup>3</sup> including the electricity cost.

The CAPEX in this measure would be the cost of all decentralized storage tanks, 1 pumps, central rain water tank, filter, pipes, valves and all related accessories.

| WCM - 01                         |                |                 |
|----------------------------------|----------------|-----------------|
| Rainwater Harvesting             |                |                 |
| ENERGY SAVINGS                   |                |                 |
| ELECTRICITY                      | Water          | CO <sub>2</sub> |
| KWh                              | m <sup>3</sup> | Kg              |
| 25,864                           | 3,982          | 6,939           |
| 0.34%                            | 3.3%           | 0.3%            |
| ECONOMIC PERFORMANCE             |                |                 |
| Annual Cost Savings [\$]         |                | USD 11,685.27   |
| Annual Share of Cost Savings [%] |                | 0.4%            |
| Implementation Cost [\$]         |                | USD 30,000.00   |
| Simple Payback [Years]           |                | 2.6             |
| Net Present Value [\$]           |                | USD 122,001.27  |
| Internal Rate of Return [%]      |                | 39%             |

## 9.2. WCM – 02: [CAPEX] Installation of a Water Leakage System

### 9.2.1. Rationale

A crucial water conservation measure in any facility is the prevention of water leaks. The water leakage system is an innovative water monitoring system designed to help identify leaks and prevent water damage. It uses advanced sensor technology to detect changes in water usage and flow, allowing property managers to quickly identify leaks and take appropriate action before significant damage occurs. By identifying areas where water is being wasted or used inefficiently, users can take steps to reduce their overall water consumption and save money on their water bills.

### 9.2.2. Proposed water conservation action

The present WCM consists of installing a /system in FSRMKH. The system consists of a set of sensors that are installed at key points in a building's water supply system, such as near the main water intake or on individual appliances like washing machines and dishwashers. These sensors are wirelessly connected to a central hub that collects and analyzes data in real-time. The hub can be accessed through a smartphone app, which provides users with a clear view of their water usage patterns and alerts them to any unusual activity or potential leaks as seen in the snapshot below.

Figure 105: Example of a Water Leakage system – Source : Smartvatten, <https://smartvatten.com/>

| Meter number or name / Device ID                         | Leak reason                                                         | Detected         | Ended            | Leak                     |
|----------------------------------------------------------|---------------------------------------------------------------------|------------------|------------------|--------------------------|
| Water 5 283003-03 (Outside Anciovi Kitchen) / 139071     | Use of conservation reservoir                                       | 19/11/2022 07:10 | 20/11/2022 07:10 | 334.8l/h, hourlimit: 50  |
| Water 5 283003-03 (Outside Anciovi Kitchen) / 139071     | Use of conservation reservoir                                       | 16/11/2022 07:10 | 17/11/2022 07:10 | 1037l/h, hourlimit: 50   |
| Water 5 283003-03 (Outside Anciovi Kitchen) / 139071     | Swimming pool fill-up                                               | 03/11/2022 07:10 | 15/11/2022 07:10 | 1116.2l/h, hourlimit: 50 |
| Water 5 283003-03 (Outside Anciovi Kitchen) / 139071     | Usage of conservation reservoir for irrigation                      | 30/08/2022 07:10 | 01/11/2022 07:10 | 1280.8l/h, hourlimit: 50 |
| Water 5 283003-03 (Outside Anciovi Kitchen) / 139071     | Cleaning of Kitchen's grease traps and drainages done in 2 kitchens | 07/07/2022 07:10 | 29/08/2022 07:10 | 1867.1l/h, hourlimit: 50 |
| Water 2 28301903 (Via Bonifacio-Staff Entrance) / 156927 | Use of conservation reservoir                                       | 03/11/2022 06:10 | 04/11/2022 06:10 | 435l/h, hourlimit: 50    |
| Water 2 28301903 (Via Bonifacio-Staff Entrance) / 156927 | Leakage plunge pools                                                | 01/11/2022 06:10 | 02/11/2022 06:10 | 182.1l/h, hourlimit: 50  |
| Water 2 28301903 (Via Bonifacio-Staff Entrance) / 156927 | Leakage in one plunge pool                                          | 28/10/2022 06:10 | 29/10/2022 06:10 | 296.4l/h, hourlimit: 50  |
| Water 2 28301903 (Via Bonifacio-Staff Entrance) / 156927 | Use of conservation reservoir                                       | 25/10/2022 06:10 | 27/10/2022 06:10 | 404.1l/h, hourlimit: 50  |
| Water 2 28301903 (Via Bonifacio-Staff Entrance) / 156927 | Use of conservation reservoir                                       | 22/10/2022 06:10 | 23/10/2022 06:10 | 392.5l/h, hourlimit: 50  |
| Water 2 28301903 (Via Bonifacio-Staff Entrance) / 156927 | Use of conservation reservoir                                       | 13/10/2022 06:10 | 14/10/2022 06:10 | 297.9l/h, hourlimit: 50  |
| Water 2 28301903 (Via Bonifacio-Staff Entrance) / 156927 | Use of conservation reservoir                                       | 11/10/2022 06:10 | 12/10/2022 06:10 | 366.2l/h, hourlimit: 50  |
| Water 2 28301903 (Via Bonifacio-Staff Entrance) / 156927 | Use of conservation reservoir                                       | 05/10/2022 06:10 | 10/10/2022 06:10 | 54.1l/h, hourlimit: 50   |
| Water 2 28301903 (Via Bonifacio-Staff Entrance) / 156927 | Plunge pools pipe broken                                            | 02/10/2022 06:10 | 03/10/2022 06:10 | 233.9l/h, hourlimit: 50  |
| Water 2 28301903 (Via Bonifacio-Staff Entrance) / 156927 | Use of conservation reservoir                                       | 28/09/2022 06:10 | 30/09/2022 06:10 | 412.1l/h, hourlimit: 50  |



One of the key benefits of the Water Leakage system is its ability to distinguish between normal water usage patterns and those that indicate a leak or other issue. For example, the system can detect a continuous flow of water when no appliances are in use, which is a sign of a leaky pipe or faucet. It can also detect sudden spikes in water usage, which may indicate a burst pipe or other emergency situation.

In addition to providing early warning of potential leaks and other issues, the water Leakage system can also help property managers optimize their water usage and reduce waste. The dashboard provides detailed insights into water usage patterns, allowing users to identify areas where they can reduce consumption and save money on their water bills.

The table below details the key financial indicators of this EEM.

| WCM - 02                             |        |                 |
|--------------------------------------|--------|-----------------|
| Installation of Water Leakage system |        |                 |
| ENERGY SAVINGS                       |        |                 |
| ELECTRICITY                          | Water  | CO <sub>2</sub> |
| KWh                                  | m3     | Kg              |
| 0                                    | 14,691 | 0               |
| 0.00%                                | 12.3%  | 0.0%            |
| ECONOMIC PERFORMANCE                 |        |                 |
| Annual Cost Savings [\$]             |        | USD 12,862.32   |
| Annual Share of Cost Savings [%]     |        | 0.5%            |
| Implementation Cost [\$]             |        | USD 50,000.00   |
| Simple Payback [Years]               |        | 3.9             |
| Net Present Value [\$]               |        | USD 117,312.20  |
| Internal Rate of Return [%]          |        | 26%             |

### 9.3. WCM – 03: [CAPEX] Replacement of faucets and Aerators in Guestrooms and Public Toilets

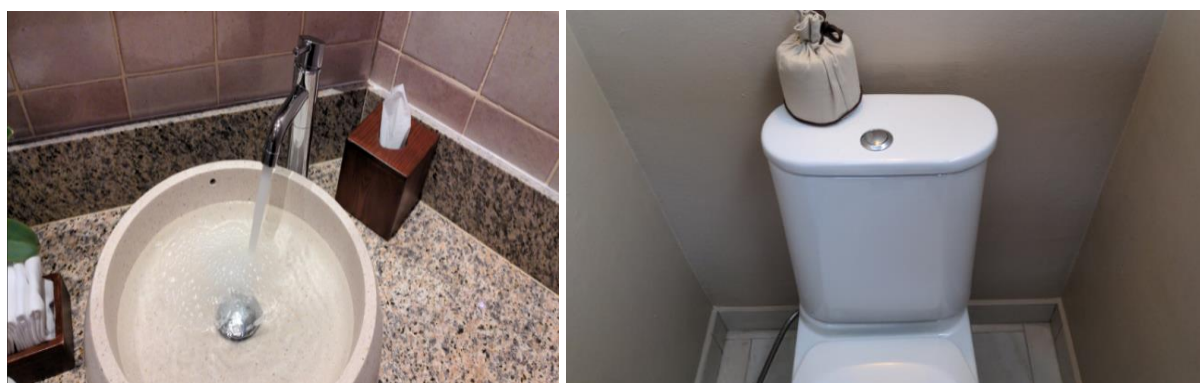
#### 9.3.1. Rationale

Resorts have a substantial water consumption which is dependent upon factors like services provided, equipment used, age of facility, and other factors, this large range indicates that some resorts are using water more efficiently than others and there is a lack of consensus on a specific benchmark per typical facility in a certain region. However, there are clearly facilities that use water more efficiently than others.

#### 9.3.2. Proposed water conservation action

In FSMKH water is mainly used in pools, guestrooms and public toilets and faucets, kitchens, and laundry. As mentioned earlier the plumbing fixtures in public restrooms and guestrooms are without sensors and toilets are all single flush - push buttons.

Figure 106: faucet (left), toilet (right)



The table below shows the total faucets and toilets inventory in FSRMKH.

Table 36: Water Inventory

|                | Locations  |            |              |          |             |       |
|----------------|------------|------------|--------------|----------|-------------|-------|
|                | 100 Series | 200 Series | Water Villas | E- Hotel | Public Area | Total |
| <b>Faucets</b> | 52         | 69         | 94           | 149      | 37          | 401   |
| <b>Toilets</b> | 27         | 35         | 51           | 150      | 47          | 310   |

The first step towards water efficiency is to review the maintenance procedures to ensure that leaks are found and repaired as they occur. Ensure janitorial staff shut off all faucets and report leak immediately. These seemingly small measures can have big returns, especially if staff understand the value of their efforts.

Here below is a list of issues that could help in the overall water management:

- 1. Water Pressure Reduction:** Review if water pressure can be reduced without affecting showers
- 2. Installation of Water Displacement device:** installing a water displacement device (or 'Save-a-Flush') as existing toilets in FSRMKH are of single flush type – these cut the amount of water used for each flush, typically by one or two litres. Alternatively use Flush Volume Restrictor which adjust the flush volume to ensure optimal flushing.
- 3. Changing Aerators to water efficient ones through:**
  - a. Automatic Sensor Faucets:** This type of faucet is installed in the public toilets and can be replaced by the metering faucets at the service areas toilets. An automatic faucet has the advantage of shutting off automatically after hand washing, thereby reducing water waste. The automatic shutoff mechanism also reduces the risk of sink overflow due to a faucet being left on either inadvertently or deliberately. Automatic sensor faucets are water saving devices that help save 70% of the water that would otherwise be unused. Other benefits of automatic faucets are found in inhibiting the spread of germs which are known to thrive on faucet handles, as well as help prevent or mitigate scalding incidents caused by hot water. A total of 401 faucets can be retrofitted.
  - b. Installation of Flow Control:** Flow control is achieved by stopping the kinetic energy at the faucet's discharge point down to less than 4 feet per second and directing water through precisely engineered and spaced perforated plates within the flow control. The application of the special Inline Flow Controls would save between 20% and 80% on every application. This can be achieved by replacing faucet aerators with new Flow Controls and installing Inline Flow Controls before showerheads.

Water faucets flow was taken during the field audit and the following results are shown in the table below.

Table 37: Water flowmeter readings

|                         | Time    | Quantity | Flow          |
|-------------------------|---------|----------|---------------|
| Unit                    | seconds | Liters   | Liters/Second |
| guestroom changing room | 20      | 2        | 0.099         |
| guestroom #108          | 13      | 2        | 0.150         |
| café Huraa male toilet  | 17      | 2        | 0.114         |
| guestroom #124          | 11      | 2        | 0.182         |
| guestroom #319          | 11      | 2        | 0.175         |

The below table showcases the savings to be achieved by using appropriate water savings aerators.

Table 38: Expected Water Savings per application (using Flow Controls)

| Plumbing Fixture | Water Used   |                   |           |
|------------------|--------------|-------------------|-----------|
|                  | With Aerator | With Flow Control | % Savings |
| Public Restroom  | 2.5 GPM      | 0.5 GPM           | 83-90%    |

Ordinary aerators don't compensate for pressure changes in inlet pressure, so the greater the pressure the more water you use. Flow Controls on the other hand, use a unique pressure compensation system that saves water and reduces the amount of energy needed to heat water, by delivering the same flow rate, regardless of water pressure.

The flow control in addition to the automatic sensor faucets can be for faucet can be applicable in all toilets in FSMRKH, which can lead to a reduction of more than 40% of the water consumed in these areas.

In the absence of a real water consumption tracking in the faucets of the E hotel (Staff housing) , it is not possible to calculate the total potential savings achieved by this measure. However, the below table provides an estimate of the potential savings in terms of water and electricity and the related CAPEX. The savings calculations will be based on an estimation of 5 hours of use daily.

Table 39: Faucets – Water Conservation Measures Project Figure

|                | Time    | Quantity   | Flow  | Flow | Total Flow Estimated | Savings | Water Savings  | Electricity savings | Water Savings (Annual) | Electricity Savings (Annual) | Electricity Savings (Annual) |
|----------------|---------|------------|-------|------|----------------------|---------|----------------|---------------------|------------------------|------------------------------|------------------------------|
| Unit           | seconds |            | LPS   | LPM  | LPM                  | LPM     | \$/Minute      | KWh/Minute          | \$                     | KWh                          | \$                           |
| 100s series    | 13      | 52         | 0.166 | 9.96 | 517.92               | 258.96  | 226.724659     | 1.683               | 165,509                | 3,072                        | 974                          |
| 200s series    | 13      | 69         | 0.217 | 13   | 897                  | 448.5   | 392.670719     | 2.915               | 286,650                | 5,320                        | 1,687                        |
| Public Toilets | 17      | 37         | 0.114 | 6.84 | 253.08               | 126.54  | 110.788301     | 0.823               | 80,875                 | 1,501                        | 476                          |
| Water Villas   | 11      | 94         | 0.175 | 10.5 | 987                  | 493.5   | 432.069119     | 3.208               | 315,410                | 5,854                        | 1,856                        |
| <b>Total</b>   |         | <b>401</b> |       |      |                      |         | <b>1,162.3</b> | <b>8.6</b>          | <b>\$ 848,445</b>      | <b>15,747</b>                | <b>\$ 4,992</b>              |

Assuming 1Liter reduction per flush from the toilets flush control installation, savings are shown in the table below :

Table 40 : Toilets Estimated Savings

| Type                          | Location         | Quantity   | Estimated Water Savings (Liter) | Estimated Water Savings (m3) | Cost Savings (\$) |
|-------------------------------|------------------|------------|---------------------------------|------------------------------|-------------------|
| toilets                       | Guest Rooms      | 113        | 628,131                         | 628.13                       | 549.94            |
| toilets                       | E Accommodations | 150        | 21,900                          | 21.90                        | 19.17             |
| toilets                       | Public Area      | 47         | 17,155                          | 17.16                        | 15.02             |
| <b>Total / <u>Average</u></b> |                  | <b>310</b> | <b>667185.84</b>                | <b>667.18584</b>             | <b>584.1345</b>   |

The table below presents the key indicators of this EEM.

| WCM - 03                                  |       |                 |
|-------------------------------------------|-------|-----------------|
| Installation of Aerators and Flow Control |       |                 |
| ENERGY SAVINGS                            |       |                 |
| ELECTRICITY                               | Water | CO <sub>2</sub> |
| KWh                                       | m3    | Kg              |
| 11,338                                    | 2,412 | 3,042           |
| 0.15%                                     | 2.0%  | 0.1%            |
| ECONOMIC PERFORMANCE                      |       |                 |
| Annual Cost Savings [\$]                  |       | USD 5,705.54    |
| Annual Share of Cost Savings [%]          |       | 0.2%            |
| Implementation Cost [\$]                  |       | USD 48,922.00   |
| Simple Payback [Years]                    |       | 8.6             |
| Net Present Value [\$]                    |       | USD 25,295.26   |
| Internal Rate of Return [%]               |       | 12%             |

## 10. WCMs Summary Figures & Implementation Notes

### 10.1. WCMs Summary – CAPEX, Cost Savings & Payback Period

The following table provides the summary of the Water Conservation Measure (WCM) with key indicators including CAPEX, % Savings from Total Utilities Cost and Payback period. As can be seen, if all EEMs are implemented, the total annual cost savings would be **\$30,253** at a CAPEX of **\$128,922** and a payback of **4.26 Years**.

Table 41: WCMs summary – CAPEX, Cost savings and Payback Period

| CODE     | WCM Name                                  | ANNUAL SAVINGS     |                     | COST         | PAYBACK |
|----------|-------------------------------------------|--------------------|---------------------|--------------|---------|
| -        | -                                         | \$                 | KGs CO <sub>2</sub> | \$           | Years   |
| WCM - 01 | Rainwater Harvesting                      | \$11,685.27        | 6,939               | \$30,000.00  | 2.57    |
| WCM - 02 | Installation of Water Leakage system      | \$12,862.32        | 0                   | \$50,000.00  | 3.89    |
| WCM - 03 | Installation of Aerators and Flow Control | \$5,705.54         | 3,042               | \$48,922.00  | 8.57    |
|          | TOTAL *                                   | \$30,253.13 (1.2%) | 9,981               | \$128,922.00 | 4.26    |

## 10.2. WCMs Summary – Electricity (KWh) and Water (m3) Savings

The below table provides the annual savings of every WCM and for each utility. Electricity savings are seen to be **25,864 KWh** per year representing **0.3%** of the actual electricity consumption of the RO while the water savings are at **21,085 m3** corresponding to **17.7%** of the total water production.

Table 42:WCM Summary – Electricity (KWh) and Water (m3) Savings

| CODE     | EEM Name                                  | ELECTRICITY   | WATER         |
|----------|-------------------------------------------|---------------|---------------|
| -        | -                                         | KWh           | M3            |
| WCM - 01 | Rainwater Harvesting                      | 25,864 (0.3%) | 3,982 6.52%   |
| WCM - 02 | Installation of Water Leakage System      | 0 (0.0%)      | 14,691 12.32% |
| WCM - 03 | Installation of Aerators and Flow Control | 0 (0.0%)      | 2,412 1.46%   |
|          | TOTAL                                     | 25,864 0.3%   | 21,085 17.7%  |

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END OF REPORT